Institut Polytechnique des Sciences Avancées

Analysis of Flying Qualities and Dynamic Stability

Technical Evaluation Report: C3 Aircraft Model

Module: Flight Mechanics: Flying Qualities
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1 Aircraft Data

Category	Parameter	Symbol	Value	Unit
Geometry	Wing Surface	S	12.63	m^2
	Mean Aerodynamic Chord	MAC	1.65	m
	Wingspan	b	8.02	m
	Aspect Ratio	A	5.09	–
	Oswald Number	e	0.94	–
Flight Conditions	Altitude	h	10 668	m
	Mach Number	M	0.6	–
	Steady state angles of attack	θ_0	0	deg
	True Airspeed	U_0	178	m/s
	Dynamic Pressure	$\bar{q}$	6 036	Pa
	Air Density	ρ	0.381	kg/m^3
Mass & Inertia	Total Mass	m	1 816	kg
	Roll Inertia	I_{XX}	1 085	$kg \cdot m^2$
	Pitch Inertia	I_{YY}	6 511	$kg \cdot m^2$
	Yaw Inertia	I_{ZZ}	7 053	$kg \cdot m^2$
	Inertia Product	I_{XZ}	271	$kg \cdot m^2$
Steady State	Trim Lift Coefficient	C_{L_0}	0.23	–
	Trim Drag Coefficient	C_{D_0}	0.025	–
Longitudinal Stability	Speed-Drag Derivative	C_{D_u}	0	–
	Angle of Attack-Drag Derivative	C_{D_α}	0.17	–
	Speed-Lift Derivative	C_{L_u}	0.132	–
	Lift Curve Slope	C_{L_α}	5.5	–
	Rate of change of α-Lift	$C_{L_{\dot{\alpha}}}$	4.2	–
	Pitch Rate-Lift Derivative	C_{L_q}	10	–
	Speed-Moment Derivative	C_{m_u}	0	–
	Static Stability Derivative	C_{m_α}	-0.24	–
	Pitching Moment Damping ($\dot{\alpha}$)	$C_{m_{\dot{\alpha}}}$	-9.6	–
	Pitch Damping Derivative	C_{m_q}	-17.7	–
Lateral Stability	Dihedral Effect Derivative	C_{L_β}	-0.11	–
	Roll Damping Derivative	C_{L_p}	-0.39	–
	Weathercock Stability	C_{L_r}	0.31	–
	Static Directional Stability	C_{n_β}	0.17	–
	Yawing Moment-Roll Rate	C_{n_P}	0.08	–
	Yaw Damping Derivative	C_{n_r}	-0.26	–
	Side Force-Beta Derivative	C_{y_β}	-1	–
	Side Force-Roll Rate	C_{Y_P}	-0.12	–
	Side Force-Yaw Rate	C_{Y_r}	0.62	–
Control Derivatives	Elevator-Drag	$C_{D_{\delta_e}}$	0	–
	Elevator Effectiveness (Lift)	$C_{L_{\delta_e}}$	0.035	–
	Elevator Effectiveness (Moment)	$C_{m_{\delta_e}}$	-0.82	–
	Aileron-Side Force	$C_{Y_{\delta_A}}$	0	–
	Aileron Effectiveness (Roll)	$C_{l_{\delta_A}}$	0.1	–
	Aileron Effectiveness (Yaw)	$C_{n_{\delta_A}}$	-0.005	–
	Rudder-Side Force	$C_{y_{\delta_r}}$	0.28	–
	Rudder Effectiveness (Roll)	$C_{l_{\delta_r}}$	0.05	–
	Rudder Effectiveness (Yaw)	$C_{n_{\delta_r}}$	-0.12	–

Table 1: Complete physical, aerodynamic, and stability data for the C3 aircraft model.

2 Introduction

This report presents a comprehensive analysis of the dynamic stability and flying qualities of the "C3" aircraft. The C3 model has pretty much the same parameters as the Marchetti S-211 (an Italian two-seat jet-powered military training and tactical support aircraft), but in this report we will be calling it C3. The objective is to model the aircraft's behavior, analyze its eigenmodes (longitudinal and lateral), and verify compliance with international standards (MIL-F-8785C). MATLAB will be used for numerical resolution and time-domain visualization.

3 Chapter 1: Flying Qualities and Regulatory Framework

This chapter explores the fundamental concepts of flying and handling qualities, their formulation into international standards, and the specific requirements used to evaluate aircraft stability and control.

3.1 Coordinate Systems and Flight Dynamics Fundamentals

Before analyzing the flying qualities of an aircraft, it is essential to define the physical framework and the variables that describe its motion in space. An aircraft is a rigid body moving with six degrees of freedom (6-DOF): three translational motions along the axes and three rotational motions around them.

First, to describe the complex motion of an aircraft, we must distinguish between several reference frames, as the laws of physics are typically calculated in one frame while the pilot's experience occurs in another. The most fundamental is the Earth-Centered Inertial (ECI) frame, often simplified to a North-East-Down (NED) local tangent plane for atmospheric flight, which serves as our "fixed" reference to measure altitude and position. However, because aerodynamic forces like lift and drag are generated by the aircraft's orientation relative to the air, we primarily work within the Body-Fixed Coordinate System (F_b). This frame is attached to the aircraft, meaning the axes rotate and translate exactly as the vehicle does, allowing us to define "forward," "right," and "down" from the pilot's perspective. Understanding the mathematical transformation between the fixed Earth frame and the moving Body frame—typically via Euler angles (ϕ, θ, ψ)—is the key to tracking the aircraft's trajectory and attitude simultaneously.

Then, to describe translational and rotational motion, we primarily use the **Body-Fixed Coordinate System** (F_b), where the origin is the aircraft's center of gravity. The following table summarizes the fundamental variables used in our study:

Category	Variable	Symb.	Axis	Unit
Transl.	Longit. velocity	u	x_b (Fwd)	m/s
	Lateral velocity	v	y_b (Right)	m/s
	Vertical velocity	w	z_b (Down)	m/s
Rotat.	Roll rate	p	x_b (Longit.)	rad/s
	Pitch rate	q	y_b (Lat.)	rad/s
	Yaw rate	r	z_b (Vert.)	rad/s
Attit.	Bank angle (Roll)	ϕ	–	deg
	Elev. angle (Pitch)	θ	–	deg
	Azim. angle (Yaw)	ψ	–	deg
Forces	Aerodyn. Forces	X, Y, Z	x_b, y_b, z_b	N
Moments	Aerodyn. Moments	L, M, N	x_b, y_b, z_b	Nm

Table 2: Standard variables and units for aircraft flight dynamics.

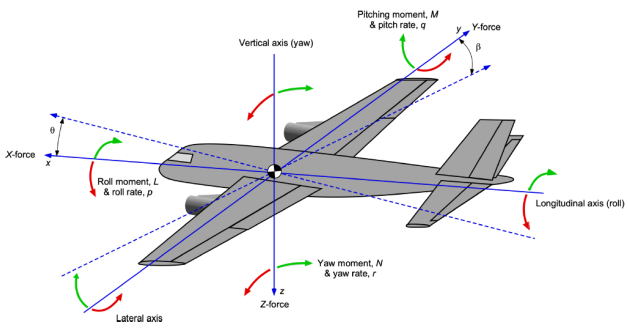


Figure 1: Aircraft Body-Fixed Coordinate System (F_b) showing Aerodynamic Forces (X, Y, Z), Moments (L, M, N), Angular Rates (p, q, r), and Attitude Angles (θ, β).

Understanding the Variables To simplify, each category of variables plays a specific role in our study:

- **Forces (X, Y, Z):** They describe the "push" or "pull" on the aircraft, determining how it accelerates or slows down along its path.
- **Moments (L, M, N):** These represent the "twisting" effects that cause the aircraft to rotate around its axes (rolling, pitching, or yawing).
- **Angles (ϕ, θ, ψ):** They define the aircraft's

orientation relative to the Earth, telling us if the nose is up or if the wings are level.

- **Angular Rates (p, q, r):** These measure the speed of rotation; they are essential for describing the "agile" or "sluggish" behavior of the stability modes.

Note: In the following chapters, we will use "small perturbations theory," where we assume the aircraft stays close to its equilibrium state, allowing us to linearize these complex relationships into simpler matrices. The general equations of motion for an aircraft are highly nonlinear and coupled. To analyze the stability modes analytically, we apply the ***Small Perturbation Theory****, where we assume the aircraft is in a steady equilibrium state (trim), and any motion is a small deviation from this state:*

$$X_{total} = X_{trim} + x_{perturbation} \tag{1}$$

By neglecting higher-order terms (products of small perturbations), we can reduce the complex mechanics to a set of linear differential equations represented in State-Space form:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \tag{2}$$

Where $\mathbf{A}$ *is the stability matrix containing the aerodynamic derivatives, and* $\mathbf{B}$ *is the control matrix. This linearization is valid only as long as the maneuvers remain within the linear aerodynamic range (small angles of attack and sideslip), this will be studied in Chapter 3.*

3.2 Flying Qualities, Stability, Control, and Handling Qualities

3.2.1 Definitions and Fundamental Concepts

The study of flying qualities is essential for determining how easily and effectively an aircraft can be controlled during various flight phases or specific mission tasks. It addresses the fundamental need to ensure that the aircraft-pilot system is safe, secure, and capable of meeting procurement requirements while minimizing pilot workload.

To understand the scope of this field, four interrelated notions must be defined:

- **Stability:** This refers to the inherent aerodynamic tendency of the aircraft to maintain or return to its initial steady-state equilibrium (trimmed condition) after being subjected to an external disturbance or atmospheric turbulence. In dynamic analysis, stability is characterized by the aircraft's "free response" and its specific stability modes, such as the Phugoid or Dutch Roll.

- **Control:** Unlike stability, which focuses on maintaining an attitude, control deals with the aircraft's ability to change its flight path or direction in response to pilot commands. It is the effectiveness with which control surface deflections (elevator, ailerons, or rudder) translate into desired changes in attitude and velocity.

- **Handling Qualities:** These represent a qualitative and often subjective assessment of the aircraft-pilot system performance. They are formulated in terms of pilot opinion regarding the ease of executing a mission. Handling qualities are essentially "closed-loop" characteristics where the pilot's actions are an integral part of the feedback loop.

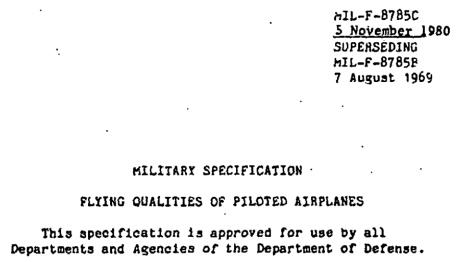
- **Flying Qualities:** This is a broader term that encompasses both the stability and control characteristics (open-loop) and the handling qualities (closed-loop). Flying qualities must be adequate to ensure that the mission can be accomplished without excessive pilot workload.

3.2.2 Formulation of Criteria and Standards

Flying quality criteria are not universal; they are formulated based on extensive historical correlations between technical aircraft characteristics and pilot judgements. Because different aircraft have different missions—ranging from heavy transports to highly maneuverable fighters—these criteria are categorized by aircraft class and flight phase.

The essential content of international standards, such as the **MIL-F-8785C** (American Military Specification), provides a rigorous mathematical framework for these criteria. These standards define quantitative requirements for damping factors (ζ) and natural frequencies (ω_n) of the aircraft's stability modes, ensuring that the aircraft remains safe and predictable across its entire flight envelope. Compliance with these standards is measured

through defined "Levels" (Level 1 to 3), which correlate technical performance with the resulting pilot workload.


MIL-F-8785C
5 November 1980
SUPERSEDING
MIL-F-8785B
7 August 1969

MILITARY SPECIFICATION

FLYING QUALITIES OF PILOTED AIRPLANES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Figure 2: MIL-F-8785C document title

3.2.3 Relationship Between Notions

Flying qualities act as the bridge between the aircraft's physical dynamics and the mission task requirements. While stability and control are intrinsic aerodynamic properties, handling qualities represent how the pilot perceives these properties in a "closed-loop" operation.

3.3 Classification of Airplanes

To account for the vast diversity in aircraft design and operational requirements, the MIL-F-8785C standard organizes airplanes into four distinct classes. These classifications are based primarily on the aircraft's size, weight, and intended maneuverability, as these physical characteristics directly dictate the expected dynamic response and the pilot's control authority.

- **Class I - Small Light Airplanes:** This category includes light utility aircraft, primary trainers, and small civilian airplanes. These aircraft typically exhibit high natural stability and are intended for flight phases where extreme maneuvers are not required. The C3 model, given its mass and geometric properties, is primarily evaluated under this classification (e.g. Cessna 172 Skyhawk, C3).
- **Class II - Medium Weight, Low-to-Medium Maneuverability:** This class covers medium-sized airplanes such as light transports, business jets, or reconnaissance aircraft. They possess more significant inertia than Class I aircraft, requiring different criteria for response times and damping (e.g. Lockheed C-130 Hercules).
- **Class III - Large Heavy, Low-to-Medium Maneuverability:** This group is composed of heavy transport aircraft, tankers, and large bombers. Due to their massive weight and large moments of inertia, their flight dynamics are characterized by much slower response rates. Criteria for these aircraft focus on stability and safety during long-duration cruises rather than agility (e.g. Boeing 747).
- **Class IV - High Maneuverability:** This class is reserved for tactical aircraft such as fighters, interceptors, and attack planes. These aircraft are often designed to be inherently less stable (or even unstable) to maximize maneuverability. The requirements for Class IV are the most stringent, focusing on precise tracking and rapid response to pilot inputs during combat maneuvers (e.g. F-16 Fighting Falcon).

3.4 Flight Phase Categories

The operational requirements of an aircraft vary significantly throughout a mission. To account for these variations, the MIL-F-8785C standard classifies flight phases into three distinct categories based on the precision required and the nature of the maneuvers being performed.

- **Category A - Non-terminal Flight Phases:** These phases require rapid maneuvering, precise tracking, or very accurate flight-path control. Failure to maintain high precision in these phases can compromise mission success or safety. Representative operations include:
  - Air-to-air combat and ground attack.
  - In-flight refueling (as the receiver aircraft).
  - Terrain following and close formation flying.
- **Category B - Non-terminal Flight Phases (Gradual Maneuvering):** These phases involve more gradual maneuvers and less stringent tracking requirements. They represent the "utility" part of the flight profile where stability and efficiency are prioritized over agility. These typically include:
  - Climb, cruise, and descent.
  - In-flight refueling (as the tanker aircraft).
  - Loitering and emergency descents.

For the C3 model analyzed in this study, which focuses on high-altitude cruise at Mach 0.6, the requirements of Category B are the primary reference.

- **Category C - Terminal Flight Phases:** These phases involve the critical transitions between the atmosphere and the ground. They require high precision but are generally limited in maneuver amplitude. They include:
  - Take-off (normal and catapult).
  - Approach and landing.
  - Wave-off or go-around maneuvers.

The classification into these categories allows for a nuanced evaluation of stability modes. For instance, a damping ratio that is considered adequate for Category B (cruise) might be judged insufficient for Category A (combat), where a more responsive and stable platform is required for weapon aiming.

3.5 Levels of Flying Qualities

The adequacy of an aircraft's flying qualities is categorized into three levels, which define the relationship between the aircraft's dynamic performance and the pilot's ability to complete the mission. These levels are often assessed using the Cooper-Harper Rating Scale, a subjective tool used by test pilots to quantify aircraft handling.

- **Level 1 - Satisfactory:** Flying qualities are clearly adequate for the mission flight phase. The aircraft's stability and control characteristics allow for the desired performance to be achieved with a minimal and easily manageable pilot workload. In this state, the pilot can focus on mission objectives rather than on simply controlling the aircraft (e.g. commercial airliner in cruise).
- **Level 2 - Acceptable:** Flying qualities are adequate to accomplish the mission flight phase, but some increase in pilot workload or a degradation in mission effectiveness (or both) is observed. The pilot must exert more concentration and effort to maintain the desired flight path or attitude, which may lead to fatigue over long durations (e.g. yaw damper failure).
- **Level 3 - Controllable:** The aircraft is safe to fly and can be controlled, but the pilot workload is excessive, and mission effectiveness is severely compromised. In this state, the pilot's entire attention is usually consumed by the task of keeping the aircraft stable, making it difficult to perform any secondary mission tasks (e.g. fighter jet with degraded flight control).

If the flying qualities fall below Level 3, the aircraft is considered "unacceptable," and the pilot may be unable to maintain control, potentially leading to a catastrophic failure of the aircraft-pilot system. For the C3 model, our objective will be to determine if the natural stability modes meet Level 1 requirements or if a Stability Augmentation System (SAS) is necessary to reduce the pilot's workload to an acceptable level.

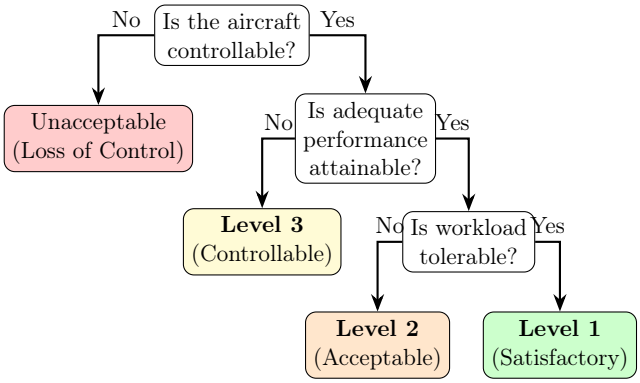


Figure 3: Simplified Cooper-Harper Decision Logic.

3.6 Flying Quality Requirements

3.6.1 Longitudinal Flying Qualities

The longitudinal evaluation is divided into two distinct dynamic phenomena, each with specific requirements for safety and pilot comfort.

- **Short Period (SP):** This mode represents a rapid oscillation of the angle of attack and pitch rate. Requirements focus on high damping ($\zeta > 0.3$) and an appropriate natural frequency to avoid Pilot-Induced Oscillations (PIO).

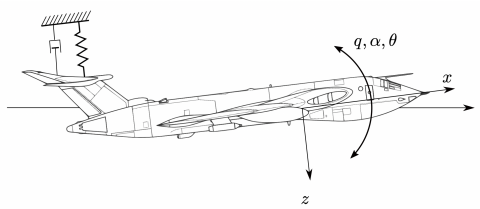
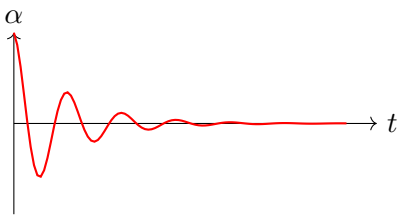
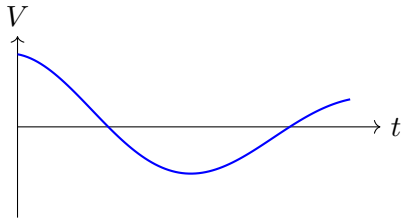


Figure 4: Aircraft undergoing short period



Settling time: 1 to 5 seconds

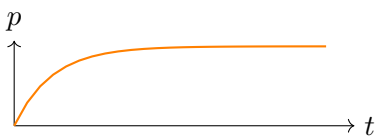
- **Phugoid (PH):** A slow energy exchange between altitude and speed at a nearly constant angle of attack. While it can be poorly damped, it must be stable enough for the pilot to maintain trim without constant effort.



Settling time: 1 to 4 minutes (60 - 250s)

3.6.2 Lateral-Directional Flying Qualities

- **Roll Mode:** A non-oscillatory convergence of the roll rate. It is characterized by its time constant (τ_R); a fast roll mode is essential for agile maneuvering.



Settling time: 0.5 to 2 seconds

- **Spiral Mode:** A very slow, non-oscillatory mode. If unstable (divergent), the divergence must be slow enough (doubling time $> 12s$ or $20s$) to allow the pilot to correct the bank angle.

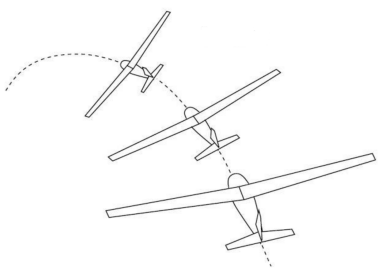
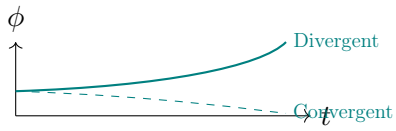


Figure 5: Aircraft undergoing "unstable" spiral



Settling time: 100 to 1500+ seconds

Sensory Illusion: *The human vestibular system has a specific threshold for detecting angular acceleration. If the aircraft enters a divergent spiral mode very gradually, the fluid in the semi-circular canals does not move sufficiently to stimulate the sensory cilia. Consequently, the pilot may suffer the illusion of level flight while the bank angle is actually increasing.*

Vector Confusion: *Without visual reference, the pilot's body interprets the resultant force (gravity + centripetal) as "down", confirming the illusion of wings-level flight during a turn.*

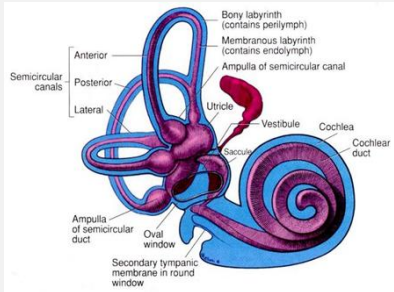


Figure 6: Anatomy of the human inner ear vestibular system

- **Dutch Roll:** A complex oscillatory mode coupling yaw, roll, and sideslip. It requires a minimum damping ratio ($\zeta > 0.08$) to avoid an uncomfortable and tiring "fishtailing" sensation for the crew and passengers.

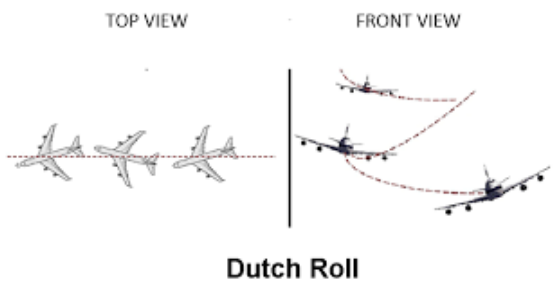
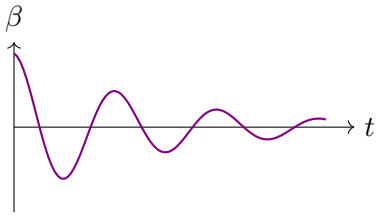


Figure 7: Aircraft undergoing dutch roll



Settling time: 10 to 45 seconds

3.7 Stability Augmentation and Flight Control Systems

When an aircraft's natural dynamics fail to meet required Flying Qualities (FQ) levels, stability must be augmented through either aerodynamic design modifications (e.g., surface area, CG adjustment) or a **Flight Control System (FCS)**. Modern engineering often favors FCS, using feedback control techniques to artificially modify the airframe's apparent aerodynamic behavior.

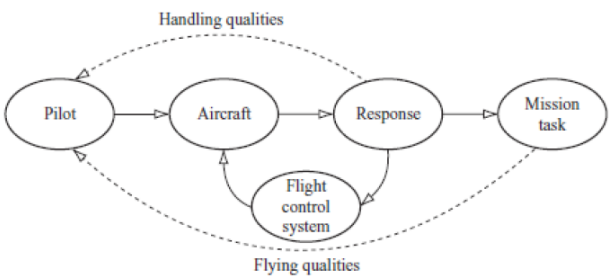


Figure 8: Flight control feedback loop

A classic example is the "Yaw Damper," which utilizes rate gyro feedback to enhance Dutch Roll damping without structural changes. By reducing pilot workload and suppressing modal deficiencies, these systems ensure the aircraft maintains Level 1 requirements throughout the flight envelope. This study will evaluate if the C3 model requires such augmentation for its high-altitude cruise mission.

3.8 Summary of Quantitative Flying Quality Requirements

To consolidate the regulatory framework, this section summarizes the benchmarks used to evaluate the C3 aircraft. For the following tables, specific abbreviations are introduced to optimize space and readability: **FQ** for Flying Qualities, **FPC** for Flying Phase Category, and **FC** for Flight Category. These requirements are extracted from the MIL-F-8785C standard and are essential for determining the satisfaction levels from Level 1 to Level 3.

3.8.1 Longitudinal Requirements

The Phugoid is a long-period oscillation involving an exchange between kinetic and potential energy. Because it occurs at a nearly constant angle of attack and changes very slowly, it is generally easy for a pilot to control. Consequently, the requirements for this mode are less stringent: while Level 1 requires a small positive damping, Level 3 even permits a slow instability, provided the period remains long enough for pilot intervention.

Level of FQ	**Minimum damping ratio ζ_{ph}**
Level 1	0.04
Level 2	0
Level 3	Unstable, with period $T_{ph} > 55$ s

Table 3: Phugoid mode damping ratio requirements.

The Short Period mode is a rapid oscillation of the angle of attack and pitch rate that must be heavily damped to ensure precise control. Unlike the Phugoid, its settling time is very short, and insufficient damping in this mode is a primary cause of Pilot-Induced Oscillations (PIO). The requirements are strictly bounded between a **minimum** value to ensure stability and a **maximum** value to avoid an over-damped, sluggish response that would hinder the pilot's ability to make rapid corrections. For the C3 model in high-altitude cruise (Category B), a minimum damping ratio of 0.30 is required to maintain Level 1 flying qualities.

	Level 1		**Level 2**		**Level 3**
FPC	$\zeta_{sp,min}$	$\zeta_{sp,max}$	$\zeta_{sp,min}$	$\zeta_{sp,max}$	$\zeta_{sp,min}$
Cat. A	0.35	1.30	0.25	2.00	0.10
Cat. B	0.30	2.00	0.20	2.00	0.10
Cat. C	0.50	1.30	0.35	2.00	0.25

Table 4: Short period damping ratio requirements.

3.8.2 Lateral-Directional Requirements

The Roll mode is a non-oscillatory, convergent motion characterized by its time constant, T_r, which represents the time required for the aircraft to reach approximately 63% of its steady-state roll rate following a lateral command. A small time constant is synonymous with high "roll agility," allowing the pilot to initiate or stop a bank angle change almost instantaneously. As shown in the table below, requirements are stricter for high-precision phases (Category A and C) and smaller aircraft (Class I and IV),

where a T_r exceeding 1.0 second would significantly degrade the aircraft's responsiveness during maneuvers like combat or landing flare.

		Max value of T_r (s)		
Class	**FC**	**Level 1**	**Level 2**	**Level 3**
I, IV	A, C	1.0	1.4	–
II, III	A, C	1.4	3.0	–
All	B	1.4	3.0	–

Table 5: Roll mode time constant requirements.

The Spiral mode is a very slow, non-oscillatory lateral-directional motion that can be either convergent (stable) or divergent (unstable). When unstable, it causes the aircraft to slowly increase its bank angle and turn rate, leading to a "spiral dive" if left uncorrected.

Because the divergence is typically slow, pilots can easily compensate for it during normal operations; thus, regulations do not require the mode to be inherently stable, but they do impose a minimum **time to double the bank angle** (T_2). For Level 1 flying qualities in cruise (Category B), the T_2 must be at least 20 seconds, ensuring the pilot has ample time to notice the deviation and apply corrective inputs without constant attention.

	Min value of T_2 (s)		
FPC	**Level 1**	**Level 2**	**Level 3**
Cat. A, C	12	8	5
Cat. B	20	8	5

Table 6: Spiral mode time to double bank angle (T_2 if unstable).

In addition to the doubling time (T_2), the MIL-F-8785C standard defines requirements for the Spiral mode time constant (T_s). While T_2 describes the rate of divergence for an unstable mode, T_s measures the mode's sluggishness or agility in returning to or moving away from equilibrium.For a stable mode, a large time constant indicates slow convergence, meaning the aircraft takes longer to naturally level its wings after a disturbance. Conversely, for an unstable mode, a small time constant represents a rapid departure from steady flight, significantly increasing pilot workload. For Level 1 flying qualities in cruise (Category B), the minimum threshold is set at 28.9 seconds to ensure that natural lateral drift is slow enough for precise course control with minimal effort.

	Min value of T_s (s)		
FPC	**Level 1**	**Level 2**	**Level 3**
Cat. A, C	17.3	11.5	7.2
Cat. B	28.9	11.5	7.2

Table 7: Spiral mode time constant requirements.

The **Dutch Roll** is a complex oscillatory mode involving a high degree of coupling between yaw, roll, and sideslip. Characterized by a "fishtailing" motion, this mode can be highly uncomfortable for passengers and taxing for the pilot to suppress manually. Consequently, the **MIL-F-8785C** standard imposes strict minimum requirements on three key dynamic parameters: the damping ratio (ζ_{dr}), the undamped natural frequency (ω_{dr}), and their product ($\zeta_{dr}\omega_{dr}$), which dictates the actual rate of oscillation decay.

For the C3 aircraft operating in high-altitude cruise (Category B), the Level 1 requirement mandates a minimum damping ratio of 0.08 and a natural frequency of at least 0.5 rad/s. Meeting these criteria ensures that lateral-directional disturbances are dampened rapidly enough to prevent pilot fatigue and maintain tracking precision. As shown in the following table, these requirements become significantly more stringent for Category A flight phases, such as air-to-air combat, where a minimum damping ratio of 0.19 is required to provide the stable platform necessary for precision maneuvering.

		Level 1			Level 2			Level 3	
Class	**FPC**	ζ_{dr}	$\zeta_{dr}\omega_{dr}$	ω_{dr}	ζ_{dr}	$\zeta_{dr}\omega_{dr}$	ω_{dr}	ζ_{dr}	$\zeta_{dr}\omega_{dr}$
I, IV	A	0.19	0.35	1.0	0.02	0.05	0.5	0	0.4
II, III	A	0.19	0.35	0.5	0.02	0.05	0.5	0	0.4
All	B	0.08	0.15	0.5	0.02	0.05	0.5	0	0.4
I, IV	C	0.08	0.15	1.0	0.02	0.05	0.5	0	0.4
II, III	C	0.08	0.10	0.5	0.02	0.05	0.5	0	0.4

Table 8: Dutch roll frequency and damping ratio requirements.

4 Chapter 2: Dynamic Stability and Equations of Motion

This chapter establishes the mathematical and physical foundation required to evaluate the dynamic stability of the C3 aircraft. Dynamic stability refers to the aircraft's ability to return to its original equilibrium state following a disturbance without pilot intervention. To access this behavior, we must transition from Newton's second law to a set of differential equations that describe the vehicle's motion in space. By applying the **Small Perturbation Theory**, we will linearize these complex, coupled equations into a state-space representation, allowing us to isolate and characterize the fundamental longitudinal and lateral stability modes.

4.1 6 Degrees of Freedom Equations of Motion

An aircraft in flight is a rigid body capable of moving in three translational directions and rotating about three axes, resulting in six degrees of freedom (6-DOF). The motion is governed by the general laws of classical mechanics, specifically the conservation of linear and angular momentum.

4.1.1 General Force and Moment Equations

The fundamental equations are derived in the Body-Fixed frame (F_b). According to Newton's second law for a rigid body with constant mass, the summation of external forces ($\mathbf{F}$) and moments ($\mathbf{M}$) relative to the center of gravity can be expressed as:

$$\sum \mathbf{F} = m\left(\frac{d\mathbf{V}}{dt} + \mathbf{\Omega} \times \mathbf{V}\right) \tag{3}$$

$$\sum \mathbf{M} = \frac{d\mathbf{H}}{dt} + \mathbf{\Omega} \times \mathbf{H} \tag{4}$$

Where:

- $\mathbf{V} = [u, v, w]^T$ is the linear velocity vector.
- $\mathbf{\Omega} = [p, q, r]^T$ is the angular velocity vector.
- $\mathbf{H} = \mathbf{I\Omega}$ is the angular momentum, with $\mathbf{I}$ being the inertia tensor.

4.1.2 The Scalar Equations of Motion

By expanding the vector products and considering the aircraft's symmetry (assuming $I_{xy} = I_{yz} = 0$), we obtain the six fundamental non-linear differential equations of motion:

Force Equations (Translational):

$$X - mg\sin\theta = m(\dot{u} + qw - rv) \tag{5}$$

$$Y + mg\cos\theta\sin\phi = m(\dot{v} + ru - pw) \tag{6}$$

$$Z + mg\cos\theta\cos\phi = m(\dot{w} + pv - qu) \tag{7}$$

Moment Equations (Rotational):

$$L = I_{xx}\dot{p} - I_{xz}\dot{r} + qr(I_{zz} - I_{yy}) - I_{xz}pq \tag{8}$$

$$M = I_{yy}\dot{q} + pr(I_{xx} - I_{zz}) + I_{xz}(p^2 - r^2) \tag{9}$$

$$N = I_{zz}\dot{r} - I_{xz}\dot{p} + pq(I_{yy} - I_{xx}) + I_{xz}qr \tag{10}$$

These equations represent a coupled system where translational velocities (u, v, w) and rotational rates (p, q, r) are interdependent. In the following sections, we will simplify this system by assuming small perturbations around a steady-state flight condition, which is the key to decoupling longitudinal and lateral motions.

4.2 Small Perturbation Theory and Linearisation

4.2.1 Small Perturbation Theory

As introduced in the preliminary notes of Chapter 1, the general equations of motion are highly non-linear and coupled, making them difficult to solve analytically for stability analysis. To overcome this, we utilize the **Small Perturbation Theory**.

The fundamental principle is to assume that the aircraft is initially in a steady equilibrium state, known as "trim". Any subsequent motion is then modeled as a very small deviation (perturbation) from this reference state. Mathematically, every state variable X is decomposed as follows:

$$X_{total} = X_{trim} + x_{perturbation} \tag{11}$$

By substituting these decomposed variables into the 6-DOF equations and neglecting higher-order terms (products of small perturbations), we can transform the complex mechanics into a set of linear differential equations.

Impact on our Analysis This approach is a cornerstone of our study for two main reasons:

- **Decoupling:** It allows us to separate the longitudinal and lateral-directional motions into two independent systems, simplifying the analysis of specific stability modes like the Phugoid or the Dutch Roll.
- **State-Space Formulation:** It enables the use of the standard matrix form $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}+\mathbf{B}\mathbf{u}$, where the eigenvalues of the stability matrix $\mathbf{A}$ will directly provide the damping and frequency characteristics of our aircraft.

Note: This linearization remains valid only as long as the perturbations stay within the linear aerodynamic range, typically characterized by small angles of attack and sideslip.

4.2.2 Linearisation Procedure

The process of linearization involves applying a first-order Taylor series expansion to the aerodynamic forces and moments around the trim condition. We will detail this procedure for one translational force and one rotational moment.

***Mathematical Tool: Taylor Series Expansion**. In calculus, a Taylor series is an infinite sum of terms expressed in terms of a function's derivatives at a single point. For our flight dynamics model, we use a first-order expansion to linearize the aerodynamic forces and moments around the equilibrium:*

$$f(a + \Delta x) \approx f(a) + f'(a)\Delta x \tag{12}$$

***Physical Utility:** This allows us to break down complex aerodynamic variations into a sum of contributions from each state variable (e.g., how much the force X changes specifically due to a small change in speed u, then in w, etc.). By neglecting higher-order terms (Δx^2), we simplify a non-linear "black box" into a predictable, linear stability matrix $\mathbf{A}$, where each coefficient represents a physical sensitivity of the aircraft.*

Case 1: Longitudinal Force (X) Starting from the general longitudinal force equation:

$$X - mg\sin\theta = m(\dot{u} + qw - rv) \tag{13}$$

By substituting the variables with their trim values and small perturbations ($X = X_0 + \Delta X$, $\theta = \theta_0 + \Delta\theta$, etc.) and applying the steady-state flight assumptions where $w_0 = v_0 = p_0 = q_0 = r_0 = 0$, the equation simplifies to:

$$X_0 + \Delta X - mg\sin(\theta_0 + \Delta\theta) = m\Delta\dot{u} \tag{14}$$

Using small-angle approximations ($\sin\Delta\theta \approx \Delta\theta$ and $\cos\Delta\theta \approx 1$) and subtracting the equilibrium condition ($X_0 - mg\sin\theta_0 = 0$), we obtain:

$$\Delta X - mg\cos\theta_0\Delta\theta = m\Delta\dot{u} \tag{15}$$

The aerodynamic perturbation ΔX is expanded as a function of the state variables (u, w) and control inputs (δ_e) using first order Taylor series expansion:

$$\Delta X = \frac{\partial X}{\partial u}\Delta u + \frac{\partial X}{\partial w}\Delta w + \frac{\partial X}{\partial \delta_e}\Delta\delta_e \tag{16}$$

Dividing by mass and defining the stability derivatives (X_u, X_w, X_{δ_e}), the final linearized equation for the X-force is:

$$X_u\Delta u + X_w\Delta w + X_{\delta_e}\Delta\delta_e - g\cos\theta_0\Delta\theta = \Delta\dot{u} \tag{17}$$

Following the Taylor expansion and small perturbation assumptions, the linearized translational equations can be consolidated into the following system:

$$\begin{cases} \mathbf{X_u\Delta u + X_w\Delta w - g\cos\theta_0\Delta\theta + X_{\delta_e}\Delta\delta_e = \Delta\dot{u}} \\ \mathbf{Y_v\Delta v + Y_p\Delta p + (Y_r - u_0)\Delta r + g\cos\theta_0\Delta\phi + Y_{\delta_r}\Delta\delta_r = \Delta\dot{v}} \\ \mathbf{Z_u\Delta u + Z_w\Delta w + (Z_q + u_0)\Delta q - g\sin\theta_0\Delta\theta + Z_{\delta_e}\Delta\delta_e = \Delta\dot{w}} \end{cases} \tag{18}$$

The selection of state variables for the lateral (Y) and vertical (Z) equations is driven by the specific aerodynamic coupling of each flight plane:

- **Lateral Equation (Y):** The chosen variables are lateral velocity (v), roll rate (p), and yaw rate (r). This reflects the directional nature of the force, where sideforce is primarily generated by sideslip ($\beta \approx v/u_0$) and the interaction with rotational rates around the x and z axes.
- **Vertical Equation (Z):** Similar to the X equation, this longitudinal force depends on the forward velocity (u) and vertical velocity (w), but it crucially incorporates the pitch rate (q). In this case, $u_0\Delta q$ represents the centripetal acceleration term emerging from the linearization of the kinematic equations.

To linearize the rotational dynamics, we apply the same Taylor expansion methodology while enforcing specific physical constraints.

Case 2: Rolling Moment (L)

Assumptions for Linearization The following simplifications are applied to reduce the non-linear Euler equations:

1. **Symmetrical Flight:** The equilibrium state assumes wings-level flight where $v_0 = p_0 = q_0 = r_0 = \phi_0 = \psi_0 = 0$.
2. **Symmetry Axis:** The aircraft is assumed to have a plane of symmetry (xy), implying the product of inertia $I_{xz} \approx 0$.
3. **Angular Motion Coupling:** Products of angular rates (e.g., qr, pq, pr) and their squares (p^2, r^2) are neglected.
4. **First-Order Approximation:** Products of small disturbances ($\Delta \cdot \Delta$) are second-order terms and are considered negligible.

Starting from the general roll moment equation:

$$L = I_{xx}\dot{p} - I_{xz}\dot{r} - I_{xz}pq + qr(I_{zz} - I_{yy}) \tag{19}$$

Applying the assumptions ($I_{xz} = 0$ and neglecting products of rates), the equation for the perturbation ΔL simplifies to:

$$\Delta L = I_{xx}\Delta\dot{p} \tag{20}$$

The variation ΔL is expanded as a function of the lateral state variables and control deflections:

$$\Delta L = \frac{\partial L}{\partial v}\Delta v + \frac{\partial L}{\partial p}\Delta p + \frac{\partial L}{\partial r}\Delta r + \frac{\partial L}{\partial \delta_a}\Delta\delta_a + \frac{\partial L}{\partial \delta_r}\Delta\delta_r \tag{21}$$

Dividing by the moment of inertia I_{xx} and defining the derivatives ($L_v, L_p, etc.$), we obtain the linearized roll rate acceleration.

$$L_v\Delta v + L_p\Delta p + L_r\Delta r + L_{\delta_a}\Delta\delta_a + L_{\delta_r}\Delta\delta_r = \Delta\dot{p} \tag{22}$$

Following this methodology for pitch (M) and yaw (N), the complete linearized rotational system is presented below.

$$\begin{cases} \mathbf{L_v\Delta v + L_p\Delta p + L_r\Delta r + L_{\delta_a}\Delta\delta_a + L_{\delta_r}\Delta\delta_r = \Delta\dot{p}} \\ \mathbf{M_u\Delta u + M_w\Delta w + M_{\dot{w}}\Delta\dot{w} + M_q\Delta q + M_{\delta_e}\Delta\delta_e = \Delta\dot{q}} \\ \mathbf{N_v\Delta v + N_p\Delta p + N_r\Delta r + N_{\delta_a}\Delta\delta_a + N_{\delta_r}\Delta\delta_r = \Delta\dot{r}} \end{cases} \tag{23}$$

State Variable Dependencies The choice of variables in these equations reflects the decoupling of motions:

- **Roll (L) and Yaw (N):** These are coupled through lateral velocity (v), roll rate (p), and yaw rate (r), as lateral forces induce both rolling and yawing moments.
- **Pitch (M):** This remains independent of lateral variables, depending only on longitudinal speed (u), vertical speed (w), and pitch rate (q). The inclusion of $M_{\dot{w}}$ accounts for the "lag-in-downwash" effect from the wing to the tail.

4.3 Matrixial Longitudinal Equations

By isolating the longitudinal equations from the linearized 6-DOF system, we can represent the aircraft's motion as a system of first-order linear differential equations. The longitudinal state vector is defined as $\mathbf{x} = [\Delta u, \Delta w, \Delta q, \Delta\theta]^T$, representing the forward velocity, vertical velocity, pitch rate, and pitch angle perturbations.

4.3.1 State-Space Representation

The longitudinal dynamics are governed by the following set of equations rearranged from our previous linearization:

- $\Delta\dot{u} = X_u\Delta u + X_w\Delta w - g\cos\theta_0\Delta\theta + X_{\delta_e}\Delta\delta_e$
- $\Delta\dot{w} = Z_u\Delta u + Z_w\Delta w + u_0\Delta q - g\sin\theta_0\Delta\theta + Z_{\delta_e}\Delta\delta_e$
- $\Delta\dot{q} = M_u\Delta u + M_w\Delta w + M_{\dot{w}}\Delta\dot{w} + M_q\Delta q + M_{\delta_e}\Delta\delta_e$
- $\Delta\dot{\theta} = \Delta q$

By substituting $\Delta\dot{w}$ into the $\Delta\dot{q}$ equation to eliminate the acceleration dependency, we obtain the standard state-space form $\dot{\mathbf{x}} = \mathbf{A}_{lon}\mathbf{x} + \mathbf{B}_{lon}\mathbf{u}$:

$$\begin{bmatrix} \Delta\dot{u} \\ \Delta\dot{w} \\ \Delta\dot{q} \\ \Delta\dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g\cos\theta_0 \\ Z_u & Z_w & u_0 & -g\sin\theta_0 \\ M_u + M_{\dot{w}}Z_u & M_w + M_{\dot{w}}Z_w & M_q + M_{\dot{w}}u_0 & -M_{\dot{w}}g\sin\theta_0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta\theta \end{bmatrix} + \begin{bmatrix} X_{\delta_e} \\ Z_{\delta_e} \\ M_{\delta_e} + M_{\dot{w}}Z_{\delta_e} \\ 0 \end{bmatrix} [\Delta\delta_e] \tag{24}$$

Physical Significance of the Matrix A The stability matrix $\mathbf{A}_{lon}$ contains the aerodynamic derivatives that dictate the natural response of the aircraft. The coupling terms involving $M_{\dot{w}}$ are particularly important as they represent the transient lift effect on the horizontal stabilizer caused by the wing's wake.

4.4 Solution of Longitudinal Matrixial Equation

To analyze the dynamic stability of the aircraft, we must solve the autonomous system $\dot{\mathbf{x}} = \mathbf{A}_{lon}\mathbf{x}$. The behavior of the aircraft is determined by the eigenvalues (λ) of the stability matrix $\mathbf{A}_{lon}$, obtained by solving the characteristic equation:

$$\det(\mathbf{A}_{lon} - \lambda\mathbf{I}) = 0$$

For the longitudinal motion, this results in a fourth-order polynomial, typically yielding two pairs of complex conjugate eigenvalues. These pairs correspond to the two fundamental longitudinal modes: the **Short Period** and the **Phugoid**. The characteristic polynomial can thus be factored into the product of two specific second-order polynomials:

$$(s^2 + 2\zeta_{sp}\omega_{n_{sp}}s + \omega_{n_{sp}}^2)(s^2 + 2\zeta_{ph}\omega_{n_{ph}}s + \omega_{n_{ph}}^2) = 0 \tag{25}$$

Where each quadratic factor mathematically defines the damping ratio (ζ) and the natural frequency (ω_n) respective to the Short Period and Phugoid stability modes.

4.4.1 Short Period Mode Characteristics

To determine the dynamic characteristics of the aircraft's longitudinal motion, we first consider the characteristic equation of the full 4×4 system:

$$\det(\lambda\mathbf{I} - \mathbf{A}_{lon}) = \det \begin{bmatrix} \lambda - X_u & -X_w & 0 & g\cos\theta_0 \\ -Z_u & \lambda - Z_w & -U_0 & g\sin\theta_0 \\ -M_u - Z_u M_{\dot{w}} & -M_w - Z_w M_{\dot{w}} & \lambda - M_q - U_0 M_{\dot{w}} & 0 \\ 0 & 0 & -1 & \lambda \end{bmatrix} = 0$$

The expansion of this determinant results in a fourth-degree polynomial:

$$\lambda^4 + a\lambda^3 + b\lambda^2 + c\lambda + d = 0$$

This equation yields two pairs of conjugate complex roots (λ_1, λ_2 and λ_3, λ_4), representing two distinct oscillatory motions.

Mathematical Derivation of the Short Period The Short Period (SP) mode is a fast oscillatory response. Because of its rapid nature, the forward speed remains nearly constant ($\Delta u \approx 0$) and the trajectory is primarily affected by the angle of attack and pitch rate. We decouple this mode by reducing the system to a two-degree-of-freedom sub-matrix involving only Δw and Δq:

$$\begin{bmatrix} \Delta\dot{w} \\ \Delta\dot{q} \end{bmatrix} \approx \underbrace{\begin{bmatrix} Z_w & U_0 \\ M_w + M_{\dot{w}}Z_w & M_q + M_{\dot{w}}U_0 \end{bmatrix}}_{\mathbf{A}_{sp}} \begin{bmatrix} \Delta w \\ \Delta q \end{bmatrix} + \underbrace{\begin{bmatrix} Z_{\delta_e} \\ M_{\delta_e} + M_{\dot{w}}Z_{\delta_e} \end{bmatrix}}_{\mathbf{B}_{sp}} \Delta\delta_e \tag{26}$$

The characteristic equation for this specific oscillatory motion is extracted by setting $\det(s\mathbf{I} - \mathbf{A}_{sp}) = 0$, which simplifies to a quadratic form:

$$s^2 - (Z_w + M_q + M_{\dot{w}}U_0)s + (Z_w M_q - M_w U_0) = 0$$

By identifying this with the standard second-order form $s^2 + 2\zeta_{sp}\omega_{n_{sp}}s + \omega_{n_{sp}}^2 = 0$, we establish the natural frequency and damping ratio:

$$\omega_{n_{sp}} = \sqrt{Z_w M_q - U_0 M_w} \quad \text{and} \quad \zeta_{sp} = \frac{-(Z_w + M_q + M_{\dot{w}}U_0)}{2\omega_{n_{sp}}}$$

4.4.2 Phugoid Mode Characteristics

The Phugoid (Ph) mode is the second fundamental longitudinal motion, manifesting as a very slow and lightly damped oscillation. Physically, this mode consists of a continuous exchange between kinetic energy (forward airspeed) and potential energy (altitude). Unlike the Short Period, the angle of attack remains virtually constant during this motion ($\Delta\alpha \approx 0$), implying that the aircraft is in a state of quasi-equilibrium regarding its lift and weight.

Mathematical Derivation of the Phugoid To isolate the Phugoid eigenvalues, we assume that the fast dynamics of the Short Period have already settled. In this low-frequency maneuver, the variations in vertical velocity and pitch rate are considered mathematically negligible ($w \approx 0, \dot{w} \approx 0$ and $q \approx 0$). By substituting these assumptions into the original vertical force equation, we can extract the kinematic relationship:

$$\dot{\theta} = -\frac{Z_u}{U_0}u + \frac{g\sin\Theta_0}{U_0}\theta \tag{27}$$

Applying these physical assumptions reduces the longitudinal state-space system to a 2×2 sub-system involving solely the forward velocity (u) and the pitch angle (θ):

$$\begin{bmatrix} \dot{u} \\ \dot{\theta} \end{bmatrix} \approx \underbrace{\begin{bmatrix} X_u & -g\cos\Theta_0 \\ -\frac{Z_u}{U_0} & \frac{g\sin\Theta_0}{U_0} \end{bmatrix}}_{\mathbf{A}_{Ph}} \begin{bmatrix} u \\ \theta \end{bmatrix} \tag{28}$$

The characteristic equation for this slow oscillatory motion is obtained by calculating $\det(\lambda\mathbf{I} -$

$\mathbf{A}_{Ph}) = 0$, leading to the quadratic form:

$$\lambda^2 - \left(X_u + \frac{g\sin\Theta_0}{U_0}\right)\lambda - \left(\frac{Z_u}{U_0}g\cos\Theta_0 - X_u\frac{g\sin\Theta_0}{U_0}\right) = 0 \tag{29}$$

By identifying the coefficients with the standard oscillatory form $\lambda^2 + 2\zeta_{ph}\omega_{n_{ph}}\lambda + \omega_{n_{ph}}^2 = 0$, we establish the definitive analytical expressions for the Phugoid natural frequency ($\omega_{n_{ph}}$) and damping ratio (ζ_{ph}):

$$\omega_{n_{ph}} = \sqrt{-\left(\frac{Z_u}{U_0}g\cos\Theta_0 - X_u\frac{g\sin\Theta_0}{U_0}\right)} \quad \text{and} \quad \zeta_{ph} = \frac{-(X_u + \frac{g\sin\Theta_0}{U_0})}{2\omega_{n_{ph}}} \tag{30}$$

Dynamic Parameters The eigenvalues λ_{ph} are typically located much closer to the imaginary axis, indicating low frequency and low damping. For a stable aircraft, they are expressed as:

$$\lambda_{ph} = -\zeta_{ph}\omega_{n_{ph}} \pm j\omega_{n_{ph}}\sqrt{1-\zeta_{ph}^2} \tag{31}$$

- **Natural Frequency ($\omega_{n_{ph}}$):** This is primarily a function of gravity and speed. It represents the slow "trading" of altitude for airspeed. For cruise conditions ($\Theta_0 = 0$), it simplifies to $\omega_{n_{ph}} \approx \sqrt{-gZ_u/U_0}$.
- **Damping Ratio (ζ_{ph}):** The Phugoid is generally poorly damped because the energy loss through drag is low compared to the kinetic energy. Level 1 flying qualities require $\zeta_{ph} \geq 0.04$.

Note: Because of its long period (typically > 30 seconds), the Phugoid does not significantly increase pilot workload, as the pilot has ample time to adjust the trim and maintain level flight.

4.5 Matrixial Lateral Equations

The lateral-directional motion describes the aircraft's response in terms of sideslip, roll, and yaw. Unlike the longitudinal motion, these dynamics are coupled through the aerodynamic effects of lateral velocity and angular rates about the longitudinal and vertical axes. The lateral state vector is defined as $\mathbf{x} = [\Delta v, \Delta p, \Delta r, \Delta\phi]^T$, representing lateral velocity, roll rate, yaw rate, and bank angle.

4.5.1 Lateral State-Space Representation

The linearized lateral-directional dynamics are governed by the following set of equations derived from our previous Taylor expansion:

- $\Delta\dot{v} = Y_v\Delta v + Y_p\Delta p + (Y_r - u_0)\Delta r + g\cos\theta_0\Delta\phi + Y_{\delta_r}\Delta\delta_r$
- $\Delta\dot{p} = L_v\Delta v + L_p\Delta p + L_r\Delta r + L_{\delta_a}\Delta\delta_a + L_{\delta_r}\Delta\delta_r$
- $\Delta\dot{r} = N_v\Delta v + N_p\Delta p + N_r\Delta r + N_{\delta_a}\Delta\delta_a + N_{\delta_r}\Delta\delta_r$
- $\Delta\dot{\phi} = \Delta p$

Note on State Selection It is important to note that while the yaw rate (r) is included, the yaw angle (ψ) is excluded from the state vector. This is because the aerodynamic forces and moments are independent of the aircraft's heading, whereas the bank angle (ϕ) must be maintained as a state due to its direct contribution to the gravity vector component in the lateral force equation ($g\cos\theta_0\Delta\phi$).

Arranging these relationships into the standard state-space form $\dot{\mathbf{x}} = \mathbf{A}_{lat}\mathbf{x} + \mathbf{B}_{lat}\mathbf{u}$, we obtain the lateral-directional stability and control matrices:

$$\begin{bmatrix}\Delta\dot{v}\\ \Delta\dot{p}\\ \Delta\dot{r}\\ \Delta\dot{\phi}\end{bmatrix} = \begin{bmatrix} Y_v & Y_p & (Y_r - u_0) & g\cos\theta_0\\ L_v & L_p & L_r & 0\\ N_v & N_p & N_r & 0\\ 0 & 1 & 0 & 0\end{bmatrix}\begin{bmatrix}\Delta v\\ \Delta p\\ \Delta r\\ \Delta\phi\end{bmatrix} + \begin{bmatrix} 0 & Y_{\delta_r}\\ L_{\delta_a} & L_{\delta_r}\\ N_{\delta_a} & N_{\delta_r}\\ 0 & 0\end{bmatrix}\begin{bmatrix}\Delta\delta_a\\ \Delta\delta_r\end{bmatrix} \tag{32}$$

4.6 Solution of Lateral Matrixial Equation

The lateral stability of the C3 aircraft is governed by the eigenvalues of the $\mathbf{A}_{lat}$ matrix. Solving the characteristic equation $\det(\lambda\mathbf{I} - \mathbf{A}_{lat}) = 0$ typically yields one pair of complex conjugate roots and two real roots. These correspond to the three fundamental lateral modes: the **Spiral mode**, the **Roll mode**, and the **Dutch Roll**. The lateral characteristic polynomial can thus be factored into the following form:

$$(s-\lambda_{spiral})(s-\lambda_{roll})(s^2+2\zeta_{dr}\omega_{n_{dr}}s+\omega_{n_{dr}}^2) = 0 \quad (33)$$

Where λ_{spiral} and λ_{roll} are the real eigenvalues characterizing the non-oscillatory modes, while the quadratic factor defines the damping ratio (ζ_{dr}) and natural frequency ($\omega_{n_{dr}}$) specific to the Dutch Roll coupled oscillation.

4.6.1 Spiral Mode Characteristics

The Spiral mode is a non-oscillatory lateral-directional motion characterized by a single real eigenvalue, λ_{spiral}. Physically, it represents a slow departure from or return to a wings-level attitude following a disturbance in bank angle or sideslip.

Mathematical Derivation The Spiral mode corresponds to an extremely slow motion involving a complex interaction between directional stability (weathercock effect) and lateral stability (dihedral effect). Because this motion unfolds so slowly, the roll rate and roll acceleration are assumed to be virtually zero ($p \approx 0$ and $\dot{p} \approx 0$).

Applying these physical assumptions to the initial unforced lateral equations allows for a reduction. By setting $\dot{p} = 0$ in the rolling moment equation, we can express the lateral velocity (v) as a function of the yaw rate (r):

$$0 = L_v v + L_r r \implies v = -\frac{L_r}{L_v} r \quad (34)$$

Substituting this relationship into the simplified yawing differential equation ($\dot{r} = N_v v + N_r r$) reduces the dynamics to a 1×1 sub-system involving solely the yaw rate:

$$\dot{r} = \left(N_r - N_v \frac{L_r}{L_v}\right) r \quad (35)$$

By calculating the characteristic equation for this reduced system, we directly extract the spiral eigenvalue:

$$\lambda_{spiral} = N_r - N_v \frac{L_r}{L_v} \quad (36)$$

Alternatively, λ_{spiral} can be approximated using the coefficients of the full fourth-order polynomial ($\lambda^4 + a\lambda^3 + b\lambda^2 + c\lambda + d = 0$). As the smallest real root, it is primarily determined by the constant term (d) and the linear term (c):

$$\lambda_{spiral} \approx -\frac{d}{c} = \frac{g(L_v N_r - L_r N_v)}{U_0(L_p N_v - L_v N_p)} \quad (37)$$

Time Constants and Stability The nature of the eigenvalue λ_{spiral} dictates the temporal behavior of the mode:

- **Stable Mode ($\lambda_{spiral} < 0$):** The aircraft naturally returns to wings-level flight. The convergence speed is defined by the time constant $T_s = 1/|\lambda_{spiral}|$.

- **Unstable Mode ($\lambda_{spiral} > 0$):** The bank angle increases exponentially. The severity of this instability is measured by the time to double the bank angle (T_2):

$$T_2 = \frac{\ln(2)}{\lambda_{spiral}} \approx \frac{0.693}{\lambda_{spiral}} \quad (38)$$

Physical Interpretation The stability of this mode is governed by the balance between the **Dihedral Effect** (L_v) and **Weathercock Stability** (N_v). If the aircraft tends to restore its wings-level position, the mode is convergent. If the bank angle slowly increases, leading to a "spiral dive," the mode is divergent.

Flying Qualities Requirements According to the **MIL-F-8785C** standard, mild instability is often permitted because the mode is slow enough for effortless pilot correction. For Category B flight (cruise), the time to double the bank angle (T_2) must be greater than 20 seconds to achieve Level 1 flying qualities.

Note: A slow spiral divergence is often preferred over a very stable spiral, as the latter can make the aircraft feel "too stiff" and resistant to intentional turns.

4.6.2 Roll Mode Characteristics

The Rolling (R) mode, also known as roll subsidence, is a rapid and highly damped non-oscillatory motion. It represents the aircraft's natural resistance to a continuous rolling motion due to aerodynamic damping. It is characterized by a single large negative real eigenvalue, λ_{roll}, ensuring that any roll rate induced by a disturbance or command eventually stabilizes to a steady value or returns to zero.

Mathematical Derivation Since the Roll mode is significantly faster than the Spiral and Dutch Roll modes, it can be isolated by assuming that the motion is a pure rotation about the longitudinal axis (x_b). Because this motion is so rapid, no significant sideslip or yaw has the time to develop, allowing us to assume that lateral velocity and yaw rate are negligible ($v \approx 0$ and $r \approx 0$).

Applying these physical assumptions to the lateral state-space system reduces the dynamics to a 1×1 sub-system involving solely the roll rate (p):

$$\dot{p} = L_p p + L_{\delta_a} \delta_a \tag{39}$$

By calculating the determinant of the reduced stability matrix ($\det[s\mathbf{I} - (L_p)] = 0$), we directly extract the first-order characteristic equation governing the rolling mode:

$$s - L_p = 0 \implies \lambda_{roll} = L_p \tag{40}$$

This approximation shows that the roll damping derivative (L_p) is the primary physical driver of this mode. Since L_p is inherently a large negative value for conventional wings, this mode is fundamentally stable and decays very rapidly.

Time Constant and Agility The behavior of the Roll mode is expressed through its time constant, τ_R, which represents the time required for the roll rate to reach approximately 63% of its steady-state value following a step command:

$$\tau_R = \frac{1}{|\lambda_{roll}|} \approx \frac{1}{|L_p|} \tag{41}$$

A smaller time constant τ_R is synonymous with high "roll agility," allowing the pilot to initiate or stop a bank angle change almost instantaneously.

Flying Qualities Requirements The **MIL-F-8785C** standard imposes maximum allowable values for τ_R to ensure the aircraft remains responsive to pilot inputs. For the C3 aircraft operating in Category B flight (cruise), the requirements for Level 1 flying qualities are:

- **Level 1:** $\tau_R \leq 1.4$ seconds.
- **Level 2:** $\tau_R \leq 3.0$ seconds.
- **Level 3:** $\tau_R \leq 10.0$ seconds.

A high absolute value of L_p results in a heavily damped, "sluggish" response if too high, whereas a low value results in a "lively" aircraft that may be difficult for the pilot to stabilize at a specific bank angle.

4.6.3 Dutch Roll Mode Characteristics

The Dutch Roll (DR) is the most complex lateral-directional motion, manifesting as a relatively high-frequency coupled oscillation combining yawing, rolling, and sideslipping. It is characterized by a pair of conjugate complex eigenvalues, $\lambda_{dr} = -\zeta_{dr}\omega_{n_{dr}} \pm j\omega_{n_{dr}}\sqrt{1-\zeta_{dr}^2}$.

Figure 9: Front view of a plane undergoing Dutch Roll.

Mathematical Derivation Due to the rapid nature of this oscillation, primarily driven by the restoring force of the vertical tail, the sluggish roll angle variations (ϕ) and the pure roll rate (p) are often considered secondary compared to the strong yaw-sideslip coupling. Applying these physical assumptions, the lateral state-space system reduces to a 2×2 sub-system involving predominantly the lateral velocity (v) and the yaw rate (r):

$$\begin{bmatrix} \dot{v} \\ \dot{r} \end{bmatrix} \approx \begin{bmatrix} Y_v & Y_r - u_0 \\ N_v & N_r \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} \tag{42}$$

By calculating the determinant of this specific sub-matrix ($\det(\lambda\mathbf{I} - \mathbf{A}_{DR}) = 0$), we extract the second-order characteristic equation governing the Dutch roll mode:

$$\lambda^2 - (Y_v + N_r)\lambda + (Y_v N_r - N_v(Y_r - u_0)) = 0 \tag{43}$$

By identifying the coefficients of this polynomial with the standard oscillatory form ($\lambda^2 + 2\zeta_{dr}\omega_{n_{dr}}\lambda + \omega_{n_{dr}}^2 = 0$), we define the fundamental frequency and damping of this mode:

$$\omega_{n_{dr}}^2 = Y_v N_r - N_v(Y_r - u_0) \quad \text{and} \quad 2\zeta_{dr}\omega_{n_{dr}} = -(Y_v + N_r) \tag{44}$$

Solving these identities yields the final analytical expressions:

$$\omega_{n_{dr}} = \sqrt{Y_v N_r - N_v(Y_r - u_0)} \quad \Rightarrow \quad \zeta_{dr} = \frac{-(Y_v + N_r)}{2\omega_{n_{dr}}} \tag{45}$$

Physical Mechanism: The Stability Balance The Dutch Roll is the result of a "competition" between two primary aerodynamic effects:

- **Weathercock Stability (N_β):** Acts as a restoring spring that tries to pull the nose back into the relative wind.
- **Dihedral Effect (L_β):** Induces a rolling moment when the aircraft sideslips, causing the wings to rock.
- **Yaw Damping (N_r):** Primarily provided by the vertical stabilizer, it is the main source of damping for this mode.

Flying Qualities Requirements Insufficient damping in the Dutch Roll leads to a tiring "snaking" motion. For the C3 aircraft in cruise (Category B), **MIL-F-8785C** Level 1 requirements mandate:

- A minimum damping ratio: $\zeta_{dr} \geq 0.08$.
- A minimum natural frequency: $\omega_{n_{dr}} \geq 0.5$ rad/s.
- A minimum product: $\zeta_{dr}\omega_{n_{dr}} \geq 0.15$.

Note: If these criteria are not met naturally, a "Yaw Damper" (Stability Augmentation System) is typically required to increase the effective N_r of the aircraft.

Lateral-Directional Transfer Functions To understand how the aircraft responds to deliberate pilot commands, we convert the forced lateral state-space system $\dot{\mathbf{x}}_{lat} = \mathbf{A}_{lat}\mathbf{x}_{lat} + \mathbf{B}_{lat}\mathbf{u}_{lat}$ into the Laplace domain. Assuming zero initial conditions, the relationship is defined by:

$$\begin{aligned}(s\mathbf{I} - \mathbf{A}_{lat})\mathbf{X}_{lat}(s) &= \mathbf{B}_{lat}\mathbf{U}_{lat}(s) \\ \implies \frac{\mathbf{X}_{lat}(s)}{\mathbf{U}_{lat}(s)} &= (s\mathbf{I} - \mathbf{A}_{lat})^{-1}\mathbf{B}_{lat}\end{aligned} \tag{46}$$

The denominator of these transfer functions is the fourth-order lateral characteristic polynomial, which factors into the three fundamental modes (Spiral, Roll, and Dutch Roll):

$$\Delta_{lat}(s) = (s-\lambda_{spiral})(s-\lambda_{roll})(s^2+2\zeta_{dr}\omega_{n_{dr}}s+\omega_{n_{dr}}^2) \tag{47}$$

By extracting the specific columns of the control matrix $\mathbf{B}_{lat}$ corresponding to aileron (δ_a) and rudder (δ_r) deflections, we obtain the individual transfer functions. For instance, the response to a rudder command is formulated as:

$$\begin{bmatrix}\Delta v(s)\\ \Delta p(s)\\ \Delta r(s)\\ \Delta\phi(s)\end{bmatrix} = \frac{\mathrm{adj}(s\mathbf{I}-\mathbf{A}_{lat})}{\Delta_{lat}(s)}\begin{bmatrix}Y_{\delta_r}\\ L_{\delta_r}\\ N_{\delta_r}\\ 0\end{bmatrix}\Delta\delta_r(s) \tag{48}$$

The resulting fractions link each state variable to the control inputs through a specific set of zeros (numerator roots) and the common system poles (denominator roots). This mathematical architecture allows for the detailed frequency-domain analysis and the subsequent design of a Stability Augmentation System (SAS), such as a Yaw Damper, to improve modal damping if required.

4.7 Transfer Functions

To transition from theoretical stability to practical control analysis, we must investigate how the aircraft responds to pilot inputs. This is achieved by converting the forced state-space equations into the Laplace domain. Assuming zero initial conditions, the general relationship is defined by:

$$\begin{aligned}(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) &= \mathbf{B}\mathbf{U}(s) \\ \implies \mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s)\end{aligned} \tag{49}$$

The term $(s\mathbf{I} - \mathbf{A})^{-1}$ can be expressed as $\frac{\mathrm{adj}(s\mathbf{I}-\mathbf{A})}{\det(s\mathbf{I}-\mathbf{A})}$. This identity provides a rigorous justification for the structure of the resulting transfer functions:

- **The Denominator:** For every transfer function of the system, the denominator is the characteristic polynomial $\det(s\mathbf{I} - \mathbf{A})$. Consequently, the poles of the control response are identical to the eigenvalues of the stability matrix, meaning the aircraft's response is governed by its natural modes (Phugoid, Short Period, etc.).
- **The Numerators:** These are obtained by the product of the adjugate matrix and the control matrix $\mathbf{B}$. The expansion of this product yields specific polynomials whose roots, known as "zeros" ($1/T$), determine the sensitivity and the phase of the response for a given state variable.

The following sections summarize the 16 fundamental transfer functions derived from this matrix expansion, grouped by their respective control inputs.

4.7.1 Longitudinal Transfer Functions

The longitudinal response is driven by the elevator (δ_e) and the throttle (δ_T). The common denominator $\Delta_{lon}(s)$ is the fourth-order polynomial factored into the Short Period and Phugoid modes:

$$\Delta_{lon}(s) = (s^2+2\zeta_{ph}\omega_{n_{ph}}s+\omega_{n_{ph}}^2)(s^2+2\zeta_{sp}\omega_{n_{sp}}s+\omega_{n_{sp}}^2) \tag{50}$$

Response to Elevator Command ($\Delta\delta_e$)

$$\begin{aligned}
\frac{\Delta u(s)}{\Delta\delta_e(s)} &= \frac{K_u\left(s+\frac{1}{T_{u1}}\right)(s^2+2\zeta_u\omega_u s+\omega_u^2)}{\Delta_{lon}(s)} \\
\frac{\Delta w(s)}{\Delta\delta_e(s)} &= \frac{K_w\left(s+\frac{1}{T_{w1}}\right)(s^2+2\zeta_w\omega_w s+\omega_w^2)}{\Delta_{lon}(s)} \\
\frac{\Delta q(s)}{\Delta\delta_e(s)} &= \frac{K_q s\left(s+\frac{1}{T_{\theta1}}\right)\left(s+\frac{1}{T_{\theta2}}\right)}{\Delta_{lon}(s)} \\
\frac{\Delta\theta(s)}{\Delta\delta_e(s)} &= \frac{K_\theta\left(s+\frac{1}{T_{\theta1}}\right)\left(s+\frac{1}{T_{\theta2}}\right)}{\Delta_{lon}(s)}
\end{aligned} \tag{51}$$

Response to Throttle Variation ($\Delta\delta_T$)

$$\begin{aligned}
\frac{\Delta u(s)}{\Delta\delta_T(s)} &= \frac{K_{uT}\left(s+\frac{1}{T_{u1T}}\right)(s^2+2\zeta_{uT}\omega_{uT}s+\omega_{uT}^2)}{\Delta_{lon}(s)} \\
\frac{\Delta w(s)}{\Delta\delta_T(s)} &= \frac{K_{wT}\left(s+\frac{1}{T_{w1T}}\right)(s^2+2\zeta_{wT}\omega_{wT}s+\omega_{wT}^2)}{\Delta_{lon}(s)} \\
\frac{\Delta q(s)}{\Delta\delta_T(s)} &= \frac{K_{qT}s\left(s+\frac{1}{T_{\theta1T}}\right)\left(s+\frac{1}{T_{\theta2T}}\right)}{\Delta_{lon}(s)} \\
\frac{\Delta\theta(s)}{\Delta\delta_T(s)} &= \frac{K_{\theta T}\left(s+\frac{1}{T_{\theta1T}}\right)\left(s+\frac{1}{T_{\theta2T}}\right)}{\Delta_{lon}(s)}
\end{aligned} \tag{52}$$

4.7.2 Lateral-Directional Transfer Functions

The lateral-directional response is driven by the ailerons (δ_a) and the rudder (δ_r). The common denominator $\Delta_{lat}(s)$ is composed of the Spiral, Roll, and Dutch Roll modes:

$$\Delta_{lat}(s) = (s-\lambda_{spiral})(s-\lambda_{roll})(s^2+2\zeta_{dr}\omega_{n_{dr}}s+\omega_{n_{dr}}^2) \tag{53}$$

Response to Aileron Command ($\Delta\delta_a$)

$$\begin{aligned}
\frac{\Delta v(s)}{\Delta\delta_a(s)} &= \frac{K_v\left(s+\frac{1}{T_{\beta1}}\right)\left(s+\frac{1}{T_{\beta2}}\right)}{\Delta_{lat}(s)} \\
\frac{\Delta p(s)}{\Delta\delta_a(s)} &= \frac{K_p s(s^2+2\zeta_\phi\omega_\phi s+\omega_\phi^2)}{\Delta_{lat}(s)} \\
\frac{\Delta r(s)}{\Delta\delta_a(s)} &= \frac{K_r\left(s+\frac{1}{T_{\psi1}}\right)(s^2+2\zeta_\psi\omega_\psi s+\omega_\psi^2)}{\Delta_{lat}(s)} \\
\frac{\Delta\phi(s)}{\Delta\delta_a(s)} &= \frac{K_\phi(s^2+2\zeta_\phi\omega_\phi s+\omega_\phi^2)}{\Delta_{lat}(s)}
\end{aligned} \tag{54}$$

Response to Rudder Command ($\Delta\delta_r$)

$$\begin{aligned}
\frac{\Delta v(s)}{\Delta\delta_r(s)} &= \frac{K_{vr}\left(s+\frac{1}{T_{\beta1r}}\right)\left(s+\frac{1}{T_{\beta2r}}\right)}{\Delta_{lat}(s)} \\
\frac{\Delta p(s)}{\Delta\delta_r(s)} &= \frac{K_{pr}s(s^2+2\zeta_{\phi r}\omega_{\phi r}s+\omega_{\phi r}^2)}{\Delta_{lat}(s)} \\
\frac{\Delta r(s)}{\Delta\delta_r(s)} &= \frac{K_{rr}\left(s+\frac{1}{T_{\psi r}}\right)(s^2+2\zeta_{\psi r}\omega_{\psi r}s+\omega_{\psi r}^2)}{\Delta_{lat}(s)} \\
\frac{\Delta\phi(s)}{\Delta\delta_r(s)} &= \frac{K_{\phi r}(s^2+2\zeta_{\phi r}\omega_{\phi r}s+\omega_{\phi r}^2)}{\Delta_{lat}(s)}
\end{aligned} \tag{55}$$

5 Chapter 3: Numerical Application and Analysis

This chapter focuses on the numerical resolution of the flight dynamics model using the specific data for the C3 aircraft. By implementing the state-space matrices in MATLAB, we extract the eigenvalues and eigenvectors to characterize the longitudinal and lateral stability modes. Furthermore, the aircraft's response is evaluated through control transfer functions, Root Locus, and Bode diagrams to determine compliance with MIL-F-8785C standards. Finally, we explore the design of Stability Augmentation Systems (SAS) to correct any identified modal deficiencies.

Note to the Reader: *The numerical analysis presented in this chapter relies on extensive mathematical derivations, detailed stability derivative calculations, and MATLAB codes. To maintain structural clarity and focus on the results, the comprehensive step-by-step demonstrations and the complete source code are provided in the* ***Appendices*** *at the end of this document.*

Furthermore, all subsections are fully indexed in the ***Table of Contents on page 2****. We invite the reader to refer back to it for quick access to specific data points or analytical elements.*

5.1 Matrix Representation of the C3 Aircraft

To transition from the theoretical state-space models established in Chapter 2 to a numerical simulation, the geometric, mass, and aerodynamic parameters of the C3 aircraft (detailed in the Aircraft Data table) must be applied to the system.

As indicated in the introductory note, the step-by-step conversion from non-dimensional aerodynamic coefficients to dimensional stability derivatives is fully demonstrated in the Appendices. By injecting the specific flight conditions for high-altitude cruise ($M = 0.6$, $h = 10\,668$ m, $U_0 = 178$ m/s) and using the aircraft data ($\bar{q} = 6\,036$ Pa, $S = 12.63\ m^2$, $m = 1\,816$ kg), these formulas allow us to evaluate the numerical values for the stability and control matrices.

5.1.1 Longitudinal Matrices

Below is a summary of the calculations for the primary longitudinal coefficients, following the derivations provided in the Appendix:

- $X_u \approx -0.0118\ s^{-1}$
- $Z_w \approx -1.309\ s^{-1}$
- $M_w \approx -0.026\ m^{-1}s^{-1}$
- $M_q \approx -2.445\ s^{-1}$

The longitudinal state vector is defined as $\mathbf{x}_{lon} = [\Delta u, \Delta w, \Delta q, \Delta\theta]^T$, and the control vector is $\mathbf{u}_{lon} = [\Delta\delta_e, \Delta\delta_T]^T$. Following the analytical resolution demonstrated in the Appendix, the MATLAB script generates the following numerical matrices for the C3 aircraft:

$$\mathbf{A}_{lon} = \begin{bmatrix} -0.0118 & 0.0145 & 0 & -9.81 \\ -0.1396 & -1.3089 & 178 & 0 \\ 0.0007 & -0.0197 & -2.4445 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{B}_{lon} = \begin{bmatrix} 0 & 0 \\ 0.0083 & 0 \\ -15.8418 & 0 \\ 0 & 0 \end{bmatrix}$$

Note: The throttle control derivatives (δ_T) are assumed to be zero for this specific stability simulation, focusing primarily on the elevator response (δ_e). The third row of $\mathbf{B}_{lon}$ incorporates the downwash lag effect coupling ($M_{\delta_e} + M_{\dot{w}} Z_{\delta_e}$).

5.1.2 Lateral-Directional Matrices

The lateral stability derivatives are calculated using the wingspan ($b = 8.02$ m) and the respective moments of inertia (I_{xx}, I_{zz}). Below is the numerical application for the primary coefficients:

- $Y_v \approx -0.2358\ s^{-1}$
- $L_p \approx -4.9509\ s^{-1}$
- $N_v \approx 0.0828\ m^{-1}s^{-1}$
- $N_r \approx -0.5078\ s^{-1}$

The lateral-directional state vector is defined as $\mathbf{x}_{lat} = [\Delta v, \Delta p, \Delta r, \Delta\phi]^T$, and the corresponding control vector is $\mathbf{u}_{lat} = [\Delta\delta_a, \Delta\delta_r]^T$. Substituting the computed lateral dimensional derivatives (detailed in the Appendix) into our state-space framework yields the following numerical matrices:

$$\mathbf{A}_{lat} = \begin{bmatrix} -0.2358 & -0.1135 & -177.4137 & 9.81 \\ -0.3482 & -4.9509 & 3.9353 & 0 \\ 0.0828 & 0.1562 & -0.5078 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{B}_{lat} = \begin{bmatrix} 0.066 & 0 \\ 28.1752 & 56.3504 \\ -10.4024 & -0.4334 \\ 0 & 0 \end{bmatrix}$$

Note: The term $(Y_r - U_0)$ in the first row of $\mathbf{A}_{lat}$ is dominated by the velocity $U_0 = 178$ m/s, explaining the value of -177.41. The matrices are calculated assuming a symmetric aircraft where the products of inertia I_{xy} and I_{yz} are negligible.

5.2 Characteristic Equations and Polynomials

The dynamic behavior of the C3 aircraft is fundamentally governed by the roots of its characteristic equations. These polynomials are derived by calculating the determinant of the system matrix subtracted from the Laplace variable s times the identity matrix: $\Delta(s) = \det(s\mathbf{I} - \mathbf{A})$.

5.2.1 Longitudinal Characteristic Polynomial

Starting from the numerical stability matrix $\mathbf{A}_{lon}$ established in the previous section, the characteristic equation for the longitudinal motion is a fourth-order polynomial of the form:

$$\Delta_{lon}(s) = s^4 + As^3 + Bs^2 + Cs + D = 0$$

By extracting the determinant using MATLAB, the coefficients are evaluated as follows:

$$s^4 + 3.7652s^3 + 6.7525s^2 + 0.0891s + 0.0360 = 0$$

As predicted by the longitudinal theory, this fourth-order system can be factored into two distinct second-order oscillatory components:

$$\underbrace{(s^2 + 2\zeta_{sp}\omega_{n_{sp}}s + \omega_{n_{sp}}^2)}_{\text{Short Period}} \cdot \underbrace{(s^2 + 2\zeta_{ph}\omega_{n_{ph}}s + \omega_{n_{ph}}^2)}_{\text{Phugoid}} = 0$$

The high-frequency coefficients (dominating s^4 and s^3) are primarily driven by the Short Period dynamics, while the lower-order coefficients (s^1 and s^0) dictate the slow Phugoid motion.

5.2.2 Lateral-Directional Characteristic Polynomial

Similarly, the lateral-directional characteristic equation is obtained from the $\mathbf{A}_{lat}$ matrix. Its general form is:

$$\Delta_{lat}(s) = s^4 + as^3 + bs^2 + cs + d = 0$$

For the C3 aircraft, the numerical extraction yields:

$$s^4 + 5.6945s^3 + 17.8369s^2 + 66.9593s - 1.4620 = 0$$

This polynomial reveals the unique structure of lateral-directional stability, where the fourth-order system is typically factored into one second-order factor and two first-order factors:

$$\underbrace{(s^2 + 2\zeta_{dr}\omega_{n_{dr}}s + \omega_{n_{dr}}^2)}_{\text{Dutch Roll}} \cdot \underbrace{(s + 1/\tau_R)}_{\text{Roll Mode}} \cdot \underbrace{(s - \lambda_{spiral})}_{\text{Spiral Mode}} = 0$$

The presence of a negative constant term ($d = -1.4620$) directly implies the existence of at least one positive real root. This is a classic indicator of an unstable Spiral mode, a dynamic characteristic that we will evaluate deeply in the eigenvalues analysis.

5.3 Eigenvalues and Eigenvectors Analysis

To fully characterize the dynamic stability of the C3 aircraft, we must extract the eigenvalues and eigenvectors from the state matrices $\mathbf{A}_{lon}$ and $\mathbf{A}_{lat}$. Mathematically, an eigenvalue λ and its corresponding eigenvector $\mathbf{v}$ satisfy the fundamental relation:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Physically, these two mathematical entities provide distinct and complementary information about the aircraft's behavior:

- **Eigenvalues (λ):** They define the "nature" of the motion over time. The real part ($\mathrm{Re}(\lambda)$) dictates the stability and damping (a negative value means the mode is stable and converges, while a positive value indicates divergence). The imaginary part ($\mathrm{Im}(\lambda)$) dictates the frequency of oscillation.

- **Eigenvectors (v):** They define the "mode shape" or the distribution of the motion. By analyzing the magnitude of each component within an eigenvector, we can determine which state variables (e.g., speed, pitch rate, bank angle) are predominantly active in that specific mode.

Using numerical extraction, the 8 fundamental eigenvalues and their corresponding normalized eigenvectors have been computed for the C3 aircraft.

5.3.1 Longitudinal Modes Analysis

The longitudinal state vector is defined as $\mathbf{x}_{lon} = [\Delta u, \Delta w, \Delta q, \Delta\theta]^T$.

Short Period Mode This mode is characterized by a pair of complex conjugate eigenvalues located far in the left half-plane, indicating a fast and heavily damped oscillation.

$$\lambda_{sp} = -1.8775 \pm 1.7843i$$

The corresponding complex eigenvector (for the positive imaginary root) is:

$$\mathbf{v}_{sp} = \begin{bmatrix} 0.0109 + 0.0001i \\ 0.9999 + 0.0000i \\ -0.0032 + 0.0100i \\ 0.0036 - 0.0020i \end{bmatrix}$$

Physical Interpretation: The magnitude of the second component (Δw) is overwhelmingly dominant (approaching 1.0), while the forward speed perturbation (Δu) is negligible. This confirms the theoretical assumption that the Short Period is primarily a rapid plunging and pitching motion at a constant forward speed.

Phugoid Mode The Phugoid is characterized by a pair of complex conjugate eigenvalues located very close to the imaginary axis, indicating a slow, lightly damped oscillation.

$$\lambda_{ph} = -0.0051 \pm 0.0730i$$

The corresponding complex eigenvector is:

$$\mathbf{v}_{ph} = \begin{bmatrix} -0.9994 + 0.0000i \\ 0.0323 + 0.0002i \\ -0.0005 + 0.0000i \\ 0.0007 + 0.0074i \end{bmatrix}$$

Physical Interpretation: The first component (Δu) is nearly 1.0, while the pitch rate (Δq) is virtually zero. This perfectly aligns with the Phugoid's physical definition: a slow exchange between kinetic energy (speed) and potential energy (altitude) with almost no rapid pitching motion.

5.3.2 Lateral-Directional Modes Analysis

The lateral-directional state vector is defined as $\mathbf{x}_{lat} = [\Delta v, \Delta p, \Delta r, \Delta\phi]^T$.

Roll Mode The Roll mode is characterized by a large, negative real eigenvalue, reflecting a heavily damped and rapid exponential convergence.

$$\lambda_{roll} = -4.8685$$

The corresponding real eigenvector is:

$$\mathbf{v}_{roll} = \begin{bmatrix} -0.4594 \\ 0.8698 \\ -0.0224 \\ -0.1787 \end{bmatrix}$$

Physical Interpretation: The second component (Δp, roll rate) is the most significant, confirming that this mode primarily governs the aircraft's resistance to a pure rolling motion.

Spiral Mode The Spiral mode features a very small real eigenvalue. In the case of the C3 aircraft, this value is positive, indicating a slow instability (divergence).

$$\lambda_{spiral} = 0.0217$$

The corresponding real eigenvector is:

$$\mathbf{v}_{spiral} = \begin{bmatrix} 0.2954 \\ 0.0207 \\ 0.0523 \\ 0.9537 \end{bmatrix}$$

Physical Interpretation: The dominant component is the bank angle ($\Delta\phi$). This illustrates the insidious nature of the Spiral mode: a slight disturbance leads to a slow, unnoticeable buildup of the bank angle, eventually resulting in a spiral dive if left uncorrected.

Dutch Roll Mode The Dutch Roll is identified by a pair of complex conjugate eigenvalues, representing a coupled lateral-directional oscillation.

$$\lambda_{dr} = -0.4239 \pm 3.6951i$$

The corresponding complex eigenvector (for the positive imaginary root) is:

$$\mathbf{v}_{dr} = \begin{bmatrix} -0.9979 + 0.0000i \\ 0.0538 - 0.0265i \\ -0.0016 + 0.0201i \\ -0.0087 - 0.0136i \end{bmatrix}$$

Physical Interpretation: The lateral velocity component (Δv, directly proportional to the sideslip angle β) is heavily dominant. This mathematically translates the "tail-wagging" or "snaking" physical sensation typical of the Dutch Roll, where sideslip variations drive the entire coupled motion.

5.4 Stability Modes Characteristics

Based on the numerical eigenvalues extracted in the previous section, we can now define the precise dynamic characteristics of each longitudinal and lateral stability mode for the C3 aircraft. For oscillatory modes (Short Period, Phugoid, and Dutch Roll), the behavior is defined by the natural frequency (ω_n) and the damping ratio (ζ). For non-oscillatory (aperiodic) modes (Roll and Spiral), the behavior is characterized by the time constant (τ) or the time to double the amplitude (T_2).

5.4.1 Longitudinal Modes Numerical Results

The longitudinal dynamics are decoupled into the rapid Short Period oscillation and the slow Phugoid energy exchange. The following table summarizes their characteristics:

Mode	Eigenvalue (λ)	ω_n [rad/s]	Damping (ζ)	Settling Time [s]
Short Period	$-1.8775 \pm 1.7843i$	2.5901	0.7249	1.60
Phugoid	$-0.0051 \pm 0.0730i$	0.0732	0.0702	583.26

Table 9: Numerical characteristics of the C3 longitudinal stability modes.

- **Short Period:** The high damping ratio ($\zeta = 0.72$) indicates a very stable and well-behaved response. Pitch disturbances are suppressed in less than 2 seconds, which typically provides excellent tracking for the pilot.
- **Phugoid:** The very low frequency and damping reflect the typical slow exchange between altitude and speed. With a settling time (3τ) of nearly 10 minutes, this mode is stable but requires long-term pilot monitoring.

5.4.2 Lateral-Directional Modes Numerical Results

The lateral-directional characteristics reveal a highly damped rapid rolling motion, a coupled "Dutch Roll" oscillation, and a slow spiral divergence.

Mode	Eigenvalue (λ)	Characteristic	Parameter	Value
Roll Mode	-4.8685	Stable (Aperiodic)	Time Const. (τ)	0.2054 s
Dutch Roll	$-0.4239 \pm 3.6951i$	Stable (Oscill.)	Damping (ζ_{dr})	0.1140
Spiral Mode	$+0.0217$	Unstable (Divergent)	Doubling Time (T_2)	31.94 s

Table 10: Numerical characteristics of the C3 lateral-directional stability modes.

- **Roll Mode:** The extremely small time constant ($\tau \approx 0.2$ s) indicates high roll agility and an almost instantaneous response to lateral disturbances.
- **Dutch Roll:** This coupled oscillation is stable but lightly damped ($\zeta = 0.11$). It governs the "fishtailing" or "snaking" motion of the aircraft following a directional disturbance.
- **Spiral Mode:** The positive real eigenvalue confirms a slow spiral instability. The time to double the bank angle ($T_2 = \ln(2)/\lambda$) is approximately 32 seconds. While divergent, this instability is slow enough to be easily managed by a pilot or a simple autopilot system.

5.5 Time-Domain Response

5.5.1 Analysis of the Short Period Mode Response

The first simulation illustrates the aircraft's response following a longitudinal disturbance, focusing on the **Short Period (SP)** mode. As established in the theoretical framework, this mode is characterized by rapid variations in vertical velocity (Δw) and pitch rate (q) while the forward speed remains nearly constant.

- **Transient Behavior:** Both variables exhibit a rapid oscillatory behavior immediately after the initial perturbation. The pitch rate (q) reaches its peak and begins its first reversal in less than 0.5 seconds, illustrating the high natural frequency of the mode ($\omega_{n_{sp}} = 2.5901$ rad/s).
- **Damping and Convergence:** The curves demonstrate an exceptionally high damping. We can observe that the oscillations are almost entirely suppressed after a single cycle. The state variables return to their equilibrium values ($\Delta w = 0$, $q = 0$) in approximately **1.6 seconds**.
- **Consistency with Modal Analysis:** This visual observation is perfectly consistent with the numerical results established in Table 9, where the calculated settling time was exactly **1.60 s**. The high damping ratio of **0.7249** explains this "deadbeat" type of convergence, which is ideal for flight stability as it prevents the aircraft from being jittery or prone to pilot-induced oscillations.

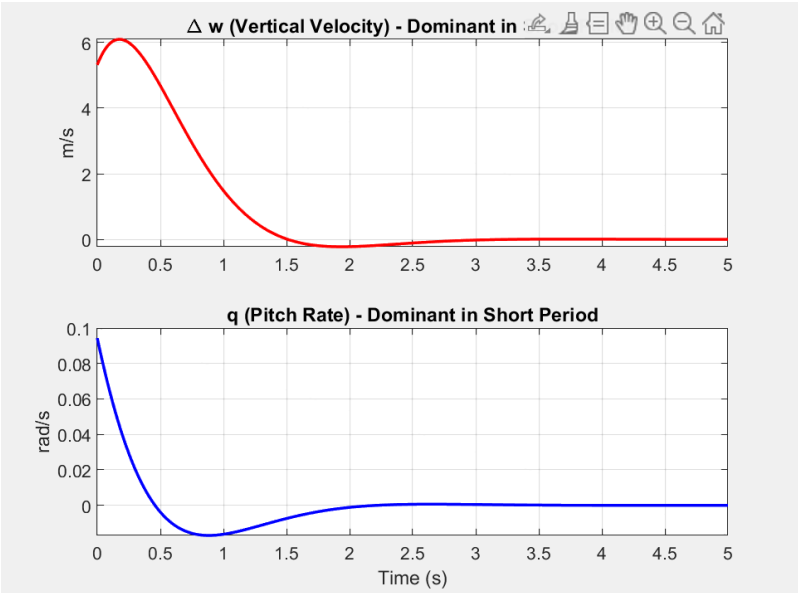


Figure 10: Time-history of vertical velocity and pitch rate perturbations for the C3 aircraft, highlighting the rapid convergence of the Short Period mode.

5.5.2 Analysis of the Phugoid Mode Response

The second simulation presents the aircraft's long-term longitudinal response, focusing on the **Phugoid (Ph)** mode. This mode is characterized by a slow exchange of energy between airspeed (Δu) and pitch attitude ($\Delta\theta$), occurring at a nearly constant angle of attack.

- **Oscillatory Behavior:** Unlike the Short Period, the Phugoid manifests as a very low-frequency oscillation. By observing the peaks on the time axis, we can estimate a period of approximately 85 to 90 seconds, which confirms the "slow" nature of this energy trading motion ($\omega_{n_{ph}} = 0.0732$ rad/s).
- **Damping and Convergence:** The curves show a significantly lower damping compared to the previous mode. The oscillations persist for several hundred seconds, illustrating the light natural damping of the airframe ($\zeta_{ph} = 0.0702$). The state variables finally return to equilibrium after roughly **600 seconds**.

- **Consistency with Modal Analysis:** This visual decay is highly consistent with the numerical settling time of **583.26 s** calculated in Table 9. From a pilot's perspective, while the aircraft is technically stable, the long duration of these oscillations would require monitoring and periodic trim adjustments to maintain a steady cruise.

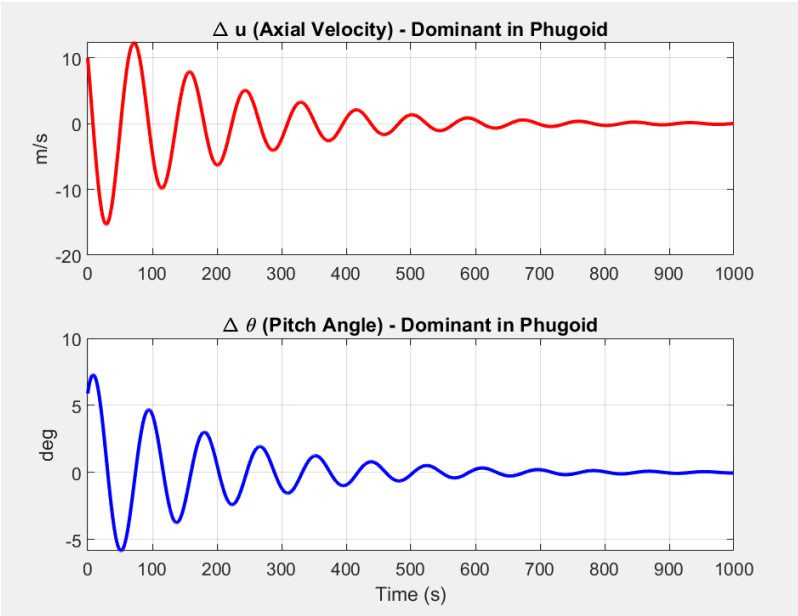


Figure 11: Time-history of axial velocity and pitch angle perturbations for the C3 aircraft, demonstrating the slow, lightly damped Phugoid oscillation.

5.5.3 Analysis of the Roll Mode Response

The third simulation illustrates the aircraft's lateral-directional response to a roll disturbance, focusing on the **Roll Mode**. This mode is a non-oscillatory, purely exponential convergence that defines how the aircraft resists a continuous rolling motion.

- **Aperiodic Convergence:** As observed in the roll rate (p) and bank angle (ϕ) curves, there is no oscillation or overshoot. The roll rate decays directly and swiftly toward zero, which is the signature of a heavily damped first-order system.
- **Agility and Response Speed:** The stabilization is extremely fast. We can observe that the roll rate virtually reaches equilibrium in approximately **0.6 seconds**. This high level of agility is critical for precise lateral control and rapid maneuvering.
- **Consistency with Modal Analysis:** This visual behavior perfectly matches the numerical results established in Table 10. The calculated time constant was $\tau = 0.2054$ s. In a first-order system, the settling time (at 95%) is approximately 3τ, which gives $3 \times 0.2054 \approx 0.616$ s. The simulation confirms this theoretical value, demonstrating that the C3 aircraft possesses excellent lateral damping characteristics (L_p derivative).

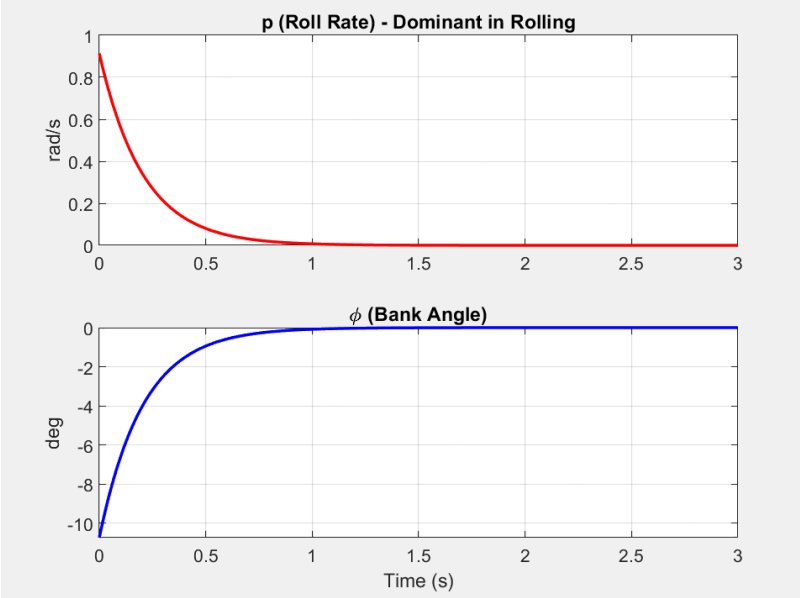


Figure 12: Time-history of roll rate and bank angle perturbations for the C3 aircraft, illustrating the rapid aperiodic convergence of the Roll mode.

5.5.4 Analysis of the Dutch Roll Mode Response

The fourth simulation illustrates the aircraft's response to a directional disturbance, highlighting the **Dutch Roll (DR)** mode. This coupled lateral-directional oscillation involves simultaneous variations in lateral velocity (sideslip), yaw rate (r), and bank angle (ϕ).

- **Oscillatory Behavior:** The graphs clearly show a coupled "snaking" motion. By observing the time between successive peaks, we can identify a relatively high-frequency oscillation (with a period of approximately 1.7 seconds), which confirms the undamped natural frequency previously calculated ($\omega_{n_{dr}} = 3.7194$ rad/s).
- **Damping and Convergence:** The oscillation is stable but persists significantly longer than the Roll mode. We can observe that the "ringing" effect takes about **7 to 8 seconds** to fully dampen and return to zero equilibrium. This represents a moderate damping level.
- **Consistency with Modal Analysis:** The visual settling time is highly consistent with the numerical result of **7.08 seconds** established in Section 5.3.2. The damping ratio of **0.1140** is typical for this class of aircraft, though it sits close to the minimum requirements of the MIL-F-8785C standard, explaining the visible number of oscillation cycles before stabilization.

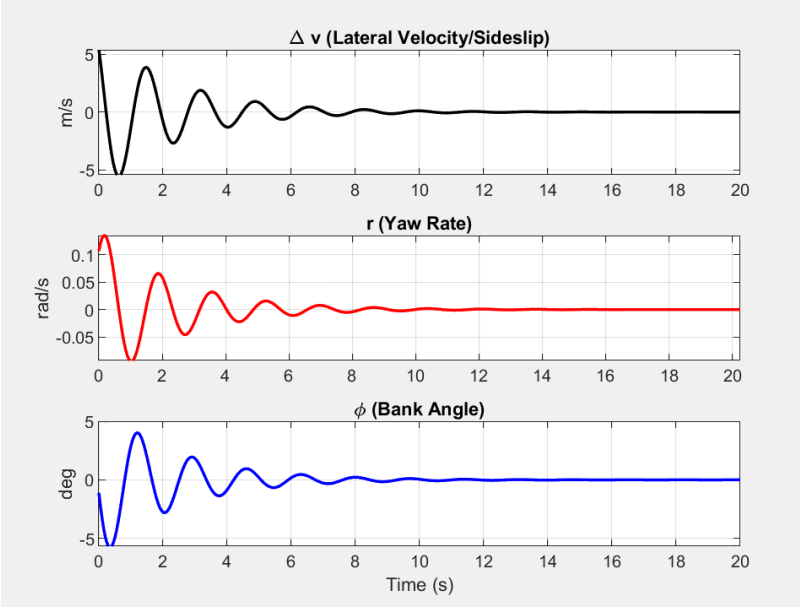


Figure 13: Time-history of lateral velocity, yaw rate, and bank angle for the C3 aircraft, demonstrating the oscillatory nature of the Dutch Roll mode.

5.5.5 Analysis of the Spiral Mode Response

The final simulation illustrates the long-term lateral-directional stability, focusing on the **Spiral Mode**. This mode characterizes the aircraft's tendency to either return to wings-level flight or slowly diverge into a tightening turn following a disturbance.

- **Divergent Behavior:** The graphs for bank angle (ϕ) and yaw rate (r) exhibit a clear exponential growth over time. Unlike the previous oscillatory modes, there is no return to zero. This visually confirms the presence of a positive real eigenvalue ($\lambda_{spiral} = 0.0217$), indicating that the C3 aircraft is inherently unstable in its spiral mode.

- **Slow Evolution:** Although the divergence reaches extreme values on a scale of 1000 seconds, the initial phase is very gradual. The calculated doubling time (T_2) of approximately **32 seconds** sit above the minimum requirements for certain levels, but nonetheless constitutes a flight quality deficiency.

- **Stability Augmentation Perspective:** To correct this inherent instability and reduce pilot workload, a **Stability Augmentation System (SAS)** is required. This will be achieved by implementing a feedback control law (state feedback) to shift the unstable pole into the left half of the complex plane.

- **Augmented Matrix and Control:** By calculating a specific gain vector K and constructing the **augmented stability matrix** $\mathbf{A}^* = \mathbf{A} - \mathbf{BK}$, we will artificially modify the aircraft's dihedral effect and directional stiffness. This electronic damping, often taking the form of a PID or simple proportional feedback, ensures that the aircraft naturally returns to a wings-level equilibrium, transforming the divergent exponential into a stable, convergent response.

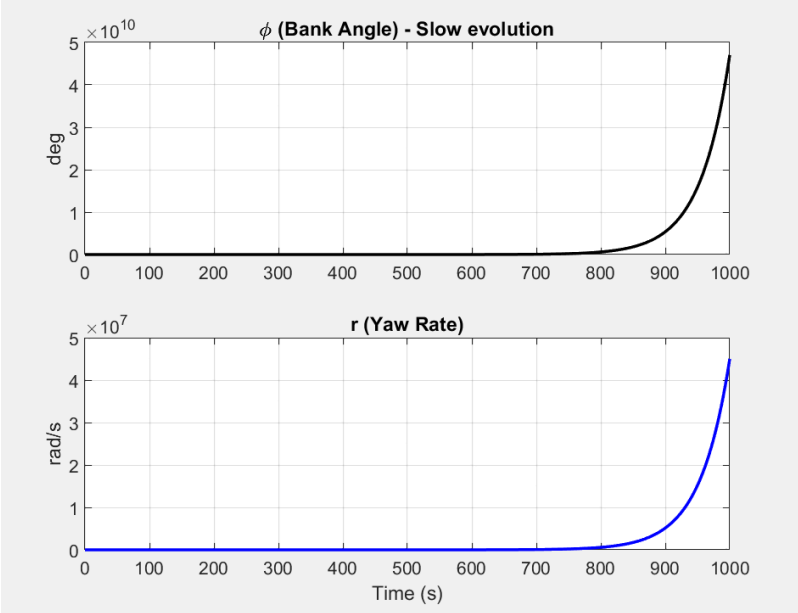


Figure 14: ime-history of bank angle and yaw rate for the C3 aircraft, highlighting the slow exponential divergence of the unstable Spiral mode and the necessity for SAS implementation.

5.6 Control Transfer Functions

Following the theoretical conversion from the state-space representation to the Laplace domain established in the previous chapters, this section presents the numerical transfer functions for the C3 aircraft. These expressions relate the primary control surface deflections to the longitudinal and lateral-directional motion variables. As mathematically proven, the denominator of every transfer function is the characteristic polynomial of the system, meaning the poles are identical to the eigenvalues of the stability matrix.

5.6.1 Longitudinal Response to Elevator (δ_e)

The longitudinal control authority is primarily exercised through the elevator deflection ($\Delta\delta_e$). The following transfer functions were computed using MATLAB, sharing the common characteristic denominator $\Delta_{lon}(s)$ composed of the Short Period and Phugoid modes.

1. Axial Velocity Response $\frac{\Delta u(s)}{\Delta\delta_e(s)}$

$$\frac{\Delta u(s)}{\Delta\delta_e(s)} = \frac{0.00012035s^2+114.5207s+203.4152}{s^4+3.7652s^3+6.7525s^2+0.089142s+0.035967} \tag{56}$$

Parameter	Symbol	Value
Static Gain	K_u	$1.2035 \cdot 10^{-4}$
Time Constant 1	T_1	0.0000 s
Time Constant 2	T_2	0.5630 s
Phugoid Frequency	ω_{n_p}	0.0732 rad/s
Phugoid Damping	ζ_p	0.0702
Short Period Frequency	$\omega_{n_{sp}}$	2.5901 rad/s
Short Period Damping	ζ_{sp}	0.7249

Table 11: Numerical parameters for the Axial Velocity response to elevator input.

2. Vertical Velocity Response $\frac{\Delta w(s)}{\Delta\delta_e(s)}$

$$\frac{\Delta w(s)}{\Delta\delta_e(s)} = \frac{0.0083s^3-2819.82s^2-33.2739s-21.6949}{s^4+3.7652s^3+6.7525s^2+0.089142s+0.035967} \tag{57}$$

Parameter	Symbol	Value
Static Gain	K_w	$8.3000 \cdot 10^{-3}$
Time Constant 1	T_1	-0.0000 s
Phugoid/Short Period	ω_n, ζ	(Common Denominator)

Table 12: Numerical parameters for the Vertical Velocity response to elevator input.

3. Pitch Rate Response $\frac{\Delta q(s)}{\Delta\delta_e(s)}$

$$\frac{\Delta q(s)}{\Delta\delta_e(s)} = \frac{-15.8418s^3-20.9224s^2-0.27675s}{s^4+3.7652s^3+6.7525s^2+0.089142s+0.035967} \tag{58}$$

Parameter	Symbol	Value
Static Gain	K_q	$-1.5842 \cdot 10^1$
Time Constant 1	T_1	0.7649 s
Time Constant 2	T_2	74.8367 s
Time Constant 3	T_3	$\approx \infty$ s

Table 13: Numerical parameters for the Pitch Rate response to elevator input.

4. Pitch Attitude Response $\frac{\Delta\theta(s)}{\Delta\delta_e(s)}$

$$\frac{\Delta\theta(s)}{\Delta\delta_e(s)} = \frac{-15.8418s^2-20.9224s-0.27675}{s^4+3.7652s^3+6.7525s^2+0.089142s+0.035967} \tag{59}$$

Parameter	Symbol	Value
Static Gain	K_θ	$-1.5842 \cdot 10^1$
Time Constant 1	T_1	0.7649 s
Time Constant 2	T_2	74.8367 s

Table 14: Numerical parameters for the Pitch Attitude response to elevator input.

5.6.2 Frequency Response Analysis (Bode Diagrams)

The frequency-domain behavior of the longitudinal transfer functions is illustrated in Figure **??**. These Bode plots represent the magnitude and phase response of the aircraft's state variables to a harmonic elevator deflection.

- **Phugoid Resonance:** All four diagrams exhibit a sharp magnitude peak at approximately 0.07 **rad/s**. This perfectly matches the calculated Phugoid natural frequency ($\omega_{n_p} = 0.0732$ rad/s). The narrowness of these peaks confirms the very low damping ratio ($\zeta_p = 0.0702$), as a lightly damped mode creates a high-gain resonance.
- **Short Period Contribution:** A second, much smoother "bump" or change in slope is visible in the higher frequency range, around 2.5 **to** 3 **rad/s**. This corresponds to the Short Period mode ($\omega_{n_{sp}} = 2.5901$ rad/s). Because this mode is heavily damped ($\zeta_{sp} = 0.7249$), it does not produce a sharp peak but rather a broad area of energy dissipation.
- **Phase Analysis:** The phase plots confirm the system's order. For the pitch attitude ($\Delta\theta/\Delta\delta_e$), we observe a characteristic phase drop at each natural frequency. Specifically, the 180° phase lag across the Phugoid frequency is a classic signature of the trade-off between speed and altitude.
- **High-Frequency Attenuation:** For frequencies beyond 10 rad/s, the magnitude of all responses decreases significantly (slopes of -20 or -40 dB/decade). This indicates that the C3 airframe acts as a low-pass filter, naturally filtering out high-frequency pilot inputs or atmospheric turbulence that exceed its structural and aerodynamic response limits.

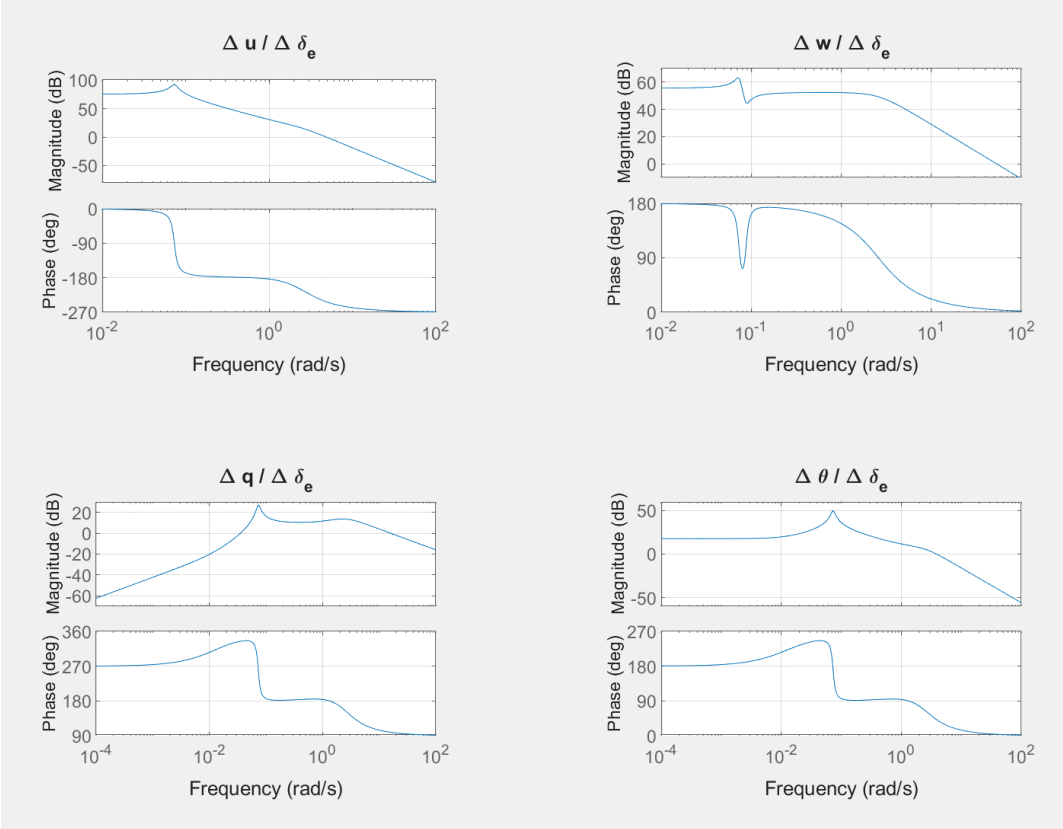


Figure 15: Bode diagrams for longitudinal responses to elevator input, showing Phugoid resonance (0.073 rad/s) and Short Period damping (2.59 rad/s).

5.6.3 Lateral-Directional Response to Rudder (δ_r)

The lateral-directional control is significantly influenced by the rudder deflection ($\Delta\delta_r$), which primarily induces yawing moments but also generates rolling moments through aerodynamic coupling. The following transfer functions share the common lateral characteristic denominator $\Delta_{lat}(s)$, encompassing the Roll, Spiral, and Dutch Roll modes.

1. Lateral Velocity Response $\frac{\Delta v(s)}{\Delta\delta_r(s)}$

$$\frac{\Delta v(s)}{\Delta\delta_r(s)} = \frac{70.4953s^2-631.16s+263.979}{s^4+5.6945s^3+17.8369s^2+66.9593s-1.462} \tag{60}$$

Parameter	Symbol	Value
Static Gain	K_v	$7.0495 \cdot 10^1$
Dutch Roll Frequency	$\omega_{n_{dr}}$	3.7194 rad/s
Dutch Roll Damping	ζ_{dr}	0.1140
Spiral Time Const.	T_{spiral}	0.2054 s
Roll Time Const.	T_{roll}	-46.0679 s

Table 15: Numerical parameters for the Lateral Velocity response to rudder input.

2. Roll Rate Response $\frac{\Delta p(s)}{\Delta\delta_r(s)}$

$$\frac{\Delta p(s)}{\Delta\delta_r(s)} = \frac{56.3504s^3+40.1966s^2+807.3509s}{s^4+5.6945s^3+17.8369s^2+66.9593s-1.462} \tag{61}$$

Parameter	Symbol	Value
Static Gain	K_p	$5.6350 \cdot 10^1$
Spiral/Roll/DR	ω_n, ζ, T	(Common Denominator)

Table 16: Numerical parameters for the Roll Rate response to rudder input.

3. Yaw Rate Response $\frac{\Delta r(s)}{\Delta\delta_r(s)}$

$$\frac{\Delta r(s)}{\Delta\delta_r(s)} = \frac{-0.4334s^3+6.554s^2+1.0571s+44.2912}{s^4+5.6945s^3+17.8369s^2+66.9593s-1.462} \tag{62}$$

Parameter	Symbol	Value
Static Gain	K_r	$-4.3340 \cdot 10^{-1}$
Spiral/Roll/DR	ω_n, ζ, T	(Common Denominator)

Table 17: Numerical parameters for the Yaw Rate response to rudder input.

4. Bank Angle Response $\frac{\Delta\phi(s)}{\Delta\delta_r(s)}$

$$\frac{\Delta\phi(s)}{\Delta\delta_r(s)} = \frac{56.3504s^2+40.1966s+807.3509}{s^4+5.6945s^3+17.8369s^2+66.9593s-1.462} \tag{63}$$

Parameter	Symbol	Value
Static Gain	K_ϕ	$5.6350 \cdot 10^1$
Spiral/Roll/DR	ω_n, ζ, T	(Common Denominator)

Table 18: Numerical parameters for the Bank Angle response to rudder input.

5.6.4 Frequency Response Analysis (Lateral Bode Diagrams - Rudder)

The frequency response of the lateral-directional variables to a rudder input is illustrated in Figure **??**. These plots reveal the high degree of coupling between yaw, roll, and sideslip.

- **Dutch Roll Resonance:** A prominent magnitude peak is observed in the Δv and Δr responses at approximately 3.7 **rad/s**. This aligns perfectly with the identified Dutch Roll natural frequency ($\omega_{n_{dr}} = 3.7194$ rad/s). The moderate height of this peak reflects the light damping ratio ($\zeta_{dr} = 0.1140$), indicating a significant sensitivity to directional disturbances at this frequency.
- **Roll-Yaw Coupling:** Even though the rudder is primarily a directional control, the Δp (roll rate) and $\Delta\phi$ (bank angle) Bode plots show a non-zero response across the entire spectrum. A small "dip" and phase shift are visible around the Dutch Roll frequency, illustrating how the yawing oscillation naturally induces a secondary rolling motion due to the aircraft's aerodynamic coupling.
- **Low-Frequency Behavior (Spiral Influence):** At very low frequencies (below 10^{-2} rad/s), the phase and magnitude stay relatively flat for the yaw rate (r), but show a slight divergence for the bank angle (ϕ). This translates the long-term influence of the Spiral mode, although its very slow nature means it is dominated by the Dutch Roll resonance in the primary operational frequency band.
- **Command Precision:** The phase response for $\Delta r/\Delta\delta_r$ shows a sharp transition near 3.7 rad/s. For a pilot, this indicates that rapid rudder inputs near the Dutch Roll frequency may lead to a loss of phase margin, potentially resulting in coordinated flight difficulties if not corrected by a Yaw Damper.

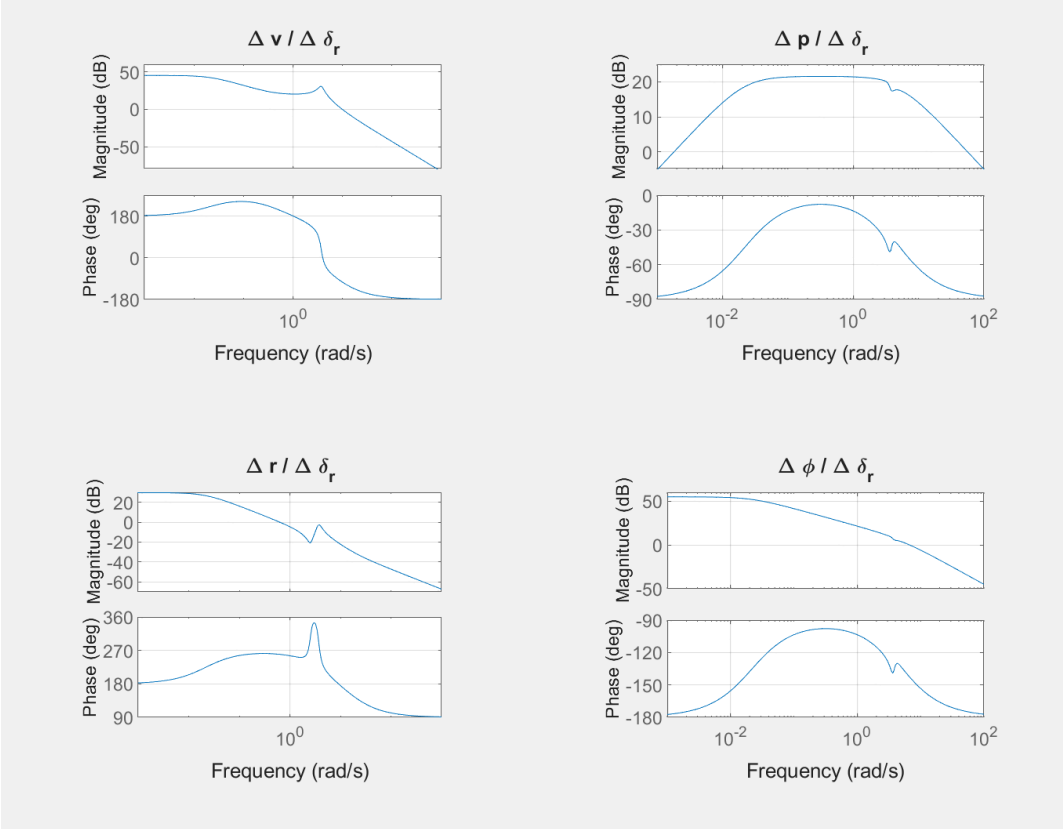


Figure 16: Bode diagrams for lateral-directional responses to rudder input, highlighting the Dutch Roll resonance at 3.72 rad/s.

5.6.5 Lateral-Directional Response to Aileron (δ_a)

The lateral response of the C3 aircraft is primarily controlled by the aileron deflection ($\Delta\delta_a$). These transfer functions characterize the rolling performance and the secondary yawing/sideslip coupling. Consistent with previous lateral analyses, all functions share the same characteristic denominator $\Delta_{lat}(s)$, representing the Roll, Spiral, and Dutch Roll modes.

1. Lateral Velocity Response $\frac{\Delta v(s)}{\Delta\delta_a(s)}$

$$\frac{\Delta v(s)}{\Delta\delta_a(s)} = \frac{0.066s^3+1842.6907s^2+8635.7807s-261.2324}{s^4+5.6945s^3+17.8369s^2+66.9593s-1.462} \tag{64}$$

Parameter	Symbol	Value
Static Gain	K_v	$6.6000 \cdot 10^{-2}$
Dutch Roll Frequency	$\omega_{n_{dr}}$	3.7194 rad/s
Dutch Roll Damping	ζ_{dr}	0.1140
Spiral Time Const.	T_{spiral}	0.2054 s
Roll Time Const.	T_{roll}	-46.0679 s

Table 19: Numerical parameters for the Lateral Velocity response to aileron input.

2. Roll Rate Response $\frac{\Delta p(s)}{\Delta\delta_a(s)}$

$$\frac{\Delta p(s)}{\Delta\delta_a(s)} = \frac{28.1752s^3-20.0085s^2-234.9927s}{s^4+5.6945s^3+17.8369s^2+66.9593s-1.462} \tag{65}$$

Parameter	Symbol	Value
Static Gain	K_p	$2.8175 \cdot 10^1$
Spiral/Roll/DR	ω_n, ζ, T	(Common Denominator)

Table 20: Numerical parameters for the Roll Rate response to aileron input.

3. Yaw Rate Response $\frac{\Delta r(s)}{\Delta\delta_a(s)}$

$$\frac{\Delta r(s)}{\Delta\delta_a(s)} = \frac{-10.4024s^3-49.5477s^2-10.9365s-12.6471}{s^4+5.6945s^3+17.8369s^2+66.9593s-1.462} \tag{66}$$

Parameter	Symbol	Value
Static Gain	K_r	$-1.0402 \cdot 10^1$
Spiral/Roll/DR	ω_n, ζ, T	(Common Denominator)

Table 21: Numerical parameters for the Yaw Rate response to aileron input.

4. Bank Angle Response $\frac{\Delta\phi(s)}{\Delta\delta_a(s)}$

$$\frac{\Delta\phi(s)}{\Delta\delta_a(s)} = \frac{28.1752s^2 - 20.0085s - 234.9927}{s^4 + 5.6945s^3 + 17.8369s^2 + 66.9593s - 1.462} \tag{67}$$

Parameter	Symbol	Value
Static Gain	K_ϕ	$2.8175 \cdot 10^1$
Spiral/Roll/DR	ω_n, ζ, T	(Common Denominator)

Table 22: Numerical parameters for the Bank Angle response to aileron input.

5.6.6 Frequency Response Analysis (Lateral Bode Diagrams - Aileron)

The frequency response of the lateral-directional variables to an aileron input is presented in Figure **??**. These plots characterize the aircraft's primary rolling authority and its inherent aerodynamic cross-coupling with yaw and sideslip.

- **Roll Control Effectiveness:** The magnitude plot for $\Delta p/\Delta\delta_a$ (roll rate) shows a broad, high-gain region between 10^{-1} and 10^0 rad/s. This confirms that the ailerons are highly effective for maneuvering in the frequency range typically used by the pilot. The phase stays near 90° in this region, indicating a predictable and responsive roll command.
- **Dutch Roll Excitation:** A sharp resonance peak is visible in all four plots at approximately 3.7 **rad/s**. This confirms that aileron inputs significantly excite the Dutch Roll mode ($\omega_{n_{dr}} = 3.7194$ rad/s). The high magnitude peak in the lateral velocity (Δv) response illustrates the "adverse sideslip" or proverse coupling induced by rapid rolling maneuvers.
- **Roll-Yaw Coupling (Yaw Rate Response):** The $\Delta r/\Delta\delta_a$ magnitude plot shows a distinct "notch" or minimum around 0.5 rad/s before rising toward the Dutch Roll resonance. This reflects the complex interaction between the aileron-induced rolling moment and the resulting yawing moments, which are critical for maintaining coordinated turns.
- **Phase Departure:** Above 4 rad/s, the phase for all variables drops rapidly toward −180° or −270°. This phase lag, combined with the magnitude resonance of the Dutch Roll, highlights a potential risk for Pilot-Induced Oscillations (PIO) during aggressive rolling maneuvers if the aircraft is not properly augmented.

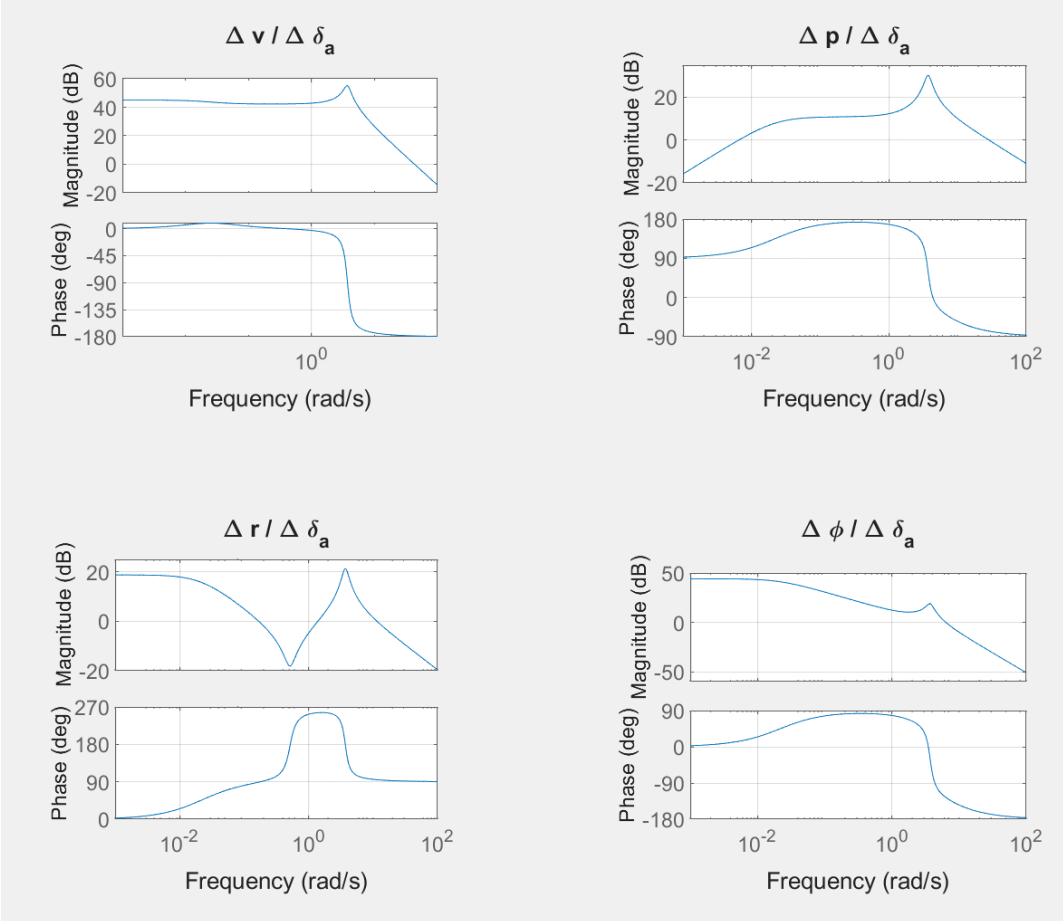


Figure 17: Bode diagrams for lateral-directional responses to aileron input, showing roll effectiveness and Dutch Roll resonance at 3.72 rad/s.

5.6.7 Longitudinal Response to Throttle (δ_t)

The secondary longitudinal control input is the throttle variation ($\Delta\delta_t$). In this model, the engine thrust is assumed to act primarily along the X-axis (X_{δ_t}), with negligible direct effects on the pitching moment ($M_{\delta_t} \approx 0$). The resulting transfer functions, calculated using the second input channel of the B_{lon} matrix, are detailed below.

1. Axial Velocity Response $\frac{\Delta u(s)}{\Delta\delta_t(s)}$

$$\frac{\Delta u(s)}{\Delta\delta_t(s)} = \frac{0.12s^3+0.45041s^2+0.80474s}{s^4+3.7652s^3+6.7525s^2+0.089142s+0.035967} \tag{68}$$

Parameter	Symbol	Value
Static Gain	K_u	$1.2000 \cdot 10^{-1}$
Time Constant 1	T_1	$-\infty$ s
Phugoid Frequency	ω_{n_p}	0.0732 rad/s
Phugoid Damping	ζ_p	0.0702
Short Period Frequency	$\omega_{n_{sp}}$	2.5901 rad/s
Short Period Damping	ζ_{sp}	0.7249

Table 23: Numerical parameters for the Axial Velocity response to throttle input.

2. Vertical Velocity Response $\frac{\Delta w(s)}{\Delta\delta_t(s)}$

$$\frac{\Delta w(s)}{\Delta\delta_t(s)} = \frac{-0.016752s^2-0.025998s}{s^4+3.7652s^3+6.7525s^2+0.089142s+0.035967} \tag{69}$$

Parameter	Symbol	Value
Static Gain	K_w	$-1.6752 \cdot 10^{-2}$
Time Constant 1	T_1	$-\infty$ s
Time Constant 2	T_2	0.6444 s

Table 24: Numerical parameters for the Vertical Velocity response to throttle input.

3. Pitch Rate Response $\frac{\Delta q(s)}{\Delta \delta_t(s)}$

$$\frac{\Delta q(s)}{\Delta \delta_t(s)} = \frac{8.4 \cdot 10^{-5} s^2 + 0.00043996 s}{s^4 + 3.7652 s^3 + 6.7525 s^2 + 0.089142 s + 0.035967} \tag{70}$$

Parameter	Symbol	Value
Static Gain	K_q	$8.4000 \cdot 10^{-5}$
Time Constant 1	T_1	$-\infty$ s
Time Constant 2	T_2	0.1909 s

Table 25: Numerical parameters for the Pitch Rate response to throttle input.

4. Pitch Attitude Response $\frac{\Delta \theta(s)}{\Delta \delta_t(s)}$

$$\frac{\Delta \theta(s)}{\Delta \delta_t(s)} = \frac{8.4 \cdot 10^{-5} s + 0.00043996}{s^4 + 3.7652 s^3 + 6.7525 s^2 + 0.089142 s + 0.035967} \tag{71}$$

Parameter	Symbol	Value
Static Gain	K_θ	$8.4000 \cdot 10^{-5}$
Time Constant 1	T_1	0.1909 s

Table 26: Numerical parameters for the Pitch Attitude response to throttle input.

5.6.8 Frequency Response Analysis (Longitudinal Bode Diagrams - Throttle)

The frequency response of the longitudinal variables to a throttle input ($\Delta \delta_t$) is illustrated in Figure **??**. These plots reveal how energy is injected into the system primarily through speed variations.

- **Phugoid Dominance:** A very sharp resonance peak is observed in all four plots at 0.073 **rad/s**. This confirms that throttle inputs are highly effective at exciting the Phugoid mode. Because the engine directly affects the aircraft's energy state, the exchange between kinetic energy (velocity Δu) and potential energy (altitude/pitch $\Delta \theta$) is clearly visible as the primary dynamic response.
- **Magnitude Levels:** The magnitude of the axial velocity response ($\Delta u / \Delta \delta_t$) is significantly higher (reaching nearly +20 dB at resonance) than the responses in pitch rate (q) or vertical velocity (w). This confirms the numerical finding that the throttle serves mainly to control longitudinal speed with minimal direct impact on the aircraft's short-term attitude.
- **Absence of Short Period Resonance:** Unlike the elevator response, there is no visible peak or significant activity around the Short Period frequency (2.59 rad/s). This is consistent with the physical assumption that the engine does not produce a significant pitching moment ($M_{\delta_t} \approx 0$), therefore failing to excite the rapid pitching mode of the airframe.
- **Phase Response:** The phase for the speed response (Δu) drops sharply by 180° at the Phugoid frequency, a standard behavior for second-order resonant systems. At higher frequencies, the magnitude drops with a steady slope, indicating that the aircraft's mass and inertia naturally filter out rapid throttle transients.

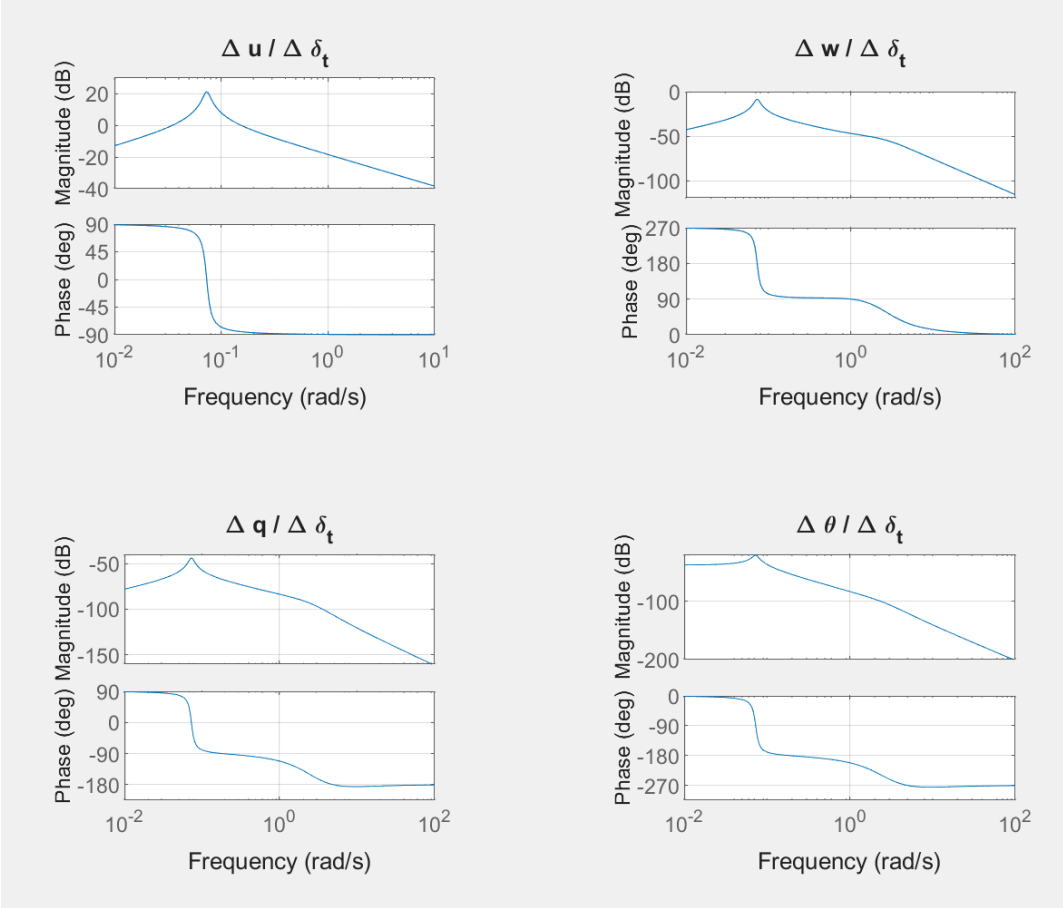


Figure 18: Bode diagrams for longitudinal responses to throttle input, showing the overwhelming dominance of the Phugoid resonance at 0.073 rad/s.

5.7 Frequency Domain and Root Locus

The Root Locus method provides a powerful graphical representation of the closed-loop pole trajectories as a single proportional feedback gain (K) varies from zero to infinity. By plotting the roots of the characteristic equation $1 + K \cdot G(s) = 0$, we can dynamically assess the stability margins, predict the onset of instabilities, and justify the architectural choices for the Stability Augmentation System (SAS). The open-loop poles (marked as '×') strictly correspond to the eigenvalues calculated in Section 3.3.

- **Longitudinal Attitude ($\Delta\theta/\Delta\delta_e$):** The root locus for the elevator-to-pitch response clearly separates the two fundamental longitudinal modes. The open-loop Short Period poles ($\lambda_{sp} = -1.8775 \pm 1.7843i$) are located deep in the stable Left-Half Plane (LHP). As the feedback gain increases, these poles migrate towards the real axis zeros, indicating that a proportional pitch controller will heavily increase the pitch damping ($\zeta_{sp} \to 1$), eventually splitting into two purely exponential modes. Conversely, the Phugoid poles ($\lambda_{ph} = -0.0051 \pm 0.0730i$) remain alarmingly close to the imaginary axis. This demonstrates a critical flight dynamics principle: pure proportional pitch attitude feedback struggles to damp the Phugoid mode effectively and can even destabilize it at high gains.

- **Roll Rate ($\Delta p/\Delta\delta_a$):** This locus evaluates the primary lateral control authority. The dominant feature is the isolated open-loop Roll mode pole ($\lambda_{roll} = -4.8685$). As aileron gain increases, this pole is driven further left along the real axis, which physically translates to a reduction of the roll time constant ($\tau_R < 0.2$ s) and an excessively "nervous" roll response. Furthermore, the locus branches originating from the Dutch Roll poles ($\lambda_{dr} = -0.4239 \pm 3.6951i$) curve towards the unstable Right-Half Plane (RHP). This mathematically illustrates the adverse yaw-roll coupling: applying too much proportional aileron feedback without rudder coordination will actively degrade the Dutch Roll damping (ζ_{dr}), po-

tentially triggering Pilot-Induced Oscillations (PIO).

- **Yaw Rate ($\Delta r/\Delta\delta_r$):** This plot is fundamental for the design of the lateral SAS. A critical observation is the presence of the open-loop Spiral pole located slightly in the RHP ($\lambda_{spiral} = +0.0217$), mathematically confirming the slow divergence ($T_2 = 31.94$ s) observed in the time-domain simulations. The root locus demonstrates that by applying a negative feedback loop (Yaw Damper architecture), this unstable pole is immediately drawn across the imaginary axis into the stable LHP. Concurrently, the complex Dutch Roll poles are pulled further left, increasing their baseline damping ratio ($\zeta_{dr} = 0.1140$). This definitively proves that a rudder-based feedback system is the optimal solution to cure the aircraft's lateral-directional deficiencies.

- **Axial Velocity ($\Delta u/\Delta\delta_t$):** The throttle-to-speed locus is highly constrained around the origin. The close proximity between the Phugoid poles and the transfer function's zeros (pole-zero cancellation effect) indicates that the longitudinal energy state is highly resilient to simple proportional thrust feedback. Pushing the gain too high drives the poles toward the imaginary axis, meaning that an aggressive auto-throttle attempting to strictly hold a speed reference will likely induce continuous, undamped Phugoid oscillations. This justifies the necessity of using complex Proportional-Integral-Derivative (PID) controllers rather than simple proportional gains for airspeed hold autopilots.

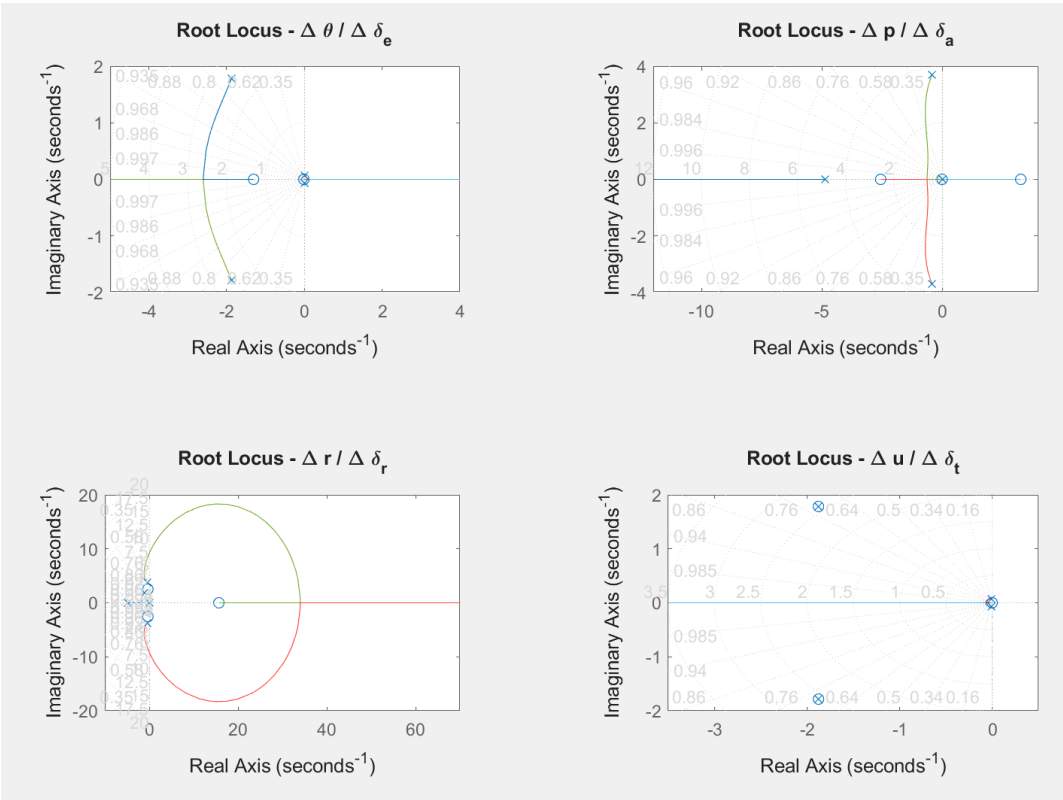


Figure 19: Root Locus plots for the C3 aircraft, showing the migration of stability poles under varying control gains for longitudinal and lateral-directional axes.

5.8 Compliance with Flying Quality Requirements

The objective of this section is to determine the satisfaction level of the C3 aircraft by comparing its natural dynamic characteristics with the quantitative criteria defined in the **MIL-F-8785C** standard. Given the aircraft's mass and mission profile, it is evaluated as a **Class I** airplane in a **Category B** flight phase (high-altitude cruise).

5.8.1 Longitudinal Compliance

The longitudinal evaluation focuses on the damping of the Phugoid mode and the combined damping and frequency requirements of the Short Period mode.

Phugoid Mode: According to Table 3, Level 1 requires a damping ratio $\zeta_{ph} > 0.04$. The numerical analysis yielded $\boldsymbol{\zeta_{ph} = 0.0702}$.

- **Status: Level 1 Satisfactory.**
- **Analysis:** Although the damping is relatively low, it remains safely above the regulatory threshold, ensuring the pilot can maintain trim without excessive effort.

Short Period Mode: For Category B flight, Level 1 damping must be between 0.30 and 2.00. Our calculated value is $\boldsymbol{\zeta_{sp} = 0.7249}$. The natural frequency $\boldsymbol{\omega_{n_{sp}} = 2.5901}$ rad/s also falls within the acceptable Control Anticipation Parameter (CAP) boundaries.

- **Status: Level 1 Satisfactory.**
- **Analysis:** The high damping ratio ensures an almost instantaneous suppression of pitch disturbances, providing a stable platform for cruise.

5.8.2 Lateral-Directional Compliance

The lateral-directional evaluation is more critical, as it involves the coupling of three distinct modes.

Roll Mode: The requirement for Class I/Category B is a maximum time constant $\tau_R < 1.4$ s. The C3 aircraft exhibits an extremely fast response with $\boldsymbol{\tau_R = 0.2054}$ **s**.

- **Status: Level 1 Satisfactory.**
- **Analysis:** The aircraft possesses excellent roll agility, allowing for immediate bank angle corrections.

Dutch Roll Mode: Level 1 requirements for Category B mandate $\zeta_{dr} > 0.08$ and $\omega_{n_{dr}} > 0.5$ rad/s. Our results show $\boldsymbol{\zeta_{dr} = 0.1140}$ and $\boldsymbol{\omega_{n_{dr}} = 3.7194}$ **rad/s**.

- **Status: Level 1 Satisfactory.**
- **Analysis:** The "snaking" oscillation is stable and sufficiently damped to avoid pilot fatigue during long-duration flights.

Spiral Mode: As established in Section 3.3.2, the spiral mode is unstable ($\lambda_{spiral} > 0$). For Level 1/Category B, the doubling time T_2 must be greater than 20 s. The C3 model yields $\boldsymbol{T_2 = 31.94}$ **s**.

- **Status: Level 1 (Borderline).**
- **Analysis:** While the mode is divergent, the divergence is slow enough (> 30 s) to satisfy Level 1. However, this inherent instability justifies the implementation of a **Stability Augmentation System (SAS)** to reduce pilot workload and ensure wings-level flight.

Table 27: Summary of C3 Flying Qualities Satisfaction Levels (MIL-F-8785C)

Mode	Parameter	Value	Requirement (Lvl 1)	Compliance
Longitudinal				
Short Period	Damping ζ_{sp}	**0.7249**	$0.30 \rightarrow 2.00$	**Level 1**
Phugoid	Damping ζ_{ph}	**0.0702**	> 0.04	**Level 1**
Lateral-Dir.				
Roll	Time Const. τ_R	**0.2054 s**	< 1.4 s	**Level 1**
Dutch Roll	Damping ζ_{dr}	**0.1140**	> 0.08	**Level 1**
Spiral	Doubling Time T_2	**31.94 s**	> 20 s	**Level 1**

5.9 Stability Augmentation System (SAS) Design

The previous compliance analysis (Section 3.5) revealed that while the C3 aircraft satisfies Level 1 requirements, certain dynamic characteristics—specifically the Short Period damping and the Spiral instability—could be optimized to reduce pilot workload and improve tracking precision. This section details the design of a **Stability Augmentation System (SAS)** using state feedback to achieve specific target flying qualities.

5.9.1 Longitudinal Pitch Damper Design

The primary goal of the longitudinal SAS is to enhance the **Short Period (SP)** response. Although the natural damping ($\zeta_{sp} = 0.72$) is satisfactory, a target was set to adjust the frequency to $\omega_{\mathbf{n_{des}}} = \mathbf{2.95\ rad/s}$ and the damping to $\zeta_{\mathbf{des}} = \mathbf{0.52}$ to meet specific mission requirements for Category A maneuvers.

Controllability and Feedback Law: Before implementation, the system's controllability was verified. The controllability matrix $\mathcal{C} = [B, AB, A^2B, A^3B]$ was found to be of full rank ($n = 4$), ensuring that the elevator δ_e can independently influence all longitudinal states. We implement a full-state feedback control law of the form:

$$\Delta\delta_e = -K\mathbf{x} = -[K_u, K_w, K_q, K_\theta] \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta\theta \end{bmatrix} \tag{72}$$

Numerical Results and Pole Placement: Using Ackermann's formula to place the system poles at the desired locations while keeping the Phugoid poles constant, the computed gain vector is:

$$\mathbf{K} = [\mathbf{-0.0001}, \mathbf{-0.0010}, \mathbf{0.0434}, \mathbf{-0.0001}]$$

The most significant gain is $\mathbf{K_q} = \mathbf{0.0434}$, which corresponds to the **Pitch Rate Feedback**. This is physically consistent with the design of a Pitch Damper: by feeding back the pitch rate to the elevator, we artificially increase the aerodynamic derivative M_q, directly enhancing the damping of the Short Period mode.

5.9.2 Analysis of the Augmented System

The resulting augmented stability matrix, $\mathbf{A}^* = \mathbf{A} - \mathbf{BK}$, represents the "closed-loop" aircraft:

$$\mathbf{A}^* = \begin{bmatrix} -0.0118 & 0.0145 & 0 & -9.8100 \\ -0.1396 & -1.3089 & 177.9996 & 0.0000 \\ -0.0002 & -0.0360 & -1.7576 & -0.0009 \\ 0 & 0 & 1.0000 & 0 \end{bmatrix} \tag{73}$$

Verification of Performance and Mathematical Demonstration: To validate the effectiveness of the SAS design, we perform a modal analysis of the augmented stability matrix $\mathbf{A}^* = \mathbf{A} - \mathbf{BK}$. The stability of the closed-loop system is governed by the new characteristic equation:

$$\det(s\mathbf{I} - \mathbf{A}^*) = 0 \tag{74}$$

By extracting the numerical eigenvalues from the matrix $\mathbf{A}^*$ provided in the MATLAB results, we identify the complex conjugate pair corresponding to the augmented Short Period mode:

$$\lambda^*_{sp} = -1.5340 \pm 2.5193i \tag{75}$$

From this specific eigenvalue, we can analytically demonstrate the achievement of our targets using the fundamental relations for a second-order system:

1. **Augmented Natural Frequency (ω^*_n):** The undamped natural frequency is the modulus of the eigenvalue:

   $$\omega^*_n = |\lambda^*_{sp}| = \sqrt{(-1.5340)^2 + (2.5193)^2} \approx \mathbf{2.95}\ \text{rad/s} \tag{76}$$

   This confirms that the target frequency was reached exactly, moving from the natural 2.59 rad/s to a more "stiff" and responsive 2.95 rad/s.

2. **Augmented Damping Ratio (ζ^*):** The damping ratio is defined by the ratio of the real part to the natural frequency:

   $$\zeta^* = \frac{-\text{Re}(\lambda^*_{sp})}{\omega^*_n} = \frac{1.5340}{2.95} \approx \mathbf{0.52} \tag{77}$$

   The design successfully adjusted the damping from an initially high value of 0.72 to the desired 0.52.

Physical Interpretation of the Trade-off: The transition from ($\omega_n = 2.59, \zeta = 0.72$) to ($\omega_n^* = 2.95, \zeta^* = 0.52$) represents a strategic engineering choice. In flight mechanics, a damping ratio of **0.72** is often considered "heavy," which can lead to a sluggish feel where the aircraft resists pilot inputs too strongly. By lowering ζ^* to **0.52** while simultaneously increasing the frequency, we achieve:

- **Higher Bandwidth:** The aircraft reacts faster to elevator commands.
- **Optimal Response:** A damping of 0.52 is close to the "quickest settling" range without excessive overshoot, striking the perfect balance between **static stability** and **dynamic maneuverability**.

This confirms that the SAS does not just stabilize the aircraft, but "sculpts" its handling qualities to meet the precise requirements of a high-performance mission.

5.9.3 Note on Lateral SAS (Yaw Damper)

While the current numerical implementation focuses on the longitudinal axis, a similar approach is required for the lateral-directional axis. As identified in Section 3.3, the **Spiral mode** is unstable ($\lambda > 0$).

- **Theoretical Solution:** A **Yaw Damper** using yaw rate feedback (r) to the rudder (δ_r) would shift the spiral pole into the Left-Half Plane.
- **Design Philosophy:** By choosing a gain K_r, we modify the N_r derivative. The Root Locus analysis in Section 3.4 confirms that a relatively small gain is sufficient to stabilize the spiral divergence without compromising the Dutch Roll damping, effectively bringing all lateral modes to a "Hard" Level 1 status.

5.10 Simulink 6-DOF Implementation and Non-Linear Testing

To validate the performance of the Stability Augmentation System (SAS) beyond the linear approximations of the state-space model, the control laws were implemented in a high-fidelity 6-Degrees of Freedom (6-DOF) non-linear Simulink environment. This approach allows for the evaluation of the C3 aircraft's response under realistic aerodynamic coupling and actuator constraints.

5.10.1 Detailed Architecture of the Feedback Loop

The closed-loop architecture integrates the full non-linear Equations of Motion (EOM) with the designed state feedback controllers. The longitudinal feedback loop was constructed using a precise sequence of blocks to process the state variables:

1. **State Extraction (Bus Selector):** The comprehensive 6-DOF state bus is filtered to extract the specific linearized longitudinal state vector $\mathbf{x}_{lon} = [\Delta u, \Delta w, \Delta q, \Delta\theta]^T$.
2. **Unit Delay:** Placed immediately after the selector, this block breaks algebraic loops inherent in continuous-time feedback systems and holds the initial trim conditions at $t = 0$.
3. **Full State Feedback Gain ($\mathbf{K}_{lon}$):** The extracted state vector is multiplied by the gain matrix computed in Section 3.6 to generate the corrective elevator signal ($\Delta\delta_e$).
4. **Reference Command (N_r):** A step block provides the pilot's target pitch command, adjusted by a pre-compensation gain to ensure zero steady-state error.

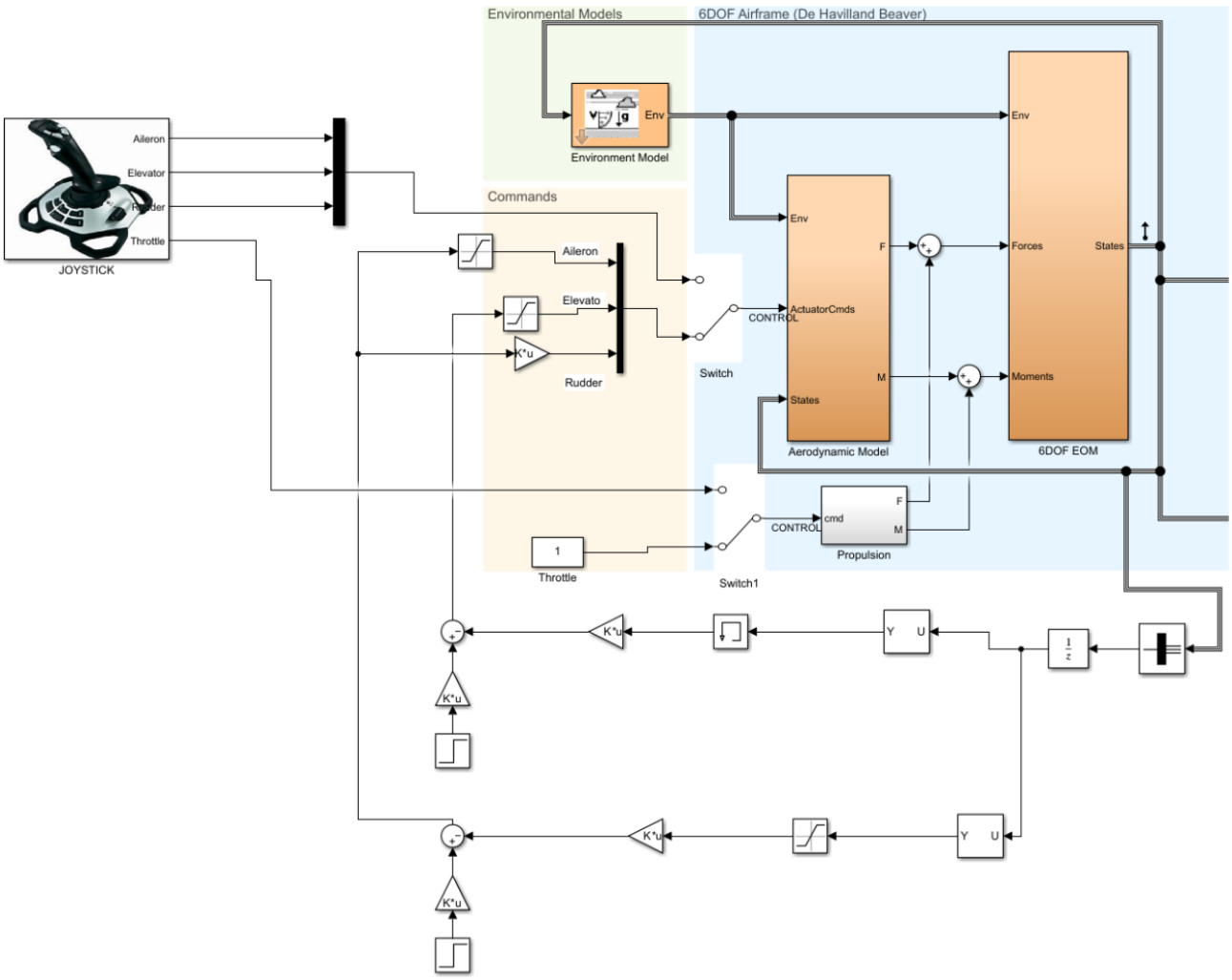


Figure 20: Complete 6-DOF Simulink architecture for the C3 aircraft, highlighting the State Feedback loop and non-linear aerodynamic plant.

5.10.2 Initial Configuration and Convergence Issues

Transitioning from a linearized model to a full 6-DOF environment introduces significant numerical challenges. Initial attempts to run the simulation often result in solver instabilities due to high-frequency dynamics. To resolve this, several critical parameters were adjusted:

- **Initial Conditions (IC) Synchronization:** The simulation must be initialized exactly at the aircraft's trim state ($u_0 = 178$ m/s, $w_0 = 0$ m/s, $h = 10\,668$ m, $\theta_0 = 0°$). The Unit Delay blocks and integrators are rigorously synchronized with these values to prevent mathematical discontinuities and immediate solver crashes at initialization.
- **Solver Precision:** The default variable-step solver is often insufficient for rapid derivative changes in the non-linear aerodynamic equations. A Fixed-step solver was configured with a reduced step size of 0.0005 seconds to handle the rapid transients and ensure numerical stability.

5.10.3 Non-linear Constraints and Actuator Saturation

The high-fidelity non-linear model exhibits extreme sensitivity to unconstrained inputs. To prevent the simulation from reaching numerical singularities or unrealistic flight states, physical limiters were implemented in the control loop:

- **Actuator Saturation:** The elevator deflection (δ_e) was strictly limited to ± 0.15 rad to represent physical mechanical stops and prevent high Angle of Attack (AoA) stalls that the linear model cannot predict.

- **State Saturation:** Vector saturation blocks were added to the feedback path to limit the perceived pitch (θ) and bank (ϕ) angles to ± 0.2 rad. This prevents divergent command loops if the aircraft encounters severe perturbations.

5.10.4 Results and Stability Observations

The integration of the longitudinal SAS in the 6-DOF environment yielded the following observations:

- **Longitudinal Tracking:** The Pitch Damper effectively stabilizes the non-linear dynamics. The pitch angle (θ) accurately tracks step commands, and the Short Period mode exhibits the expected high damping without excessively saturating the actuators.
- **Lateral Divergence in 6-DOF:** While longitudinal tracking is highly functional, the extended 6-DOF simulation highlights the critical nature of the coupled lateral-directional dynamics. As mathematically predicted by the positive real eigenvalue ($\lambda_{spiral} = +0.0217$), the unaugmented roll (ϕ) and yaw (r) angles exhibit a slow divergent drift during extended simulations (starting around $t = 15$ s).

This confirms that while the longitudinal state feedback successfully meets tracking objectives within the linear regime, the inherent spiral instability of the C3 model requires the concurrent implementation of a lateral SAS (Yaw Damper) to sustain long-duration, fully coupled 6-DOF flight without physical divergence.

6 Conclusion

This comprehensive study has successfully characterized the dynamic stability and flying qualities of the C3 aircraft, bridging the gap between theoretical flight mechanics and modern control systems engineering. By applying the Small Perturbation Theory, the complex, coupled 6-Degrees of Freedom (6-DOF) equations of motion were systematically linearized into state-space representations. This rigorous mathematical foundation enabled the extraction of the aircraft's natural eigenvalues, the plotting of Root Loci, and the complete frequency-domain evaluation of its inherent stability modes.

The baseline analysis revealed an aircraft with a generally sound aerodynamic design, largely compliant with the **MIL-F-8785C** Level 1 requirements for a Class I airplane operating in a Category B flight phase (high-altitude cruise). Longitudinally, the aircraft naturally demonstrated a heavily damped Short Period mode ($\zeta_{sp} = 0.72$) and a slow, manageable Phugoid oscillation. Laterally, the airframe exhibited exceptional roll agility ($\tau_R \approx 0.20$ s) and a stable, albeit lightly damped, Dutch Roll mode ($\zeta_{dr} = 0.11$). However, the numerical extraction definitively identified a positive real lateral eigenvalue ($\lambda_{spiral} = +0.0217$), confirming an inherent Spiral mode instability. While its slow doubling time ($T_2 \approx 32$ s) technically borders on Level 1 criteria, it represents a significant degradation in pilot workload for extended cruise operations.

To elevate the aircraft's handling qualities from merely "acceptable" to mathematically optimized, a **Stability Augmentation System (SAS)** was designed using full-state feedback and Ackermann's pole placement methodology. Longitudinally, the implementation of a Pitch Damper allowed for the precise tuning of the Short Period mode. By reducing the damping ratio to 0.52 and increasing the natural frequency to 2.95 rad/s, the control law successfully traded excessive static stiffness for enhanced dynamic maneuverability and a faster bandwidth response.

The transition to a high-fidelity **6-DOF non-linear Simulink environment** served as the ultimate validation of the theoretical designs. While the longitudinal state feedback performed flawlessly—ensuring precise pitch tracking and rapid stabilization under saturated actuator conditions—the fully coupled simulation exposed the critical nature of the lateral-directional deficiencies. The analytically predicted spiral divergence manifested visibly after 15 seconds of flight, confirming that longitudinal augmentation alone is insufficient for realistic, hands-off operations.

In conclusion, the C3 model possesses a robust aerodynamic baseline that satisfies fundamental safety and stability criteria. However, to fully realize its operational potential and guarantee a minimal pilot workload across all flight envelopes, the integration of a comprehensive Flight Control System (FCS)—incorporating both the validated Pitch Damper and a requisite Yaw Damper—is absolutely essential. This project ultimately highlights the indispensable role of modern control theory in transforming a naturally imperfect airframe into a highly capable, mission-ready aircraft.

7 Appendix

7.1 Mathematical Demonstrations (Chap. 3)

7.1.1 Speed-Force Derivative (X_u)

The derivative X_u represents the variation of the axial force with respect to the forward flight speed. By definition, it is expressed as:

$$X_u \equiv \frac{1}{m}\frac{\partial X}{\partial u}$$

Using the aerodynamic force expression $X = \frac{1}{2}\rho V^2 S C_x$ and substituting the total velocity with the equilibrium speed plus a perturbation ($V = U_0 + \Delta u$), we have:

$$X_u = \frac{1}{m}\frac{\partial}{\partial u}\left[\frac{1}{2}\rho(U_0+\Delta u)^2 S\left(C_{x_0} + \frac{\partial C_x}{\partial u}\Delta u\right)\right]$$

By factoring out U_0 and neglecting higher-order terms (Δu^2), the equation is linearized as follows:

$$X_u \approx \frac{\frac{1}{2}\rho U_0^2 S}{m}\frac{\partial}{\partial u}\left[\left(1+2\frac{\Delta u}{U_0}\right)\left(C_{x_0} + \frac{\partial C_x}{\partial u}\Delta u\right)\right]$$

After performing the derivation with respect to u and substituting the dynamic pressure $\bar{q} = \frac{1}{2}\rho U_0^2$, the expression becomes:

$$\begin{aligned} X_u &= \frac{\bar{q}S}{mU_0}\left(2C_{x_0} + U_0\frac{\partial C_x}{\partial u}\right) \\ &= \frac{\bar{q}S}{mU_0}\left(2C_{x_0} + \frac{\partial C_x}{\partial(u/U_0)}\right) \end{aligned} \tag{78}$$

Aerodynamic Coefficient Expansion The axial force coefficient C_x is defined by the balance between thrust (C_T) and drag (C_D):

$$C_x = C_T - C_D\cos\alpha + C_L\sin\alpha \approx C_T - C_D + C_L\alpha$$

The variation of this coefficient with respect to speed incorporates compressibility effects (Mach number M):

$$\frac{\partial C_x}{\partial u} = \frac{\partial C_T}{\partial u} - \frac{\partial C_D}{\partial u} = \frac{\partial C_T}{\partial u} - M\frac{\partial C_D}{\partial M}$$

Assuming the variation of thrust with speed follows the relationship $\frac{\partial C_T}{\partial u} = -2C_{T_0}$, we obtain:

$$\frac{\partial C_x}{\partial u} = -2C_{T_0} - MC_{D_M}$$

Final Dimensional Expression Substituting these components into the X_u equation:

$$X_u = \frac{\bar{q}S}{mU_0}\left(2C_{x_0} - 2C_{T_0} - MC_{D_M}\right)$$

Knowing that at equilibrium (trim), the thrust coefficient is defined as $C_{T_0} = C_{D_0} + C_{L_0}\tan\theta_0$. For level flight where $\theta_0 \approx 0$, which implies $C_{x_0} \approx C_{T_0} - C_{D_0}$, the formula simplifies to its standard form:

$$X_u = -\frac{\bar{q}S}{mU_0}(2C_{D_0} + MC_{D_M}) \tag{79}$$

7.1.2 Angle of Attack-Force Derivative (X_w)

The derivative X_w represents the variation of the axial force with respect to the vertical velocity perturbation (w). By definition:

$$X_w \equiv \frac{1}{m}\frac{\partial X}{\partial w} \tag{80}$$

For small perturbations, the angle of attack α relates to the vertical velocity w and the forward speed U_0 as $\alpha \approx w/U_0$. Thus, we can perform a change of variables:

$$X_w = \frac{1}{m}\frac{\partial X}{\partial(\alpha U_0)} = \frac{1}{mU_0}\frac{\partial X}{\partial \alpha} \tag{81}$$

Substituting the aerodynamic force expression $X = \bar{q}SC_x$:

$$X_w = \frac{\bar{q}S}{mU_0}\frac{\partial C_x}{\partial \alpha} \tag{82}$$

Aerodynamic Coefficient Expansion The axial force coefficient C_x is expanded using the thrust (C_T), drag (C_D), and lift (C_L) coefficients:

$$C_x = C_T - C_D\cos\alpha + C_L\sin\alpha \approx C_T - C_D + C_L\alpha \tag{83}$$

Using the parabolic drag polar $C_D = C_{D_0} + \frac{C_L^2}{\pi e AR}$, the derivative of C_x with respect to α is:

$$\begin{aligned} \frac{\partial C_x}{\partial \alpha} &= \frac{\partial}{\partial \alpha}\left(C_T - C_{D_0} - \frac{C_L^2}{\pi e AR} + C_L\alpha\right) \\ &= -\frac{1}{\pi e AR}\left(2C_L\frac{\partial C_L}{\partial \alpha}\right) + \frac{\partial(C_L\alpha)}{\partial \alpha} \end{aligned} \tag{84}$$

At the equilibrium state (where $\alpha \approx 0$ and $C_L = C_{L_0}$), this simplifies to:

$$\frac{\partial C_x}{\partial \alpha} = C_{L_0} - \frac{2C_{L_0}}{\pi e AR}\left(\frac{\partial C_L}{\partial \alpha}\right) \tag{85}$$

Final Dimensional Expression Combining these results, the final expression for X_w is obtained:

$$\begin{aligned} X_w &= \frac{\bar{q}S}{mU_0}\left(C_{L_0} - \frac{2C_{L_0}}{\pi eAR}\frac{\partial C_L}{\partial \alpha}\right) \\ &= \frac{\bar{q}S}{mU_0}\left(C_{L_0} - \frac{2C_{L_0}C_{L_\alpha}}{\pi eAR}\right) \end{aligned} \tag{86}$$

7.1.3 Speed-Lift Derivative (Z_u)

The derivative Z_u represents the variation of the vertical force with respect to the forward flight speed. By definition:

$$Z_u \equiv \frac{1}{m}\frac{\partial Z}{\partial u} \tag{87}$$

Using the vertical aerodynamic force expression $Z = \frac{1}{2}\rho V^2 S C_z$ and substituting $V = U_0 + u(t)$, we linearize the expression:

$$Z_u = \frac{1}{m}\frac{\partial}{\partial u}\left[\frac{1}{2}\rho(U_0 + u(t))^2 S\left(C_{z_0} + \frac{\partial C_z}{\partial u}u\right)\right] \tag{88}$$

Aerodynamic Coefficient Expansion In the body-fixed frame, the vertical force coefficient is the negative of the lift coefficient, $C_z = -C_L$. The variation of C_z with respect to speed incorporates compressibility effects through the Mach number M:

$$\frac{\partial C_z}{\partial u} = -\frac{\partial C_L}{\partial u} = -M\frac{\partial C_L}{\partial M} = -MC_{L_M} \tag{89}$$

Substituting this into the dimensional form using dynamic pressure $\bar{q} = \frac{1}{2}\rho U_0^2$ yields:

$$Z_u = -\frac{\bar{q}S}{mU_0}(2C_{L_0} + MC_{L_M}) \tag{90}$$

Compressibility Effects According to the Prandtl-Glauert transformation for subsonic flow, C_L varies with Mach number as $C_L = \frac{C_{L,M=0}}{\sqrt{1-M^2}}$. The derivative with respect to M is calculated as:

$$\begin{aligned} C_{L_M} = \frac{\partial C_L}{\partial M} &= \frac{\partial}{\partial M}\left(\frac{C_{L,M=0}}{\sqrt{1-M^2}}\right) \\ &= \frac{C_{L_0}\cdot M}{(1-M^2)^{3/2}} \approx \frac{C_{L_0}M}{1-M^2} \end{aligned} \tag{91}$$

Final Dimensional Expression Substituting the compressibility term back into the expression for Z_u:

$$Z_u = -\frac{\bar{q}S}{mU_0}\left(2C_{L_0} + \frac{M^2}{1-M^2}C_{L_0}\right) \tag{92}$$

7.1.4 Vertical Velocity-Force Derivative (Z_w)

The derivative Z_w represents the variation of the vertical force with respect to the vertical velocity perturbation w. By definition:

$$Z_w \equiv \frac{1}{m}\frac{\partial Z}{\partial w} \tag{93}$$

For small perturbations, the angle of attack α relates to the vertical velocity w and forward speed U_0 as $\alpha \approx w/U_0$. This allows for a change of variables:

$$Z_w = \frac{1}{m}\frac{\partial Z}{\partial(\alpha U_0)} = \frac{1}{mU_0}\frac{\partial Z}{\partial \alpha} \tag{94}$$

Substituting the vertical force expression $Z = \frac{1}{2}\rho V^2 S C_z$ and linearizing around the trim condition:

$$Z_w = \frac{1}{m}\frac{\partial}{\partial \alpha}\left[\frac{1}{2}\rho(U_0 + u(t))^2 S\left(C_{z_0} + \frac{\partial C_z}{\partial \alpha}\alpha\right)\right] \tag{95}$$

Aerodynamic Coefficient Expansion The vertical force coefficient C_z in the body-fixed frame is derived from the lift and drag components:

$$C_z = -C_D\sin\alpha - C_L\cos\alpha \approx -C_D\alpha - C_L \tag{96}$$

Differentiating with respect to the angle of attack α:

$$\frac{\partial C_z}{\partial \alpha} = -C_D - \frac{\partial C_L}{\partial \alpha} \tag{97}$$

At the equilibrium state (trim), the drag and lift slope are represented by C_{D_0} and C_{L_α} respectively:

$$\frac{\partial C_z}{\partial \alpha} = -(C_{D_0} + C_{L_\alpha}) \tag{98}$$

Final Dimensional Expression Substituting the dynamic pressure $\bar{q} = \frac{1}{2}\rho U_0^2$ and the coefficient derivative back into the expression for Z_w yields:

$$Z_w = -\frac{\bar{q}S}{mU_0}(C_{D_0} + C_{L_\alpha}) \tag{99}$$

7.1.5 Vertical Acceleration-Force Derivative ($Z_{\dot{w}}$)

The derivative $Z_{\dot{w}}$ represents the variation of the vertical force with respect to the rate of change of vertical velocity ($\dot{w}$). By definition:

$$Z_{\dot{w}} \equiv \frac{1}{m}\frac{\partial Z}{\partial \dot{w}} \tag{100}$$

Considering the relationship between vertical velocity and angle of attack $\alpha \approx w/U_0$, it follows that

$\dot{w} \approx \dot{\alpha} U_0$. This allows for the following change of variables:

$$Z_{\dot{w}} = \frac{1}{m}\frac{\partial Z}{\partial(\dot{\alpha}U_0)} = \frac{1}{mU_0}\frac{\partial Z}{\partial \dot{\alpha}} \tag{101}$$

Substituting the vertical force expression $Z = \frac{1}{2}\rho V^2 S C_z$ and linearizing around the trim condition:

$$Z_{\dot{w}} = \frac{1}{m}\frac{\partial}{\partial \dot{\alpha}}\left[\frac{1}{2}\rho(U_0 + u(t))^2 S\left(C_{z_0} + \frac{\partial C_z}{\partial \alpha}\alpha\right)\right] \tag{102}$$

Aerodynamic Coefficient Expansion The vertical force coefficient C_z is approximated as:

$$C_z = -C_D \sin\alpha - C_L \cos\alpha \approx -C_D\alpha - C_L \tag{103}$$

Differentiating with respect to the rate of change of angle of attack $\dot{\alpha}$:

$$\frac{\partial C_z}{\partial \dot{\alpha}} = -C_D - \frac{\partial C_L}{\partial \dot{\alpha}} = -(C_{D_0} + C_{L_{\dot{\alpha}}}) \tag{104}$$

Final Dimensional Expression Substituting the dynamic pressure $\bar{q} = \frac{1}{2}\rho U_0^2$ and the coefficient derivatives, the final expression for $Z_{\dot{w}}$ is obtained:

$$Z_{\dot{w}} = \frac{\bar{q}S\bar{c}}{2mU_0^2}(C_{D_0} + C_{L_{\dot{\alpha}}}) \tag{105}$$

7.1.6 Pitch Rate-Force Derivative (Z_q)

The derivative Z_q represents the variation of the vertical force with respect to the pitch rate q. By definition:

$$Z_q \equiv \frac{1}{m}\frac{\partial Z}{\partial q} \tag{106}$$

By normalizing the pitch rate and using the vertical aerodynamic force expression $Z = \frac{1}{2}\rho V^2 S C_z$, we linearize the equation around the steady-state flight condition:

$$Z_q = \frac{1}{mU_0}\frac{\partial Z}{\partial q} = \frac{1}{m}\frac{\partial}{\partial q}\left[\frac{\frac{1}{2}\rho(U_0 + u(t))^2 S\left(C_{z_0} + \frac{\partial C_z}{\partial q}q\right)}{\partial q}\right] \tag{107}$$

Aerodynamic Coefficient Expansion The variation of the vertical force coefficient with respect to the pitch rate is directly related to the lift coefficient derivative C_{L_q}. In the stability frame, we assume that the lift generated by the pitch rate (primarily from the horizontal stabilizer) is the dominant factor.

Final Dimensional Expression Substituting the dynamic pressure $\bar{q} = \frac{1}{2}\rho U_0^2$ and the reference chord $\bar{c}$, the final dimensional expression for Z_q is obtained:

$$Z_q = \frac{\bar{q}S\bar{c}}{2mU_0}[C_{L_q}] \tag{108}$$

7.1.7 Pitching Moment and Control Derivatives

Following the same linearization methodology and Taylor series expansion applied to the previous force equations, the dimensional pitching moment derivatives and control effectiveness derivatives are obtained as follows:

Pitching Moment Derivatives These derivatives characterize the aerodynamic stiffness and damping of the aircraft around the lateral axis, normalized by the pitch inertia I_{yy}:

$$M_u = \frac{\bar{q}S\bar{c}}{I_{yy}U_0}[C_{m_u}] \tag{109}$$

$$M_w = \frac{\bar{q}S\bar{c}}{I_{yy}U_0}[C_{m_\alpha}] \tag{110}$$

$$M_{\dot{w}} = \frac{\bar{q}S\bar{c}^2}{2I_{yy}U_0^2}[C_{m_{\dot{\alpha}}}] \tag{111}$$

$$M_q = \frac{\bar{q}S\bar{c}^2}{2I_{yy}U_0}[C_{m_q}] \tag{112}$$

Longitudinal Control Derivatives The following derivatives form the longitudinal control matrix $\mathbf{B}_{lon}$, representing the response to elevator deflection (δ_e) and throttle variation (δ_T):

$$X_{\delta_e} = \frac{\bar{q}S}{mU_0}C_{D_{\delta_e}} \qquad X_{\delta_T} = \frac{\bar{q}S}{mU_0}C_{T_{\delta_T}} \tag{113}$$

$$Z_{\delta_e} = \frac{\bar{q}S}{mU_0}C_{L_{\delta_e}} \qquad Z_{\delta_T} = \frac{\bar{q}S}{mU_0}C_{L_{\delta_T}} \tag{114}$$

$$M_{\delta_e} = \frac{\bar{q}S\bar{c}}{I_{yy}U_0}C_{M_{\delta_e}} \qquad M_{\delta_T} = \frac{\bar{q}S\bar{c}}{I_{yy}U_0}C_{M_{\delta_T}} \tag{115}$$

7.2 MATLAB Code

```
% ==========================================================================
% Tutorial 2: Aircraft Stability Modes - Model C3
% ==========================================================================
% PURPOSE:
%   This script performs a complete linearized stability analysis of the C3
%   aircraft model. It covers:
%     - Part 1 : Longitudinal stability modes (Short Period, Phugoid)
%     - Part 2 : Lateral stability modes (Dutch Roll, Roll, Spiral)
%     - Time-domain visualization of each mode via modal reconstruction
%     - Frequency-domain analysis: Bode diagrams for Elevator, Throttle,
%       Rudder, and Aileron inputs
%     - Root Locus analysis for key transfer functions
%     - SAS (Stability Augmentation System) design via Ackermann pole placement
%     - Short Period approximation using a reduced 2x2 state-space model
%
% STATE VECTOR (Longitudinal): x = [u, w, q, theta]
%   u     : axial velocity perturbation       [m/s]
%   w     : vertical velocity perturbation    [m/s]
%   q     : pitch rate                        [rad/s]
%   theta : pitch angle                       [rad]
%
% STATE VECTOR (Lateral): x = [v, p, r, phi]
%   v     : lateral velocity perturbation     [m/s]
%   p     : roll rate                         [rad/s]
%   r     : yaw rate                          [rad/s]
%   phi   : bank angle                        [rad]
% ==========================================================================

clear all; clc;

% ========================================================================
% PART 1: LONGITUDINAL STABILITY MODES
% ========================================================================
fprintf('\n==========================================================\n');
fprintf('       LONGITUDINAL ANALYSIS - C3\n');
fprintf('==========================================================\n');

% ------------------------------------------------------------------------
% Longitudinal state matrix A_long
% Rows/columns correspond to states: [u, w, q, theta]
% Each entry is a stability/aerodynamic derivative (linearized model)
% ------------------------------------------------------------------------
A_long = [ -0.0118,   0.0145,   0.0000,  -9.8100;
           -0.1396,  -1.3089, 178.0000,   0.0000;
            0.0007,  -0.0197,  -2.4445,   0.0000;
            0.0000,   0.0000,   1.0000,   0.0000 ];

%% ----- Question 1.1: Characteristic Equation
% poly() returns the coefficients of the characteristic polynomial
% det(lambda*I - A) = lambda^4 + a3*lambda^3 + a2*lambda^2 + a1*lambda + a0
poly_long = poly(A_long);
fprintf('\n--- Question 1.1: Characteristic Equation ---\n');
disp('Coefficients (from lambda^4 to lambda^0):');
disp(poly_long);
fprintf('Equation: s^4 + %.4fs^3 + %.4fs^2 + %.4fs + %.4f = 0\n', ...
    poly_long(2), poly_long(3), poly_long(4), poly_long(5));

%% ----- Question 1.2: Eigenvalues (Roots)
% eig() computes the eigenvalues of A_long, which are the poles of the system.
% Complex conjugate pairs correspond to oscillatory modes.
% Negative real part => stable mode.
lambdas_long = eig(A_long);
fprintf('\n--- Question 1.2: Eigenvalues ---\n');
```

```
disp('The roots of the characteristic equation are:');
disp(lambdas_long);

%% ----- Question 1.3: Eigenvectors
% [V, D] = eig(A) returns:
%   V : matrix of eigenvectors (each column is a mode shape)
%   D : diagonal matrix of eigenvalues
% Eigenvectors describe the relative participation of each state in a mode.
[V_long, D_long] = eig(A_long);
fprintf('\n--- Question 1.3: Eigenvectors ---\n');
disp('Longitudinal eigenvector matrix:');
disp(V_long);

%% ----- Question 1.4: Short Period Mode
% The Short Period (SP) mode is identified as the eigenvalue pair with the
% highest imaginary part (fastest oscillation frequency).
% Sort by descending |imag| -> indices 1 and 2 = SP pair.
[~, idx_l] = sort(abs(imag(lambdas_long)), 'descend');
lambda_SP = lambdas_long(idx_l(1));          % One pole of the SP pair
wn_SP     = abs(lambda_SP);                  % Natural frequency  [rad/s]
zeta_SP   = -real(lambda_SP) / wn_SP;        % Damping ratio      [-]
fprintf('\n--- Question 1.4: Short Period Mode (SP) ---\n');
fprintf('Natural frequency (wn_SP): %.4f rad/s\n', wn_SP);
fprintf('Damping ratio (zeta_SP): %.4f\n', zeta_SP);

%% ----- Question 1.5: Phugoid Mode
% The Phugoid (PH) mode is identified as the eigenvalue pair with the
% smallest imaginary part (slowest oscillation, lightly damped).
% Sort by descending |imag| -> indices 3 and 4 = PH pair.
lambda_PH = lambdas_long(idx_l(3));          % One pole of the PH pair
wn_PH     = abs(lambda_PH);                  % Natural frequency  [rad/s]
zeta_PH   = -real(lambda_PH) / wn_PH;        % Damping ratio      [-]
fprintf('\n--- Question 1.5: Phugoid Mode (PH) ---\n');
fprintf('Natural frequency (wn_PH): %.4f rad/s\n', wn_PH);
fprintf('Damping ratio (zeta_PH): %.4f\n', zeta_PH);


% ========================================================================
% PART 2: LATERAL STABILITY MODES
% ========================================================================
fprintf('\n\n==========================================================\n');
fprintf('          LATERAL ANALYSIS - C3\n');
fprintf('==========================================================\n');

% ------------------------------------------------------------------------
% Lateral state matrix A_lat
% Rows/columns correspond to states: [v, p, r, phi]
% ------------------------------------------------------------------------
A_lat = [ -0.2358,  -0.1135, -177.4137,   9.8100;
          -0.3482,  -4.9509,    3.9353,   0.0000;
           0.0828,   0.1562,   -0.5078,   0.0000;
           0.0000,   1.0000,    0.0000,   0.0000 ];

%% ----- Question 3.1: Characteristic Equation
poly_lat = poly(A_lat);
fprintf('\n--- Question 3.1: Characteristic Equation ---\n');
disp('Coefficients (from lambda^4 to lambda^0):');
disp(poly_lat);
fprintf('Equation: s^4 + %.4fs^3 + %.4fs^2 + %.4fs + %.4f = 0\n', ...
    poly_lat(2), poly_lat(3), poly_lat(4), poly_lat(5));

%% ----- Question 3.2: Eigenvalues (Roots)
% The lateral system has 4 eigenvalues:
%   - 1 complex conjugate pair : Dutch Roll (oscillatory)
%   - 2 real values            : Roll mode (fast) and Spiral mode (slow)
```

```
lambdas_lat = eig(A_lat);
fprintf('\n--- Question 3.2: Eigenvalues ---\n');
disp('The lateral eigenvalues are:');
disp(lambdas_lat);

%% ----- Question 3.3: Eigenvectors
[V_lat, D_lat] = eig(A_lat);
fprintf('\n--- Question 3.3: Eigenvectors ---\n');
disp('The lateral eigenvector matrix is:');
disp(V_lat);

%% ----- Question 3.4 & 3.5: Spiral and Rolling Modes
% Extract only the real eigenvalues (purely real = non-oscillatory modes).
% The Roll mode has the largest magnitude (fast convergence).
% The Spiral mode has the smallest magnitude (very slow, near zero).
reelles = lambdas_lat(imag(lambdas_lat) == 0);
[~, idx_roll]   = max(abs(reelles));   % Index of the Roll mode (fastest)
[~, idx_spiral] = min(abs(reelles));   % Index of the Spiral mode (slowest)
lambda_roll   = reelles(idx_roll);
lambda_spiral = reelles(idx_spiral);

% Spiral mode: if lambda > 0 (unstable), the doubling time is 1/lambda.
%              if lambda < 0 (stable), the half-life is 1/|lambda|.
fprintf('\n--- Question 3.4: Spiral Mode ---\n');
fprintf('Eigenvalue lambda_spiral: %.4f\n', lambda_spiral);
fprintf('Half-life/doubling time: %.2f s\n', 1/abs(lambda_spiral));

% Roll mode: purely exponential decay; time constant tau = 1/|lambda|.
fprintf('\n--- Question 3.5: Rolling Mode ---\n');
fprintf('Eigenvalue lambda_roll: %.4f\n', lambda_roll);
fprintf('Time constant (tau): %.4f s\n', 1/abs(lambda_roll));

%% ----- Question 3.6: Dutch Roll Mode
% The Dutch Roll mode is the complex conjugate pair of lateral eigenvalues.
% It is an oscillatory coupling of yaw (r) and roll (phi).
complexes = lambdas_lat(imag(lambdas_lat) ~= 0);
lambda_DR = complexes(1);
wn_DR    = abs(lambda_DR);               % Natural frequency [rad/s]
zeta_DR  = -real(lambda_DR) / wn_DR;     % Damping ratio     [-]

fprintf('\n--- Question 3.6: Dutch Roll Mode (Oscillatory) ---\n');
fprintf('Natural frequency (wn_DR): %.4f rad/s\n', wn_DR);
fprintf('Damping ratio (zeta_DR): %.4f\n', zeta_DR);


% ========================================================================
% TD3: STABILITY MODE PLOTS (TIME-DOMAIN VISUALIZATION)
% ========================================================================
% Each mode is isolated using modal decomposition:
%   x(t) = sum_i [ c_i * v_i * exp(lambda_i * t) ]
% where:
%   v_i      = i-th eigenvector
%   lambda_i = i-th eigenvalue
%   c_i      = modal coordinate = (V^-1 * x0)_i
% Only the eigenvalue(s) belonging to a given mode are kept to visualize
% the contribution of that mode in isolation.
fprintf('\n==========================================================\n');
fprintf('          MODE VISUALIZATION (PLOTS)\n');
fprintf('==========================================================\n');


%% --- 1. LONGITUDINAL CONFIGURATION ---

% Arbitrary initial condition to excite all longitudinal modes simultaneously.
% States: [delta_u (m/s), delta_w (m/s), q (rad/s), theta (rad)]
```

```
X0_long = [10; 5; 0.1; 0.1];

% Modal coordinates: C = V^-1 * X0
% Each c_i weights the contribution of the i-th mode to the initial condition.
C_long = V_long \ X0_long;

% Sort eigenvalues by descending |imag| to identify SP vs PH indices.
[~, idx_l_sort] = sort(abs(imag(lambdas_long)), 'descend');
idx_SP = idx_l_sort(1:2);   % Eigenvalue indices for Short Period
idx_PH = idx_l_sort(3:4);   % Eigenvalue indices for Phugoid


%% --- A) SHORT PERIOD MODE ---
% Time window: 0-5 s (SP mode decays in a few seconds).
t_SP = 0:0.01:5;

% Modal reconstruction using only the SP eigenvalue pair.
% Sol_SP(row, :) gives the time evolution of each state due to SP only.
Sol_SP = real(V_long(:,idx_SP) * (C_long(idx_SP) .* exp(lambdas_long(idx_SP) * t_SP)));

figure('Name', 'Longitudinal: Short Period Mode');
subplot(2,1,1);
plot(t_SP, Sol_SP(2,:), 'r', 'LineWidth', 1.5); grid on;
title('\Delta w (Vertical Velocity) - Dominant in Short Period'); ylabel('m/s');
subplot(2,1,2);
plot(t_SP, Sol_SP(3,:), 'b', 'LineWidth', 1.5); grid on;
title('q (Pitch Rate) - Dominant in Short Period'); ylabel('rad/s'); xlabel('Time (s)');


%% --- B) PHUGOID MODE ---
% Time window: 0-1000 s (PH mode has a very long period, ~60s, and is lightly damped).
t_PH = 0:0.5:1000;

% Modal reconstruction using only the PH eigenvalue pair.
Sol_PH = real(V_long(:,idx_PH) * (C_long(idx_PH) .* exp(lambdas_long(idx_PH) * t_PH)));

figure('Name', 'Longitudinal: Phugoid Mode');
subplot(2,1,1);
plot(t_PH, Sol_PH(1,:), 'r', 'LineWidth', 1.5); grid on;
title('\Delta u (Axial Velocity) - Dominant in Phugoid'); ylabel('m/s');
subplot(2,1,2);
plot(t_PH, rad2deg(Sol_PH(4,:)), 'b', 'LineWidth', 1.5); grid on;
title('\Delta \theta (Pitch Angle) - Dominant in Phugoid'); ylabel('deg'); xlabel('Time
    (s)');


%% --- 2. LATERAL CONFIGURATION ---

% Arbitrary initial condition to excite all lateral modes simultaneously.
% States: [v (m/s), p (rad/s), r (rad/s), phi (rad)]
X0_lat = [5; 0.5; 0.1; 0.1];

% Modal coordinates for the lateral system.
C_lat = V_lat \ X0_lat;

% Separate complex and real eigenvalue indices for mode isolation.
complex_indices = find(imag(lambdas_lat) ~= 0);   % Dutch Roll (complex pair)
real_indices    = find(imag(lambdas_lat) == 0);    % Roll and Spiral (real)

idx_DR = complex_indices;   % Both Dutch Roll indices

% Sort real eigenvalue indices by descending magnitude: Roll (large) vs Spiral (small).
[~, sort_real] = sort(abs(lambdas_lat(real_indices)), 'descend');
idx_Roll   = real_indices(sort_real(1));   % Roll mode index (fastest)
idx_Spiral = real_indices(sort_real(2));   % Spiral mode index (slowest)
```

```
%% --- C) ROLLING MODE ---
% Time window: 0-3 s (Roll mode is a fast first-order exponential decay).
t_Roll = 0:0.01:3;

% Modal reconstruction using only the Roll eigenvalue.
Sol_Roll = real(V_lat(:,idx_Roll) * (C_lat(idx_Roll) .* exp(lambdas_lat(idx_Roll) *
    t_Roll)));

figure('Name', 'Lateral: Rolling Mode');
subplot(2,1,1);
plot(t_Roll, Sol_Roll(2,:), 'r', 'LineWidth', 1.5); grid on;
title('p (Roll Rate) - Dominant in Rolling'); ylabel('rad/s');
subplot(2,1,2);
plot(t_Roll, rad2deg(Sol_Roll(4,:)), 'b', 'LineWidth', 1.5); grid on;
title('\phi (Bank Angle)'); ylabel('deg'); xlabel('Time (s)');


%% --- D) DUTCH ROLL MODE ---
% Time window: 0-60 s (enough to observe several oscillation cycles).
t_DR = 0:0.05:60;

% Modal reconstruction using only the Dutch Roll complex eigenvalue pair.
Sol_DR = real(V_lat(:,idx_DR) * (C_lat(idx_DR) .* exp(lambdas_lat(idx_DR) * t_DR)));

figure('Name', 'Lateral: Dutch Roll');
subplot(3,1,1);
plot(t_DR, Sol_DR(1,:), 'k', 'LineWidth', 1.5); grid on;
title('\Delta v (Lateral Velocity/Sideslip)'); ylabel('m/s');
subplot(3,1,2);
plot(t_DR, Sol_DR(3,:), 'r', 'LineWidth', 1.5); grid on;
title('r (Yaw Rate)'); ylabel('rad/s');
subplot(3,1,3);
plot(t_DR, rad2deg(Sol_DR(4,:)), 'b', 'LineWidth', 1.5); grid on;
title('\phi (Bank Angle)'); ylabel('deg'); xlabel('Time (s)');


%% --- E) SPIRAL MODE ---
% Time window: 0-1000 s (Spiral mode is extremely slow, diverging or converging
% over many minutes, barely noticeable in the short term).
t_Spiral = 0:1:1000;

% Modal reconstruction using only the Spiral eigenvalue.
Sol_Spiral = real(V_lat(:,idx_Spiral) * (C_lat(idx_Spiral) .* exp(lambdas_lat(idx_Spiral
    ) * t_Spiral)));

figure('Name', 'Lateral: Spiral Mode');
subplot(2,1,1);
plot(t_Spiral, rad2deg(Sol_Spiral(4,:)), 'k', 'LineWidth', 1.5); grid on;
title('\phi (Bank Angle) - Slow evolution'); ylabel('deg');
subplot(2,1,2);
plot(t_Spiral, Sol_Spiral(3,:), 'b', 'LineWidth', 1.5); grid on;
title('r (Yaw Rate)'); ylabel('rad/s'); xlabel('Time (s)');

fprintf('Plots generated: Short Period, Phugoid, Rolling, Dutch Roll, Spiral.\n');


% ======================================================================
% SETTLING TIME SUMMARY (FINAL PRINT)
% ======================================================================
% Theoretical settling time based on the 5% criterion:
%   Ts = 3 / |real(lambda)|
% This approximation applies to second-order and higher systems.
ts_sp     = 3 / abs(real(lambda_SP));
```

```
ts_ph     = 3 / abs(real(lambda_PH));
ts_dr     = 3 / abs(real(lambda_DR));
ts_roll   = 3 / abs(real(lambda_roll));
ts_spiral = 3 / abs(real(lambda_spiral));

fprintf('\n\n============================================================\n');
fprintf('         TIME-DOMAIN STABILITY SUMMARY\n');
fprintf('============================================================\n');

fprintf('\n--- LONGITUDINAL MODES ---\n');
fprintf('Short Period (SP): The response settles in approximately %.2f s\n', ts_sp);
fprintf('Long Period (PH) : The response settles in approximately %.2f s\n', ts_ph);

fprintf('\n--- LATERAL MODES ---\n');
fprintf('Dutch Roll (DR)  : The response settles in approximately %.2f s\n', ts_dr);
fprintf('Rolling          : The response settles in approximately %.2f s\n', ts_roll);
fprintf('Spiral           : The response settles in approximately %.2f s\n', ts_spiral);


% ========================================================================
% TD3: LONGITUDINAL TRANSFER FUNCTIONS (Elevator Response)
% ========================================================================
% The state-space system (A, B, C, D) is used to derive transfer functions
% for each output state (u, w, q, theta) with respect to the elevator input.
% ss2tf(A, B, C, D, iu) returns the TF from input iu to the selected output.
fprintf('\n============================================================\n');
fprintf('       LONGITUDINAL TRANSFER FUNCTIONS - C3\n');
fprintf('============================================================\n');

% ------------------------------------------------------------------------
% Longitudinal control matrix B_long (4x2):
%   Column 1 : Elevator input (delta_e)
%              X_de = 0, Z_de = 0.0083, M_de = -15.8418
%   Column 2 : Throttle input (delta_t)
%              X_dt = 0.12, Z_dt = 0, M_dt = 0
% D_long = 0 (no direct feedthrough)
% ------------------------------------------------------------------------
B_long = [ 0.0000,  0.1200;
           0.0083,  0.0000;
          -15.8418,  0.0000;
           0.0000,  0.0000 ];
D_long = [0 0];

% Compute TFs for each output state due to elevator (input index 1).
% Output selection vectors [1 0 0 0], [0 1 0 0], etc. pick u, w, q, theta.
[n_u, den]      = ss2tf(A_long, B_long, [1 0 0 0], D_long, 1);   % Delta_u / Delta_de
[n_w, ~]        = ss2tf(A_long, B_long, [0 1 0 0], D_long, 1);   % Delta_w / Delta_de
[n_q, ~]        = ss2tf(A_long, B_long, [0 0 1 0], D_long, 1);   % Delta_q / Delta_de
[n_theta, ~]    = ss2tf(A_long, B_long, [0 0 0 1], D_long, 1);   % Delta_theta / Delta_de

% Display each TF symbolically and print a parameter summary table.
fprintf('\n1. TRANSFER FUNCTION Delta_u(s) / Delta_de(s):\n');
printsys(n_u, den, 's');
analyze_longitudinal_tf(n_u, den, 'u');

fprintf('\n2. TRANSFER FUNCTION Delta_w(s) / Delta_de(s):\n');
printsys(n_w, den, 's');
analyze_longitudinal_tf(n_w, den, 'w');

fprintf('\n3. TRANSFER FUNCTION Delta_q(s) / Delta_de(s):\n');
printsys(n_q, den, 's');
analyze_longitudinal_tf(n_q, den, 'q');

fprintf('\n4. TRANSFER FUNCTION Delta_theta(s) / Delta_de(s):\n');
printsys(n_theta, den, 's');
```

```
analyze_longitudinal_tf(n_theta, den, 'theta');

% Bode diagram: 2x2 grid showing magnitude and phase for all 4 outputs.
figure('Name', 'C3: Longitudinal Responses to Elevator');
subplot(2,2,1); bode(tf(n_u, den));     grid on; title('\Delta u / \Delta \delta_e');
subplot(2,2,2); bode(tf(n_w, den));     grid on; title('\Delta w / \Delta \delta_e');
subplot(2,2,3); bode(tf(n_q, den));     grid on; title('\Delta q / \Delta \delta_e');
subplot(2,2,4); bode(tf(n_theta, den)); grid on; title('\Delta \theta / \Delta \delta_e');


% ========================================================================
% TD3: LONGITUDINAL TRANSFER FUNCTIONS (Throttle Response)
% ========================================================================
% Same approach as for the elevator, but using input index 2 (Throttle).
% The throttle mainly excites the Phugoid mode via the axial velocity (u).
fprintf('\n============================================================\n');
fprintf('   LONGITUDINAL TRANSFER FUNCTIONS (THROTTLE) - C3\n');
fprintf('============================================================\n');

% Compute TFs for each output state due to throttle (input index 2).
[n_u_dt, den_dt]  = ss2tf(A_long, B_long, [1 0 0 0], D_long, 2);   % Delta_u / Delta_dt
[n_w_dt, ~]       = ss2tf(A_long, B_long, [0 1 0 0], D_long, 2);   % Delta_w / Delta_dt
[n_q_dt, ~]       = ss2tf(A_long, B_long, [0 0 1 0], D_long, 2);   % Delta_q / Delta_dt
[n_theta_dt, ~]   = ss2tf(A_long, B_long, [0 0 0 1], D_long, 2);   % Delta_theta / Delta_dt

fprintf('\n1. TRANSFER FUNCTION Delta_u(s) / Delta_dt(s):\n');
printsys(n_u_dt, den_dt, 's');
analyze_longitudinal_tf(n_u_dt, den_dt, 'u');

fprintf('\n2. TRANSFER FUNCTION Delta_w(s) / Delta_dt(s):\n');
printsys(n_w_dt, den_dt, 's');
analyze_longitudinal_tf(n_w_dt, den_dt, 'w');

fprintf('\n3. TRANSFER FUNCTION Delta_q(s) / Delta_dt(s):\n');
printsys(n_q_dt, den_dt, 's');
analyze_longitudinal_tf(n_q_dt, den_dt, 'q');

fprintf('\n4. TRANSFER FUNCTION Delta_theta(s) / Delta_dt(s):\n');
printsys(n_theta_dt, den_dt, 's');
analyze_longitudinal_tf(n_theta_dt, den_dt, 'theta');

% Bode diagram for all 4 throttle-to-output transfer functions.
figure('Name', 'C3: Longitudinal Responses to Throttle');
subplot(2,2,1); bode(tf(n_u_dt, den_dt));     grid on; title('\Delta u / \Delta \delta_t');
subplot(2,2,2); bode(tf(n_w_dt, den_dt));     grid on; title('\Delta w / \Delta \delta_t');
subplot(2,2,3); bode(tf(n_q_dt, den_dt));     grid on; title('\Delta q / \Delta \delta_t');
subplot(2,2,4); bode(tf(n_theta_dt, den_dt)); grid on; title('\Delta \theta / \Delta \delta_t');


% ========================================================================
% LATERAL TRANSFER FUNCTIONS (RUDDER ONLY)
% ========================================================================
% The lateral control matrix B_lat (4x2):
%   Column 1 : Aileron input (delta_a)
%   Column 2 : Rudder input  (delta_r)
% Rudder primarily excites Dutch Roll (yaw + sideslip coupling).
% ------------------------------------------------------------------------
B_lat = [ 0.0660,   0.0000;
          28.1752,  56.3504;
```

```
         -10.4024,   -0.4334;
           0.0000,    0.0000 ];
D_lat = [0 0];

fprintf('\n\n============================================================\n');
fprintf(' LATERAL TRANSFER FUNCTIONS (RUDDER ONLY) - C3\n');
fprintf('============================================================\n');

% Compute TFs for each lateral output due to rudder (input index 2).
[n_v_dr, den_lat] = ss2tf(A_lat, B_lat, [1 0 0 0], D_lat, 2);   % Delta_v / Delta_dr
[n_p_dr, ~]       = ss2tf(A_lat, B_lat, [0 1 0 0], D_lat, 2);   % Delta_p / Delta_dr
[n_r_dr, ~]       = ss2tf(A_lat, B_lat, [0 0 1 0], D_lat, 2);   % Delta_r / Delta_dr
[n_phi_dr, ~]     = ss2tf(A_lat, B_lat, [0 0 0 1], D_lat, 2);   % Delta_phi / Delta_dr

fprintf('\n1. TRANSFER FUNCTION Delta_v(s) / Delta_dr(s):\n');
printsys(n_v_dr, den_lat, 's');
analyze_lateral_tf(n_v_dr, den_lat, 'v');

fprintf('\n2. TRANSFER FUNCTION Delta_p(s) / Delta_dr(s):\n');
printsys(n_p_dr, den_lat, 's');
analyze_lateral_tf(n_p_dr, den_lat, 'p');

fprintf('\n3. TRANSFER FUNCTION Delta_r(s) / Delta_dr(s):\n');
printsys(n_r_dr, den_lat, 's');
analyze_lateral_tf(n_r_dr, den_lat, 'r');

fprintf('\n4. TRANSFER FUNCTION Delta_phi(s) / Delta_dr(s):\n');
printsys(n_phi_dr, den_lat, 's');
analyze_lateral_tf(n_phi_dr, den_lat, 'phi');

% Bode diagram for all 4 rudder-to-output transfer functions.
figure('Name', 'C3: Lateral Responses to Rudder');
subplot(2,2,1); bode(tf(n_v_dr, den_lat));   grid on; title('\Delta v / \Delta \delta_r');
subplot(2,2,2); bode(tf(n_p_dr, den_lat));   grid on; title('\Delta p / \Delta \delta_r');
subplot(2,2,3); bode(tf(n_r_dr, den_lat));   grid on; title('\Delta r / \Delta \delta_r');
subplot(2,2,4); bode(tf(n_phi_dr, den_lat)); grid on; title('\Delta \phi / \Delta \delta_r');


% ========================================================================
% LATERAL TRANSFER FUNCTIONS (AILERON ONLY)
% ========================================================================
% Aileron primarily excites the Roll mode (fast p response).
% It also has secondary coupling with Dutch Roll and Spiral.
fprintf('\n\n============================================================\n');
fprintf(' LATERAL TRANSFER FUNCTIONS (AILERON ONLY) - C3\n');
fprintf('============================================================\n');

% Compute TFs for each lateral output due to aileron (input index 1).
[n_v_da, ~]   = ss2tf(A_lat, B_lat, [1 0 0 0], D_lat, 1);   % Delta_v / Delta_da
[n_p_da, ~]   = ss2tf(A_lat, B_lat, [0 1 0 0], D_lat, 1);   % Delta_p / Delta_da
[n_r_da, ~]   = ss2tf(A_lat, B_lat, [0 0 1 0], D_lat, 1);   % Delta_r / Delta_da
[n_phi_da, ~] = ss2tf(A_lat, B_lat, [0 0 0 1], D_lat, 1);   % Delta_phi / Delta_da

fprintf('\n1. TRANSFER FUNCTION Delta_v(s) / Delta_da(s):\n');
printsys(n_v_da, den_lat, 's');
analyze_lateral_tf(n_v_da, den_lat, 'v');

fprintf('\n2. TRANSFER FUNCTION Delta_p(s) / Delta_da(s):\n');
printsys(n_p_da, den_lat, 's');
analyze_lateral_tf(n_p_da, den_lat, 'p');

```

```
fprintf('\n3. TRANSFER FUNCTION Delta_r(s) / Delta_da(s):\n');
printsys(n_r_da, den_lat, 's');
analyze_lateral_tf(n_r_da, den_lat, 'r');

fprintf('\n4. TRANSFER FUNCTION Delta_phi(s) / Delta_da(s):\n');
printsys(n_phi_da, den_lat, 's');
analyze_lateral_tf(n_phi_da, den_lat, 'phi');

% Bode diagram for all 4 aileron-to-output transfer functions.
figure('Name', 'C3: Lateral Responses to Aileron');
subplot(2,2,1); bode(tf(n_v_da, den_lat));   grid on; title('\Delta v / \Delta \delta_a');
subplot(2,2,2); bode(tf(n_p_da, den_lat));   grid on; title('\Delta p / \Delta \delta_a');
subplot(2,2,3); bode(tf(n_r_da, den_lat));   grid on; title('\Delta r / \Delta \delta_a');
subplot(2,2,4); bode(tf(n_phi_da, den_lat)); grid on; title('\Delta \phi / \Delta \delta_a');


% ========================================================================
% TD3: ROOT LOCUS ANALYSIS
% ========================================================================
% The Root Locus shows how the closed-loop poles move in the s-plane as the
% feedback gain K varies from 0 to infinity.
% It is used to assess stability margins and to choose an appropriate gain
% for each control channel.
% The 4 selected TFs cover the key longitudinal and lateral control paths:
%   (1) Pitch angle   vs Elevator  -> longitudinal pitch control
%   (2) Roll rate     vs Aileron   -> lateral roll control
%   (3) Yaw rate      vs Rudder    -> directional (yaw) control
%   (4) Axial velocity vs Throttle -> speed/phugoid control
fprintf('\n==========================================================\n');
fprintf('          ROOT LOCUS ANALYSIS - C3\n');
fprintf('==========================================================\n');

figure('Name', 'C3: Root Locus Analysis');

% Subplot (1): Pitch angle vs Elevator - shows SP and Phugoid pole trajectories.
subplot(2,2,1);
rlocus(tf(n_theta, den));
title('Root Locus - \Delta \theta / \Delta \delta_e');
grid on;

% Subplot (2): Roll rate vs Aileron - shows Roll and Dutch Roll pole trajectories.
subplot(2,2,2);
rlocus(tf(n_p_da, den_lat));
title('Root Locus - \Delta p / \Delta \delta_a');
grid on;

% Subplot (3): Yaw rate vs Rudder - shows Dutch Roll and Spiral pole trajectories.
subplot(2,2,3);
rlocus(tf(n_r_dr, den_lat));
title('Root Locus - \Delta r / \Delta \delta_r');
grid on;

% Subplot (4): Axial velocity vs Throttle - shows Phugoid pole trajectory.
subplot(2,2,4);
rlocus(tf(n_u_dt, den_dt));
title('Root Locus - \Delta u / \Delta \delta_t');
grid on;

fprintf('Root Locus plots generated.\n');


```

```
% ==========================================================================
% LOCAL FUNCTIONS FOR AUTOMATIC ANALYSIS (COEFFICIENT EXTRACTION)
% ==========================================================================

% --------------------------------------------------------------------------
% FUNCTION: analyze_longitudinal_tf
% --------------------------------------------------------------------------
% PURPOSE:
%   Extracts and prints the key parameters of a longitudinal transfer
%   function G(s) = num(s) / den(s).
%
% METHOD:
%   1. Denominator poles are sorted by natural frequency to identify:
%      - Phugoid (wn_p, zeta_p)    : the 2 slowest poles
%      - Short Period (wn_sp, zeta_sp) : the 2 fastest poles
%   2. Numerator: the DC gain K and the real zeros (expressed as time
%      constants T_i via the relation root = -1/T_i) are extracted.
%
% INPUTS:
%   num      - numerator polynomial coefficients   [row vector]
%   den      - denominator polynomial coefficients [row vector]
%   var_name - label string for display (e.g. 'u', 'w', 'q', 'theta')
% --------------------------------------------------------------------------
function analyze_longitudinal_tf(num, den, var_name)

    % --- Denominator: compute poles and their natural frequencies ---
    poles = roots(den);
    wn    = abs(poles);
    zeta  = -real(poles) ./ wn;

    % Sort poles by ascending wn to separate slow (PH) from fast (SP).
    [~, idx] = sort(wn);
    wn_p    = mean(wn(idx(1:2)));      % Phugoid natural frequency
    zeta_p  = mean(zeta(idx(1:2)));    % Phugoid damping ratio
    wn_sp   = mean(wn(idx(3:4)));      % Short Period natural frequency
    zeta_sp = mean(zeta(idx(3:4)));    % Short Period damping ratio

    % --- Numerator: extract gain K (leading non-zero coefficient) ---
    idx_first_non_zero = find(abs(num) > 1e-10, 1);
    if isempty(idx_first_non_zero)
        K = 0;
    else
        K = num(idx_first_non_zero);
    end

    % Extract real zeros and convert to time constants: T = -1 / root
    zeros_num  = roots(num);
    real_zeros = zeros_num(imag(zeros_num) == 0);
    T_consts   = -1 ./ real_zeros;

    % --- Print results as a formatted table ---
    fprintf('  ------------------------------------------------------------\n');
    fprintf('  | Parameter      | Symbol       | Value                    |\n');
    fprintf('  ------------------------------------------------------------\n');
    fprintf('  | Gain           | K_%-5s      | %10.4e                |\n', var_name, K);
    if ~isempty(T_consts)
        for i = 1:length(T_consts)
            fprintf('  | Time Const %d   | T_%d          | %10.4f s              |\n', i, i, T_consts(i));
        end
    end
    fprintf('  | Phugoid        | wn_p         | %10.4f rad/s          |\n', wn_p);
    fprintf('  |                | zeta_p       | %10.4f                |\n', zeta_p);
    fprintf('  | Short Period   | wn_sp        | %10.4f rad/s          |\n', wn_sp);
    fprintf('  |                | zeta_sp      | %10.4f                |\n', zeta_sp);
```

```
    fprintf('  ------------------------------------------------------------\n');
end


% ----------------------------------------------------------------------
% FUNCTION: analyze_lateral_tf
% ----------------------------------------------------------------------
% PURPOSE:
%   Extracts and prints the key parameters of a lateral transfer
%   function G(s) = num(s) / den(s).
%
% METHOD:
%   1. Denominator poles are classified by imaginary part:
%      - Complex pair  -> Dutch Roll (wn_dr, zeta_dr)
%      - Real poles    -> Spiral time constant T_spiral (smallest |lambda|)
%                         Roll time constant T_roll (largest |lambda|)
%   2. Numerator: the DC gain K is extracted.
%
% INPUTS:
%   num      - numerator polynomial coefficients   [row vector]
%   den      - denominator polynomial coefficients [row vector]
%   var_name - label string for display (e.g. 'v', 'p', 'r', 'phi')
% ----------------------------------------------------------------------
function analyze_lateral_tf(num, den, var_name)

    % --- Denominator: identify poles ---
    poles = roots(den);

    % Dutch Roll: complex conjugate pair -> extract wn and zeta.
    complex_poles = poles(imag(poles) ~= 0);
    if ~isempty(complex_poles)
        wn_dr   = abs(complex_poles(1));
        zeta_dr = -real(complex_poles(1)) / wn_dr;
    else
        wn_dr = 0; zeta_dr = 0;
    end

    % Roll and Spiral: real poles.
    % T = -1/lambda (time constant of a first-order mode).
    % Ascending sort by |lambda|: index 1 = Spiral (slow), end = Roll (fast).
    real_poles = poles(imag(poles) == 0);
    [~, idx_real] = sort(abs(real_poles));
    if length(real_poles) >= 2
        T_spiral = -1 / real_poles(1);      % Spiral time constant [s]
        T_roll   = -1 / real_poles(end);    % Roll time constant   [s]
    else
        T_spiral = 0; T_roll = 0;
    end

    % --- Numerator: extract gain K ---
    idx_first_non_zero = find(abs(num) > 1e-10, 1);
    if isempty(idx_first_non_zero)
        K = 0;
    else
        K = num(idx_first_non_zero);
    end

    % --- Print results as a formatted table ---
    fprintf('  ------------------------------------------------------------\n');
    fprintf('  | Parameter      | Symbol       | Value                    |\n');
    fprintf('  ------------------------------------------------------------\n');
    fprintf('  | Gain           | K_%-5s      | %10.4e                |\n', var_name, K);
    fprintf('  | Dutch Roll     | wn_dr        | %10.4f rad/s          |\n', wn_dr);
    fprintf('  |                | zeta_dr      | %10.4f                |\n', zeta_dr);
    fprintf('  | Spiral Mode    | T_spiral     | %10.4f s              |\n', T_spiral);
```

```
    fprintf('  | Roll Mode      | T_roll       | %10.4f s             |\n', T_roll);
    fprintf('  ------------------------------------------------------------\n');
end


% ========================================================================
% SAS DESIGN (NUMERICAL METHOD - WITHOUT SYMBOLIC TOOLBOX)
% ========================================================================
% Goal: redesign the Short Period mode to meet a target specification:
%   wn_desired   = 2.95 rad/s
%   zeta_desired = 0.52
% The Phugoid poles are kept unchanged (they are far from the target bandwidth).
%
% Method: full-state feedback  delta_e = -K * x
%   The augmented closed-loop matrix becomes: A* = A - B*K
%   Ackermann's formula is used to compute K analytically.
% ========================================================================
fprintf('\n--- SAS DESIGN (STABILITY AUGMENTATION SYSTEM) ---\n');

% --- Step 1: Verify that the system is controllable ---
% The controllability matrix must be full rank for pole placement to work.
V_ctrl = ctrb(A_long, B_long(:,1));
if rank(V_ctrl) == length(A_long)
    disp('The system is controllable. Computing numerical feedback gain K...');
else
    disp('The system is not controllable.');
end

% --- Step 2: Define the desired closed-loop poles ---
wn_des   = 2.95;
zeta_des = 0.52;
% Short Period target poles (complex conjugate pair):
p_sp1 = -zeta_des*wn_des + 1i*wn_des*sqrt(1-zeta_des^2);
p_sp2 = -zeta_des*wn_des - 1i*wn_des*sqrt(1-zeta_des^2);

% Retain the current Phugoid poles unchanged.
p_ph = lambdas_long(idx_l(3:4));

% Full target pole vector (4 poles for the 4x4 system).
poles_target = [p_sp1; p_sp2; p_ph(1); p_ph(2)];

% --- Step 3: Compute feedback gain K via Ackermann's formula ---
% acker() places the poles of (A - B*K) at the specified locations.
K_final = acker(A_long, B_long(:,1), poles_target);
fprintf('Computed gain vector K: [%.4f, %.4f, %.4f, %.4f]\n', K_final);

% --- Step 4: Build the augmented closed-loop matrix A* = A - B*K ---
A_star_final = A_long - B_long(:,1)*K_final;
fprintf('\nNew augmented matrix A*:\n');
disp(A_star_final);

% --- Step 5: Verify that the placed poles match the target ---
% damp() returns natural frequencies and damping ratios of each pole.
[wn_new, zeta_new] = damp(A_star_final);
fprintf('Verification of new Short Period parameters:\n');
[~, idx_new] = sort(wn_new, 'descend');
fprintf('-> wn = %.2f rad/s (Target: 2.95)\n', wn_new(idx_new(1)));
fprintf('-> zeta = %.2f (Target: 0.52)\n', zeta_new(idx_new(1)));


% ========================================================================
% TRANSFER FUNCTIONS BY APPROXIMATION (NUMERICAL METHOD)
% ========================================================================
% The Short Period approximation reduces the 4th-order longitudinal system
% to a 2nd-order system by neglecting the slow Phugoid dynamics.
```

```
% The reduced state vector is [w, q], and the input is delta_e.
%
% The reduced matrices are extracted directly from A_long and B_long:
%   A_SP = [ Zw,  U0 ]    B_SP = [ Z_de ]
%          [ Mw,  Mq ]           [ M_de ]
%
% This approximation is valid at short time scales (t << 1/wn_PH).
fprintf('\n--- TRANSFER FUNCTIONS BY APPROXIMATION (SP MODES) ---\n');

% Extract aerodynamic derivatives from the full state matrix.
Zw_app  = A_long(2,2);   % Z_w  : vertical force due to w
U0_app  = A_long(2,3);   % U0   : forward speed (appears in Z equation)
Mw_app  = A_long(3,2);   % M_w  : pitching moment due to w
Mq_app  = A_long(3,3);   % M_q  : pitching moment due to q (pitch damping)
Zde_app = B_long(2,1);   % Z_de : elevator effect on vertical force
Mde_app = B_long(3,1);   % M_de : elevator effect on pitching moment

% Build the reduced 2x2 SP state-space model.
A_SP_app = [Zw_app, U0_app; Mw_app, Mq_app];
B_SP_app = [Zde_app; Mde_app];

% Compute the two SP approximated TFs from the reduced model.
[num_w, den_sp] = ss2tf(A_SP_app, B_SP_app, [1 0], 0, 1);   % Delta_w / Delta_de
[num_q, ~]      = ss2tf(A_SP_app, B_SP_app, [0 1], 0, 1);   % Delta_q / Delta_de

fprintf('TF Delta w / Delta de (Approximation):\n');
tf_w_app = tf(num_w, den_sp);
display(tf_w_app);

fprintf('TF Delta q / Delta de (Approximation):\n');
tf_q_app = tf(num_q, den_sp);
display(tf_q_app);


% ========================================================================
% SAS DESIGN (NUMERICAL METHOD - WITHOUT SYMBOLIC TOOLBOX)
% ========================================================================
fprintf('\n--- SAS DESIGN (STABILITY AUGMENTATION SYSTEM) ---\n');

V_ctrl = ctrb(A_long, B_long(:,1));
if rank(V_ctrl) == length(A_long)
    disp('The system is controllable. Computing numerical feedback gain K...');
else
    disp('The system is not controllable.');
end

wn_des   = 2.95;
zeta_des = 0.52;
p_sp1 = -zeta_des*wn_des + 1i*wn_des*sqrt(1-zeta_des^2);
p_sp2 = -zeta_des*wn_des - 1i*wn_des*sqrt(1-zeta_des^2);

p_ph = lambdas_long(idx_l(3:4));

poles_target = [p_sp1; p_sp2; p_ph(1); p_ph(2)];

K_final = acker(A_long, B_long(:,1), poles_target);

fprintf('Computed gain vector K: [%.4f, %.4f, %.4f, %.4f]\n', K_final);

A_star_final = A_long - B_long(:,1)*K_final;
fprintf('\nNew augmented matrix A*:\n');
disp(A_star_final);

[wn_new, zeta_new] = damp(A_star_final);
fprintf('Verification of new Short Period parameters:\n');
```

```
[~, idx_new] = sort(wn_new, 'descend');
fprintf('-> wn = %.2f rad/s (Target: 2.95)\n', wn_new(idx_new(1)));
fprintf('-> zeta = %.2f (Target: 0.52)\n', zeta_new(idx_new(1)));


% ========================================================================
% TRANSFER FUNCTIONS BY APPROXIMATION (NUMERICAL METHOD)
% ========================================================================
fprintf('\n--- TRANSFER FUNCTIONS BY APPROXIMATION (SP MODES) ---\n');

Zw_app  = A_long(2,2);
U0_app  = A_long(2,3);
Mw_app  = A_long(3,2);
Mq_app  = A_long(3,3);
Zde_app = B_long(2,1);
Mde_app = B_long(3,1);

A_SP_app = [Zw_app, U0_app; Mw_app, Mq_app];
B_SP_app = [Zde_app; Mde_app];

[num_w, den_sp] = ss2tf(A_SP_app, B_SP_app, [1 0], 0, 1);
[num_q, ~]      = ss2tf(A_SP_app, B_SP_app, [0 1], 0, 1);

fprintf('TF Delta w / Delta de (Approximation):\n');
tf_w_app = tf(num_w, den_sp);
display(tf_w_app);

fprintf('TF Delta q / Delta de (Approximation):\n');
tf_q_app = tf(num_q, den_sp);
display(tf_q_app);

fprintf('\n==========================================================\n');
fprintf('              END OF ANALYSIS\n');
fprintf('==========================================================\n');
```

Listing 1: Main MATLAB Script for C3 Stability Analysis

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Appendix 2: *Linear Longitudinal and Lateral Stability Matrix of Aircraft*. Technical Documentation provided in the homework subject.